

Endogenously (Non-)Ricardian Beliefs*

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Abstract

This paper revisits the impact of fiscal shocks on inflation dynamics. Traditionally, such relationships have been attributed to exogenous shifts in policy regimes. In contrast, we focus on a novel belief formation mechanism in which Ricardian equivalence, or its failure, arises endogenously. Agents choose between two parsimonious forecasting models that generate qualitatively different responses to fiscal shocks, and select the model that best fits the data. Ricardian equivalence is fragile, arising only under knife-edge conditions on beliefs and the fiscal environment. Employing least squares learning in an estimated New Keynesian model, we identify endogenous fluctuations between Ricardian and non-Ricardian regimes in the U.S., with non-Ricardian beliefs driving the high inflation of the 1970s and 2021-22, and temporary escapes to Ricardian beliefs accounting for the intervening disinflation.

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1 Introduction

This paper offers a new perspective on the role of fiscal shocks in shaping inflation dynamics by investigating the influence of endogenous changes in beliefs. Much of the existing literature examines policy regime changes or explicit assumptions on beliefs, that violate Ricardian equivalence. We develop a model instead that highlights the importance of evolving beliefs as a key driver of departures from Ricardian equivalence. With the exception of an edge case, households typically exhibit non-Ricardian beliefs that impact consumption and, in turn, inflation. In doing so, we provide new insights into the complex relationship between fiscal shocks, inflation, and the beliefs held by economic agents. Our main finding, both theoretical and empirical, is as follows: evolving beliefs on their own generate fluctuations between non-Ricardian and Ricardian regimes.

Our approach adopts a “restricted perceptions” framework, in which economic agents possess imperfect knowledge of the economy and its dynamics (see, [Evans and Honkapohja, 2001](#); [Branch and McGough, 2018](#); [Woodford, 2013](#)). This framework allows us to explore the implications of agents’ imperfect knowledge on inflation dynamics and the emergence of beliefs that do not necessarily correspond to the true state of the world. By focusing on the role of endogenous beliefs in driving inflation patterns, we contribute to a deeper understanding of the mechanisms through which fiscal shocks impact inflation within a conventional policy stance of a monetary authority focused on price stability and a fiscal authority adhering to long-run government solvency.

In our analysis, we utilize a standard New Keynesian framework as a laboratory for our ideas. The starting point builds on [Evans et al. \(2012\)](#), [Eusepi and Preston \(2018\)](#), and [Woodford \(2013\)](#), who show that Ricardian Equivalence can fail in models where agents have imperfect knowledge about aggregate variables and whether the path of future primary budget surpluses is sufficient to satisfy the government’s intertemporal budget constraint. In these papers, imperfect knowledge and non-Ricardian beliefs arise because least-squares learning generates temporary fluctuations around the rational expectations equilibrium. In contrast, our agents form subjective expectations about payoff-relevant aggregate variables based on parsimonious forecasting models and update these expectations using least-squares learning. Crucially, agents observe headline fiscal quantities – either the primary surplus or the stock of debt – but not the underlying drivers of fiscal policy. Following [Woodford \(2013\)](#), this incomplete fiscal information is the source of misspecification. We allow agents to choose between two

parsimonious forecasting models – one that includes the stock of debt and is conditionally consistent with Ricardian beliefs, and another that includes the primary budget surplus but is inherently inconsistent with Ricardian beliefs. Agents select the model that performs best in terms of mean-squared error.

A theory of predictor choice between parsimonious models achieves two aims. First, it captures that rational expectations may be prohibitively expensive in the presence of significant computational and cognitive limitations. Still, with least-squares learning and endogenous model choice, beliefs satisfy a set of cross-equation restrictions, a salient feature of rational expectations. Second, the two forecasting models generate qualitatively different responses to fiscal shocks. Our theoretical analysis shows that the debt-based model is conditionally Ricardian. When fiscal shocks are transitory, it delivers weak Ricardian equivalence in which predetermined debt does not enter the decision rules for output and inflation. The surplus-based model, by contrast, is inherently non-Ricardian. The endogenous distribution of agents across models governs the strength of the non-Ricardian channel, and the impulse responses to a fiscal shock can range from nearly, or quantitatively, Ricardian to strongly non-Ricardian depending on this distribution. Estimating our model on U.S. data suggests that a non-Ricardian equilibrium is a stable limit point to agents’ learning and model selection. Occasionally, though, the economy escapes the non-Ricardian equilibrium and quantitatively Ricardian beliefs emerge for a stretch of time.

Thus, a central theoretical and quantitative result is the fragility of Ricardian equivalence. It holds if and only if all agents use the debt-based model *and* fiscal shocks are serially uncorrelated. The presence of even an infinitesimal fraction of surplus-based forecasters, *or* any persistence in the fiscal shock, renders beliefs non-Ricardian. This fragility suggests that a non-Ricardian equilibrium is a plausible outcome of the learning process. However, least-squares learning can also generate endogenous fluctuations between Ricardian and non-Ricardian regimes.

Our empirical investigation, conducted using Bayesian estimation techniques, reveals the presence of non-Ricardian beliefs in the U.S. economy, with the extent of such beliefs fluctuating over time. Notably, we find that inflationary periods during the late 1960s and 1970s were driven by a growing prevalence of non-Ricardian beliefs, while disinflation and subsequent low inflation periods resulted from temporary shifts to a Ricardian regime. This finding suggests that the key question is not why fiscal shocks matter at certain times, but rather why inflation is not always a fiscal phenomenon.

Drawing parallels to the “Conquest” model by [Sargent \(1999\)](#), [Cho et al. \(2002\)](#), and [Sargent et al. \(2006\)](#), our study underscores the pivotal role of evolving beliefs and regime shifts in inflation dynamics. These earlier models demonstrate how a central bank’s time consistency problem, combined with occasional shocks that alter the policymaker’s view on the inflation-unemployment trade-off, can temporarily reduce inflation to its efficient rate. Our findings suggest a similar dynamic, though with emphasis on private sector expectations. The non-Ricardian equilibrium is sustained by a feedback loop. Surplus-based forecasters generate inflation responses to fiscal variables that, in turn, confirm those forecasts. An escape to the quantitatively Ricardian regime occurs when a sustained sequence of shocks – the “most likely unlikely” path – disrupts this loop by pushing inflation in the opposite direction from what the surplus-based model predicts and towards the path predicted by the debt-based model. The estimated model implies that model selection responds faster than coefficient updating, so the population share shifts rapidly toward the debt-based model, weakening the non-Ricardian channel in a self-reinforcing dynamic. The escape is temporary: eventually, the mean learning dynamics reassert themselves and non-Ricardian beliefs re-emerge. The estimated shock paths confirm that this mechanism was at work during the late 1970s to early 1980s disinflation, and the resurgence of non-Ricardian beliefs in 2021-22 is consistent with the predicted return to the non-Ricardian equilibrium.

A deliberate modeling choice is to hold the monetary and fiscal policy rules fixed throughout. Competing explanations for the 1970’s and 1980’s inflation experiences range from indeterminacy under passive monetary policy ([Clarida et al., 2000](#)) to exogenous Markov switching in policy regimes ([Davig and Leeper, 2006](#); [Bianchi and Ilut, 2017](#)) to policymaker learning about the Phillips curve tradeoff ([Cho et al., 2002](#); [Primiceri, 2006](#)). A common feature of these approaches is that the source of regime change lies on the policy side, whether through exogenous switches in policymaker preferences or through the policymaker’s own learning process. We follow the approach in [Eusepi and Preston \(2018\)](#) and hold policy constant in order to highlight that private-sector belief dynamics alone generate endogenous fluctuations between Ricardian and non-Ricardian regimes. This is a sharper expression of the proposed belief mechanism as any regime-like behavior in the estimates arises from the evolution of endogenous private-sector expectations, not from assumed or estimated changes in the policy rule. A rolling-window exercise confirms that non-Ricardian belief dynamics are robust to allowing for time-varying policy coefficients.

The remainder of the paper is organized as follows. Section 2 presents the economic environment and the theory of restricted perceptions. Section 3 develops the fragility of Ricardian equivalence, characterizes the qualitatively different impulse responses under the two forecasting models, and identifies the conditions under which least-squares learning reaches non-Ricardian limit points. Section 4 presents the Bayesian model estimates and discusses the main empirical findings. Finally, sections 5 and 6 review the related literature and provide concluding remarks.

2 Model: specification and estimation

We begin by describing the economic environment, beliefs, and equilibrium concept.

2.1 Woodford's (2013) model

The setting is a New Keynesian model, based on [Woodford \(2013\)](#), where households and firms have subjective beliefs about payoff-relevant aggregate variables.¹ Given these beliefs, households choose consumption, leisure, and one-period government debt, the only asset available to households, to solve their intertemporal optimization problem. In [Woodford's \(2013\)](#) framework, households turn over wage-setting and labor supply decisions to a union and are obligated to supply labor to a firm on the union's terms. Households also receive a lump-sum transfer of their share in firm profits.² This is a stylized assumption that renders the household's consumption rule analogous to the one in a model where the household receives a stochastic endowment. However, because firms are monopolistically competitive and face a [Calvo \(1983\)](#) nominal pricing friction, there is endogenous variation in hours and output. All exogenous shocks follow stationary AR(1) processes.

HOUSEHOLDS. [Woodford \(2013\)](#) derives an individual's consumption function,

$$c_t^i = (1 - \beta)b_t^i + \sum_{T=t}^{\infty} \beta^{T-t} E_t^i \{ (1 - \beta)(Y_T - \tau_T) - \beta\sigma(i_T - \pi_{T+1}) + (1 - \beta)s_b(\beta i_T - \pi_T) - \beta(\bar{c}_{T+1} - \bar{c}_T) \}, \quad (1)$$

or, written recursively as

$$c_t^i = (1 - \beta) [b_t^i + (Y_t - \tau_t) - s_b \pi_t] - \beta[\sigma - (1 - \beta)s_b]i_t + \beta\bar{c}_t + \beta E_t^i v_{t+1}^i, \quad (2)$$

where v_t^i is a subjective composite variable that comprises all payoff-relevant aggregate

¹The model environment is a simplified version of [Eusepi and Preston \(2018\)](#) where there are two assets, one-period government bonds in zero net-supply and longer maturity bonds. [Eusepi and Preston \(2018\)](#) demonstrate the role that maturity structure, combined with imperfect knowledge and learning, can play in generating non-Ricardian wealth effects. Payoff relevant variables are those that directly enter an agent's decision problem.

²The shares in firms are illiquid, which makes government debt the only storable good. [Eusepi and Preston \(2018\)](#) show that this assumption is consequential for non-Ricardian beliefs. Though we abstract from these issues, it is worth bearing in mind that the issue is relevant within our non-Ricardian equilibrium.

variables over which a household formulates subjective beliefs:

$$v_t^i = (1 - \beta)(Y_t - \tau_t) - [\sigma - (1 - \beta)s_b](\beta i_t - \pi_t) - (1 - \beta)\bar{c}_t + \beta E_t^i v_{t+1}^i.$$

The variables, written as log-deviations from steady-state, $b_t^i, Y_t, \pi_t, i_t, \tau_t, \bar{c}_t$ are, respectively, the individual's holdings of real government debt, aggregate output, the inflation rate, the nominal interest rate, lump-sum taxes, and a preference shock. The government uses lump-sum taxes and debt to finance its consumption of an exogenous sequence G_t . The parameter $0 < \beta < 1$ is the discount rate, σ is the elasticity of intertemporal substitution, and $s_b \equiv \bar{b}/\bar{Y}$ is the steady-state debt-to-GDP ratio.

Consumption function (1) takes a standard permanent income formulation relating consumption to the annuitized present-value of financial and non-financial income.³ The fiscal policy instrument is the real primary surplus $s_t \equiv \tau_t - G_t$. Defining effective (net-of-interest) surpluses $\hat{s}_t \equiv s_t - s_b(\beta i_t - \pi_t)$ and rearranging (1) yields

$$\begin{aligned} c_t^i = \bar{c}_t + \sum_{T=t}^{\infty} \beta^{T-t} E_t^i \{ (1 - \beta)(Y_T - \bar{c}_T - G_T) - \beta \sigma (i_T - \pi_{T+1}) \} \\ + (1 - \beta) \left[b_t^i - \sum_{T=t}^{\infty} \beta^{T-t} E_t^i \{ \hat{s}_T \} \right]. \end{aligned} \quad (3)$$

The first line in (3) is the Ricardian component capturing the effects of the expected present value of disposable income and intertemporal substitution resulting from variations in the (perceived) *ex-ante* real interest rate (parameterized by σ). The second line is the non-Ricardian component capturing the wealth effect of bond holdings and income effect resulting from the difference in the expected present values of primary surpluses and *ex-post* real returns on government bond holdings.

Following Eusepi and Preston (2018), Woodford (2013) derives equation (1) without assuming that individuals have structural knowledge about the government's intertemporal budget constraint. Even though there is passive fiscal policy, individuals do not necessarily know this or the other structural features of the economy, so that they may have imperfect knowledge about the structural form of the government's endogenously determined budget constraint. Instead, they form subjective beliefs over the evolution of aggregate variables.⁴ If they get those beliefs right, they will properly account

³Derivation of (1) is entirely standard. We refer readers to the supplementary appendix and Woodford (2013) for details.

⁴On the surface, formulating expectations over future v_t^i seems to be adopting the Euler equa-

for the evolution of debt and effective surpluses, and their beliefs will be Ricardian. Otherwise, beliefs may be non-Ricardian.

Ricardian beliefs arise with the following condition on *beliefs*

$$E_t^i \sum_{T=t}^{\infty} \beta^{T-t} \{\hat{s}_T\} = b_t. \quad (4)$$

By imposing Ricardian beliefs onto the consumption rule (1), (4) directly imposes that the household properly forecasts the path for future effective surpluses. Without assuming (4), there are potentially non-Ricardian effects or net-wealth effects. Ricardian beliefs, therefore, lead to a consumption rule that depends only on the household's subjective beliefs about future paths for disposable income and real interest rates. Conversely, by not *a priori* imposing Ricardian beliefs, households may perceive their current bond holdings and anticipated future real returns as real wealth, and a change in the expected path for future surpluses can have a real effect on consumption. See Appendix 6 or Woodford (2013) for details.⁵

FIRMS. Monopolistically competitive firms face a nominal pricing friction based on Calvo (1983). An individual firm j produces a differentiated good. With probability $0 < \alpha < 1$, it will adjust its previous price by the long-run target rate of inflation, assumed to be zero, and with probability $1 - \alpha$, a firm receives an idiosyncratic signal to (optimally) reset the price. A firm j that can optimally reset price $p_t^*(j)$, relative to the previous aggregate price level p_{t-1} , will do so to satisfy the log-linear approximated first-order condition, written recursively,

$$p_t^*(j) = (1 - \alpha\beta) (E_t^j p_t^{\text{opt}} - p_{t-1}) + (\alpha\beta) E_t^j p_{t+1}^*(j) + (\alpha\beta) \pi_t,$$

tion approach of one-step ahead forecasting and decision-making. However, the derivation of the consumption function and v_t^i is based on the infinite-horizon approach where the household's consumption/savings decisions solve their entire sequence of Euler equations, flow budget constraints, and transversality condition given their subjective beliefs. We show below how these consumption rules can be aggregated with heterogeneous agents.

⁵In all of the analysis below, the fiscal rule is *ex post* Ricardian, i.e., real primary surpluses will satisfy the government's intertemporal constraint. However, out of equilibrium, non-Ricardian beliefs could be consistent with explosive debt. The consequences of this, and its implications for strategic behavior, is an old issue in the fiscal theory of the price level literature (cf., Bassetto, 2002).

where $E_t^j p_T^{\text{opt}}$ is the perceived optimal price in T . The aggregate inflation dynamics are

$$\pi_t = (1 - \alpha)p_t^*, \quad \text{where} \quad p_t^* \equiv \int p_t^*(j) dj. \quad (5)$$

POLICY. Monetary policy is described by a [Taylor \(1993\)](#) rule,

$$i_t = \phi_\pi \pi_t + \phi_y y_t + w_t, \quad (6)$$

where the monetary policy shock follows $w_t = \rho_w w_{t-1} + \varepsilon_{wt}$, $\varepsilon_{wt} \sim N(0, \sigma_w^2)$.

A [Leeper \(1991\)](#) rule for the real primary surplus characterizes fiscal policy:

$$s_t = \phi_b b_t + z_t, \quad (7)$$

where the surplus shock is $z_t = \rho_z z_{t-1} + \varepsilon_{zt}$, $\varepsilon_{zt} \sim N(0, \sigma_z^2)$. The government also faces a flow budget constraint

$$b_{t+1} = \beta^{-1}[b_t - s_b \pi_t - s_t] + s_b i_t. \quad (8)$$

The steady-state debt-to-GDP ratio s_b plays a role in the results presented below. When $s_b = 0$, the bond and primary surplus paths are exogenous, while $s_b > 0$ implies that they are endogenous and affected, in part, by monetary policy.⁶

Throughout, the analysis focuses on the *active* monetary and *passive* fiscal policy regime:

$$1 < \phi_\pi + \frac{1 - \beta}{\kappa} \phi_y \\ (1 - \beta) < \phi_b < 1.$$

Under the benchmark *rational expectations hypothesis*, there is local determinacy (see, [Leeper, 1991](#)) implying that this locally unique *rational expectations equilibrium* displays Ricardian equivalence and is stable under least-squares learning (see, [Evans and Honkapohja, 2007](#)).

⁶This formulation arises in a cashless environment that allows us to abstract from the effect of monetary aggregates appearing in the consolidated budget constraint.

2.2 Temporary equilibrium with heterogeneous beliefs

The income-expenditure identity is given by

$$Y_t = \int c_t^i di + G_t. \quad (9)$$

Combining (2) and (9) with the bond-market clearing condition $b_t \equiv \int b_t^i di$, computing $v_t \equiv \int v_t^i di$, and averaging over expectations, allows us to express aggregate demand as the “IS equation” without *a priori* imposing Ricardian beliefs:

$$y_t = g_t - \sigma i_t + (1 - \beta)b_{t+1} + \hat{E}_t v_{t+1}, \quad (10)$$

where $y_t \equiv Y_t - Y_t^n$, $g_t \equiv \bar{c}_t + G_t - Y_t^n$ is a composite exogenous disturbance, such that $g_t = \rho_g g_{t-1} + \varepsilon_{gt}$, $\varepsilon_{gt} \sim N(0, \sigma_g^2)$. The aggregate expectations operator $\hat{E}$ is defined as $\hat{E}_t(x) = \int E_t^i(x) di$, for any variable x .

Given that heterogeneous beliefs lead to a non-degenerate cross-sectional wealth and consumption distribution, some readers may be surprised that individual household bond holdings do not appear in the aggregate demand equation. However, this results from several simplifying assumptions in [Woodford \(2013\)](#). First, assumptions about the labor market and the distribution of firm profits imply that future non-financial income is a proportion of aggregate output beyond the agent’s control. So, household consumption decisions depend on expectations about variables beyond their control. Second, in this setting, a temporary equilibrium path consists of local perturbations around a non-stochastic steady-state in which all agents hold identical beliefs. In this sense, households’ beliefs are not *too* heterogeneous. Finally, household debt holdings enter linearly in the approximated economy, and, as a result, individual bond holdings do not matter for the aggregate output path.⁷

On the firm side, applying the law of iterated expectations and aggregating across all firms results in an aggregate New Keynesian Phillips Curve:

$$\pi_t = (1 - \alpha) \beta \hat{E}_t p_{t+1}^* + \kappa y_t + u_t,$$

where $\kappa \equiv [(1 - \alpha)(1 - \alpha\beta)\zeta]/\alpha$, and the cost-push shock is $u_t \equiv \{[(1 - \alpha)(1 - \alpha\beta)]/\alpha\}\mu_t$,

⁷An extension to a setting where heterogeneous expectations give rise to a non-trivial aggregate role to the cross-sectional wealth distribution is potentially important but beyond the scope of the present study.

with $u_t = \rho_u u_{t-1} + \varepsilon_{ut}$, $\varepsilon_{ut} \sim N(0, \sigma_u^2)$.

We now define a *temporary equilibrium* for this economy.

Definition 1 *Given a distribution of beliefs $(E_t^i v_{t+1}, E_t^i p_{t+1}^*)_i$, a temporary equilibrium is a triple (b_{t+1}, π_t, y_t) and a policy (s_t, i_t) so that the bond and goods markets clear and the government budget constraint is satisfied. In particular, the following equations are satisfied*

$$\begin{aligned} b_{t+1} &= \beta^{-1} [b_t - s_b \pi_t - s_t] + s_b i_t \\ \pi_t &= (1 - \alpha) \beta \hat{E}_t p_{t+1}^* + \kappa y_t + u_t \\ y_t &= g_t - \sigma i_t + (1 - \beta) b_{t+1} + \hat{E}_t v_{t+1} \\ v_t &= (1 - \beta) (b_{t+1} - b_t) - \sigma (i_t - \pi_t) + \hat{E}_t v_{t+1}. \end{aligned}$$

2.3 Model misspecification

This section details expectation formation.

2.3.1 A restricted perceptions approach

Under full-information rational expectations, the equilibrium law of motion takes the form

$$\begin{bmatrix} \pi_t \\ v_t \\ y_t \end{bmatrix} = A \begin{bmatrix} b_t \\ s_t \end{bmatrix} + B \begin{bmatrix} g_t \\ u_t \\ w_t \end{bmatrix},$$

where A and B are conformable coefficient matrices. It follows that to formulate rational expectations, the agents adopt linear forecast rules that depend on the stock of beginning-of-period debt, b_t and the primary surplus, s_t .

In our framework, agents navigate their information landscape, optimizing their statistical forecasts based on their available information and abilities. Drawing from the econometric learning literature and the ‘cognitive consistency principle’ as outlined by [Evans and Honkapohja \(2001\)](#), we model agents as skilled economists: they favor well-specified yet parsimonious econometric models.⁸ Parsimony is an appropriate response to the complexity of economic forecasting and the inherent limitations on degrees of freedom. Moreover, as [Andre et al. \(2022\)](#) show, forecasters often adopt nar-

⁸The principle ‘holds that economic agents should be about as smart as economists’ ([Evans and McGough, 2020](#)).

ratives that guide their selective use of information, i.e., they do not use all information even when it is available. More broadly, [Molavi \(2024\)](#) shows that agents constrained to use simple forecasting models optimally anchor their decisions to a limited set of state variables, generating systematic forecast biases closely related to our restricted perceptions approach. Our model captures this dynamic, allowing for the endogenous evolution of these narratives over time.

Explicitly, we assume that computational and cognitive limitations render full information rational expectations prohibitively expensive.⁹ Consequently, we assume that agents forecast using one of two parsimonious models, each incorporating a single fiscal variable – either s_t or b_t . This approach not only enables a straightforward formalization of endogenously (non-)Ricardian beliefs but also aligns with empirical studies that utilize a limited number of fiscal indicators ([Favero and Giavazzi, 2012](#)).

By adopting this parsimonious modeling approach, we facilitate the emergence of a restricted perceptions equilibrium (RPE) wherein agents’ expectations come from the linear projection of aggregate variables onto the space spanned by their parsimonious models. In a misspecification equilibrium (ME), the population distribution across the two forecasting models is determined endogenously by agents’ discrete choices. Combined, the equilibrium distribution allows for the possibility of Ricardian equivalence emerging endogenously from agents’ model choices.

While our equilibrium concept relaxes the model consistency of rational expectations, it maintains many cross-equation restrictions characteristic of rational expectations models. Our approach is particularly suited to the study of fiscal shocks and inflation, given its ability to capture endogenous regime shifts and its roots in the long tradition of empirical research using a limited set of fiscal indicators.

Admittedly, the restriction on regressors in agents’ econometric models is somewhat ad hoc. However, alternative specifications for misspecified beliefs would require agents to possess extensive knowledge about the structural features of the economy – knowledge that extends beyond their models and includes the beliefs, constraints, and decision rules of other agents, such as the government’s commitment to satisfying solvency constraints. Our approach, by contrast, offers a more natural and tractable way to model the influence of endogenous beliefs on inflation dynamics.

⁹It would be straightforward, but very costly to estimate, an extension that includes a third choice which consists of all variables but at a cost. In equilibrium the three models would produce mean-square forecast errors of a similar magnitude. Thus, for a sufficiently large cost there would continue to be a distribution across all models and each model, including the correctly specified model, would be non-Ricardian. This latter fact is a consequence of the model’s self-referentiality.

The two parsimonious models differ in which fiscal variable agents condition on but share a common information structure. All agents directly observe the non-fiscal shocks (g_t, u_t, w_t) . The composite demand disturbance g_t includes preference shifts and government spending that households experience directly while the cost-push shock u_t reflects markup variation observable to price-setting firms. The monetary policy shock w_t is the deviation from the Taylor rule, also naturally observable given central bank communication and transparency. By contrast, the exogenous fiscal shock z_t – the component of the primary surplus not explained by the systematic response to debt – is not directly observable. Following [Woodford \(2013\)](#), agents observe fiscal aggregates but not their structural decomposition. The agents see either the surplus s_t or the stock of debt b_t , but not the decomposition $s_t = \phi_b b_t + z_t$.

2.3.2 Equilibrium

Expectations come from one of the following forecasting models, or, perceived laws of motion (PLM):

$$\begin{aligned} PLM_s : \mathbf{Z}_t &= \boldsymbol{\psi}^{s'} X_{t-1}^s + \eta_t^s \Rightarrow E_t^s \mathbf{Z}_{t+1} = \boldsymbol{\psi}^{s'} X_t^s \\ PLM_b : \mathbf{Z}_t &= \boldsymbol{\psi}^{b'} X_{t-1}^b + \eta_t^b \Rightarrow E_t^b \mathbf{Z}_{t+1} = \boldsymbol{\psi}^{b'} X_t^b, \end{aligned}$$

where $\mathbf{Z}_t' = (v_t, p_t^*, b_{t+1})$, $X_t^s = (s_t, g_t, u_t, w_t)'$, $X_t^b = (b_t, g_t, u_t, w_t)'$, and η_t^k is a model-specific perceived noise term for $k = \{s, b\}$.¹⁰ The coefficient matrix is

$$\boldsymbol{\psi}^k = (\psi^k, \Gamma^k),$$

$\psi^k = (\psi_v^k, \psi_p^k)$ and Γ^k is the coefficient for b_{t+1} .¹¹ In a *restricted perceptions equilibrium* (RPE) the coefficients will satisfy the least-squares orthogonality condition:

$$E X_{t-1}^k \left(\mathbf{Z}_t - \boldsymbol{\psi}^{k'} X_{t-1}^k \right)' = 0.$$

Beliefs, parameterized by $\boldsymbol{\psi}^k$, are derived from the optimal projection of the aggre-

¹⁰Here we make the conventional assumption that agents observe the non-fiscal shocks contemporaneously. Since the non-fiscal shocks are mutually uncorrelated with z_t , whether agents observe (g_t, u_t, w_t) contemporaneously or with a one-period lag does not impact the results. With lagged observation, forecasts of, e.g., g_{t+1} would be $\rho_g g_{t-1}$ rather than $\rho_g g_t$. This is an additional layer of inertia that does not impact the Ricardian/non-Ricardian mechanism, which depends entirely on the unobservability of z_t and the resulting omitted variable bias in agents' fiscal forecasts.

¹¹Following [Woodford \(2013\)](#), agents project onto *lagged* regressors X_{t-1}^k , so that period- t observables yield forecasts of period- $(t+1)$ aggregates. This convention simplifies the analytic expressions and, in the real-time learning model below, avoids a potential multicollinearity problem.

gate variables $\mathbf{Z}_t$ onto the restricted explanatory variable x_t^k . It follows that

$$\psi^k = \left[EX_{t-1}^k X_{t-1}^{k'} \right]^{-1} EX_{t-1}^k \mathbf{Z}_t' \equiv S(\psi^k).$$

A key implication is that agents are unable to detect their misspecification within the context of their own forecasting model. The orthogonality condition ensures that forecast residuals are uncorrelated with the included regressors, so standard within-model specification tests would not reject.¹²

Definition 2 *A restricted perceptions equilibrium is a fixed point $\psi_*^k = S(\psi_*^k)$.*

We do not impose *a priori* which of the PLM’s individuals and firms use to form expectations. Instead, we confront them with a discrete choice: they can forecast by including s_t (“model-s”) or b_t (“model-b”), and like the selection of model parameters, they will do so to minimize their forecast errors. We adopt the rationally heterogeneous expectations approach first pioneered by Brock and Hommes (1997), extended to stochastic environments by Branch and Evans (2006). Agents make a predictor selection in a random-utility setting, and the agents will only select the best-performing statistical models in the limit of vanishingly small noise. Branch and Evans (2006) showed that equilibria with heterogeneous expectations can be characterized by a fixed point of a multinomial logit mapping. In particular, heterogeneous beliefs can emerge through two mechanisms. First, the equilibrium can feature heterogeneous misspecified forecasting models that perform identically in a mean-square forecast error sense. Second, a concern for model misspecification in the discrete forecast model selection can lead to agents being distributed across misspecified models. The empirical estimates favor the latter mechanism.

Let n denote the fraction of agents selecting model-s, leaving $1 - n$ of the population forecasting with model-b.¹³ They rank these choices by calculating the relative mean

¹²A more subtle form of misspecification is revealed in Section 3: when fiscal shocks are persistent ($\rho_z > 0$), the b -model forces an AR(1) structure onto a process that is AR(2), so tests for serial correlation in the forecast residuals could in principle detect the misspecification. However, as Sargent (1999) emphasizes, restricted perceptions is an appropriate equilibrium concept precisely in environments where the data reveals the misspecification slowly. With constant-gain learning and the estimated gain parameters, agents’ effective sample sizes are short enough that such tests have limited power.

¹³For simplicity, we assume that households and firms are distributed across models identically, a simplification that could be relaxed. Instead, there could be a distribution n_h of households across models and a fraction n_f of firms. It would be straightforward to generalize this way at the cost of an expanded state vector.

square error (MSE):

$$EU^k = -E \left[(\mathbf{Z}_t - E_t^k[\mathbf{Z}_t^k])' \times \mathbf{W} \times E \left[(\mathbf{Z}_t - E_t^k[\mathbf{Z}_t^k]) \right] \right], \quad k = \{s, b\}, \quad (11)$$

where $\mathbf{W}$ is a weighting matrix.¹⁴ We define relative predictor performance $F(n) : [0, 1] \rightarrow \mathbb{R}$ as $F(n) \equiv EU^s - EU^b$.

The distribution of agents across the two forecasting models, n , is pinned down according to the multinomial logit (MNL) map (see, e.g., [Branch and Evans, 2006](#))

$$n = \frac{1}{2} \left\{ \tanh \left[\frac{\omega}{2} F(n) \right] + 1 \right\} \equiv T_\omega(n),$$

where ω denotes the “intensity of choice”. The MNL map – also, the “T-map” – states that the fraction of agents adopting model- s , n , is an increasing function of its relative forecast accuracy, measured by the function $F(n)$.

Definition 3 *A misspecification equilibrium is a fixed point $n_* = T_\omega(n_*)$.*

An immediate consequence of the continuity of $T_\omega : [0, 1] \rightarrow [0, 1]$ is that there exists a misspecification equilibrium: see Appendix B for analytic details on existence. The neoclassical case $\omega \rightarrow \infty$ warrants special attention. In this case, agents only select the best-performing statistical models. It turns out that, in this case, one can learn quite a bit about the set of misspecification equilibria by studying the endpoints to $F(n)$. For instance, when $F(0) < 0, F(1) < 0$ – that is, the model- b forecasts best when all agents use model- b , or, if they all use model- s – then $n_* = 0$ is a misspecification equilibrium. Conversely, when $F(0) > 0$ and $F(1) > 0$, then $n_* = 1$ is a misspecification equilibrium. Outside of these polar cases, there is also the possibility of multiple misspecification equilibria, $n = \{0, \hat{n}, 1\}$, for some $0 < \hat{n} < 1$, that arises when $F(0) < 0, F(1) > 0$. As we will see, the $n = 0$ misspecification equilibrium is a self-confirming equilibrium with *weakly Ricardian beliefs*, and the $n = 1$ will correspond to homogenous *non-Ricardian beliefs*.¹⁵

The neoclassical limiting case, $\omega \rightarrow \infty$, helps build intuition, but a finite ω is relevant in practice. The multinomial approach, i.e., a finite ω , has a venerable history in discrete decision-making because it provides an elegant way of introducing randomness

¹⁴For simplicity, and without loss of generality, we set $W = I$.

¹⁵For discussion of self-confirming equilibria see [Sargent \(1999\)](#). A self-confirming equilibrium is a stronger concept than RPE as it requires that agents’ beliefs are correct, though they may be misspecified off the equilibrium path.

into discrete decision-making. [Young \(2004\)](#) shows that randomness in forecasting, much like mixed strategies in actions, provides robustness against model uncertainty and flexibility in self-referential economies. The intensity of choice parameter ω is inversely related to the idiosyncratic random utility innovation and, thereby, parameterizes model uncertainty. In particular, larger values of ω parameterize less model uncertainty with the neoclassical case $\omega \rightarrow \infty$ representing no uncertainty at all. Extensive literature tests for dynamic predictor selection using empirical MNL models and typically finds finite values for ω (c.f., [Branch, 2004](#)). The parameter ω is an object in our estimation below. The model estimates suggest that ω is finite and positive.

2.3.3 Learning

The rational expectations hypothesis posits that subjective expectations do not have an independent role in the data-generating process, as cross-equation restrictions enforce consistency between agents' perceived and true laws of motion. A misspecification equilibrium also comprises a set of cross-equation restrictions that emerge through the least squares orthogonality condition. However, this paper focuses on the evolution of beliefs between (non-)Ricardian regimes, and thus, we develop a real-time learning model that converges to a misspecification equilibrium as its limit point.

Our learning model is a recursive adaptive algorithm with a constant gain, which assigns geometrically declining weights to past observations (see [Marcet and Sargent, 1989](#); [Evans and Honkapohja, 2001](#)). This learning rule exhibits several notable features. First, it is a robust estimator in the presence of model misspecification and uncertainty ([Evans et al., 2010](#)), and it performs well econometrically when coefficients drift. Real-time learning and model selection generate both of these characteristics. Second, the learning model extends the vector of state variables to include beliefs, enabling it to account plausibly for the persistence observed in U.S. macroeconomic data. Many modelers use medium-scale New Keynesian models that incorporate persistence mechanically through habits and price indexation. However, it has been shown (see [Milani, 2007](#)) that estimated New Keynesian models favor specifications where inertia arises endogenously through beliefs.

Finally, the real-time learning approach captures endogenous regime-switching beliefs in our empirical model while adhering to our restricted perceptions perspective. The key to interpreting the model estimates lies in understanding how learning can generate an "escape," as described by [Sargent \(1999\)](#), [Cho et al. \(2002\)](#), and [Williams \(2019\)](#). This feature allows our model to capture the dynamic nature of agents' beliefs

as they respond to evolving macroeconomic conditions.

Extending the two restricted forecasting models to this more general environment, we can write

$$E_t^k x_{j,t+1} = (\psi_{j,t-1}^k)' X_{k,t},$$

where, for $j = \{v, p\}$ and $k = \{s, b\}$, $x_{j,t} \in \{v_t, p_t^*\}$, and $X_{k,t}' = (k_t, g_t, u_t, w_t)$. The shocks follow uncorrelated stationary AR(1) processes with parameters ρ_j, σ_j and $\sigma_{ij} = 0$ for all $i, j \in \{g, u, w, z\}$. The econometric learning process is a recursive Bayesian model based on [Evans and Honkapohja \(2001\)](#):

$$\psi_{j,t}^k = \psi_{j,t-1}^k + \gamma_1 \Gamma X_{k,t-1} \left(x_{j,t} - (\psi_{j,t-1}^k)' X_{k,t-1} \right) \quad (12)$$

$$MSE_{j,t}^k = MSE_{j,t-1}^k + \gamma_2 \left[\left(x_{j,t} - (\psi_{j,t-1}^k)' X_{k,t-1} \right)^2 - MSE_{j,t-1}^k \right] \quad (13)$$

$$EU_t^k = -MSE_{v,t}^k - MSE_{p,t}^k$$

$$n_t = \frac{1}{2} \left\{ \tanh \left[\frac{\omega}{2} (EU_t^s - EU_t^b) \right] + 1 \right\}.$$

Equation (12) is a generalized stochastic gradient algorithm that emerges from a Bayesian time-varying parameter model where the coefficients follow a random walk.¹⁶ The parameter $0 < \gamma_1 < 1$ is the “constant gain” as it governs the responsiveness of parameter updating to recent forecast errors. The parameter Γ controls the direction of drift in the parameters. In our empirical application, we adopt the standard stochastic gradient with $\Gamma = I$.¹⁷ It is evident from the stochastic gradient algorithm that it nests the restricted perceptions equilibrium. Equation (13) is a simple recursive estimator of the mean-squared forecast errors that geometrically discounts at rate $(1 - \gamma_2)$. The learning gains, γ_1, γ_2 , are critical objects in the estimation as they control the relative speed of coefficient updating and model selection. If these estimated gain coefficients are significantly different, the dynamical system is a “fast-slow” system, a feature studied by the literature on large deviations ([Dupuis and Ellis, 1997](#)). The idea is that slow-moving variables can generate large fluctuations that occur infrequently but with high probability.

¹⁶A constant gain least-squares algorithm is another commonly employed version of econometric learning. This algorithm looks similar but replaces the parameter Γ with a recursively estimated covariance matrix for the regressors. The latter substantially raises the computational cost of the empirical procedure.

¹⁷As in [Sargent and Williams \(2005\)](#), it is an avenue for future research to consider the role different priors about the direction of drift impacts time-varying Ricardian beliefs.

2.3.4 Timing of the temporary equilibrium

The temporary equilibrium unfolds in four stages within each period. First, the period opens with predetermined debt b_t and realized exogenous shocks (g_t, u_t, w_t, z_t) . These determine the primary surplus $s_t = \phi_b b_t + z_t$ and enter the interest-rate rule $i_t = \phi_\pi \pi_t + \phi_y y_t + w_t$. Agents observe the non-fiscal shocks (g_t, u_t, w_t) contemporaneously but do not observe the fiscal innovation z_t directly. Second, agents form forecasts. Model- s agents observe (s_t, g_t, u_t, w_t) and model- b agents observe (b_t, g_t, u_t, w_t) . Each group projects next-period aggregate (payoff-relevant) variables $\mathbf{Z}_{t+1}$ onto their current observables, so that $\hat{E}_t^k \mathbf{Z}_{t+1} = \boldsymbol{\psi}^k X_t^k$. The key difference between types of agents is that s -model agents access the fiscal innovation only indirectly, through its impact on the observed surplus, while b -model agents access it only through its eventual effect on debt. Third, given these forecasts and the policy rule, markets clear and the temporary equilibrium determines (π_t, y_t, b_{t+1}) . Fourth, agents observe realized outcomes $\mathbf{Z}_t$, update their coefficient estimates $\boldsymbol{\psi}^k$ via recursive least squares, evaluate relative forecast performance, and may switch models according to the predictor selection rule.

3 (Non-)Ricardian Beliefs: mechanics

An important component to the empirical findings below is that, outside an edge case, the model features a non-Ricardian equilibrium. Moreover, the learning dynamics feature occasional departures (escapes) between belief regimes with very different implications for how the economy responds to fiscal shocks. This section establishes the theoretical mechanism behind that interpretation. First, our central theoretical finding is that Ricardian beliefs are fragile in the sense that non-Ricardian effects arise for any heterogeneity in forecasting models, i.e. for any $n > 0$. Second, allowing for persistence in fiscal shocks strengthens this fragility. Even the debt-based forecasting model produces non-Ricardian beliefs when $\rho_z > 0$. Third, we compare how the different parsimonious forecasting models generate differentiated economic responses to fiscal shocks. Finally, we connect this theoretical analysis to impulse responses in the estimated New Keynesian model.

3.1 Fragility of Ricardian beliefs

We establish that Ricardian beliefs arise as a knife-edge result. To make the analysis transparent, we work with a simplified version of the model in which the only exogenous disturbance is the fiscal shock z_t (setting $g_t = u_t = w_t = 0$), the output gap does not enter the interest rate rule ($\phi_y = 0$), and the steady-state debt-to-GDP ratio s_b is small so that government debt is approximately exogenous. Section 3.3 relaxes these restrictions sequentially through impulse responses in the calibrated model.

We begin with a definition of (weak) Ricardian equivalence.

Definition 4 *Fix parameters, policy rules, and a belief regime. In the (log-)linear equilibrium, write the output gap and inflation as functions of the predetermined real debt stock b_t and exogenous states:*

$$y_t = y_b b_t + y'_x x_t, \quad \pi_t = \pi_b b_t + \pi'_x x_t,$$

where x_t collects exogenous states/innovations (including the fiscal innovation). We say the equilibrium satisfies weak Ricardian equivalence if

$$y_b = 0.$$

That is, conditional on exogenous disturbances, inherited government debt has no effect

on the output gap in the linear equilibrium.

Remark 1 *In the class of linear equilibria considered below, $y_b = 0$ holds if and only if $\pi_b = 0$ (see equation (14) in the proof of Proposition 1). Hence weak Ricardian equivalence can equivalently be stated as invariance of (y_t, π_t) to predetermined debt in the linear decision rules.*

Proposition 1 shows that weak Ricardian equivalence is knife-edge. The proof isolates two mechanisms through which it fails: (i) persistence in fiscal shocks ($\rho_z > 0$) makes b_t an insufficient statistic for forecasting payoff-relevant objects, and (ii) heterogeneity in forecasting models ($n > 0$) introduces an additional insufficient-statistic channel even when fiscal shocks are i.i.d.

Proposition 1 *(Weak) Ricardian equivalence obtains if and only if $(n, \rho_z) = (0, 0)$. In this knife-edge case, $z_t = \varepsilon_{z,t}$ and hence (y_t, π_t) depend on the fiscal shock only through its contemporaneous innovation:*

$$y_t = y_z \varepsilon_{z,t}, \quad \pi_t = \pi_z \varepsilon_{z,t}.$$

For any $n > 0$ or any $\rho_z > 0$, we have $y_b \neq 0$ (equivalently $\pi_b \neq 0$): fiscal shocks propagate through beliefs and/or government debt becomes net wealth.

The fragility result has important implications for the estimated learning and model selection dynamics in Section 4. Since the empirical estimates favor persistent fiscal shocks ($\rho_z > 0$), mechanism (i) implies that even the b -model is misspecified in the empirically relevant case. Moreover, as we show in Section 3.4, the $n = 0$ misspecification equilibrium is not a stable limit point of the learning dynamics for finite intensity of choice ω . Although the $n = 0$ equilibrium with $\rho_z > 0$ is not weakly Ricardian in the sense of Definition 4, Section 3.3 shows that an active interest rate rule effectively offsets the non-Ricardian wealth effects, delivering a “quantitatively” Ricardian outcome at $n = 0$. This means that the economy will generically feature non-Ricardian beliefs, with only occasional, temporary escapes to this quantitatively Ricardian regime driven by unusual sequences of shocks.

Proof.

The proof proceeds in two steps. Step 1 shows that when all agents forecast with model- b ($n = 0$), weak Ricardian equivalence holds if and only if fiscal shocks are i.i.d.

($\rho_z = 0$). Step 2 shows that even when $\rho_z = 0$, any heterogeneity in forecasting models ($n > 0$) forces $y_b \neq 0$ (and hence $\pi_b \neq 0$), so weak Ricardian equivalence fails.

Step 1: When $n = 0$, weak Ricardian equivalence holds if and only if $\rho_z = 0$.

Let $n = 0$. We show that weak Ricardian equivalence implies $\rho_z = 0$. When s_b is small, government debt is exogenous and can be represented as

$$b_{t+1} = Ab_t + Bz_t$$

where $A = (1 - \phi_b)/\beta$ and $B = -\beta^{-1}$. The weak Ricardian equivalence implies that the coefficient on b_t in the RPE process for y_t equals zero, i.e. allowing only for potentially transitory effects from z_t .

Notice that

$$\begin{aligned} y_t &= v_t - \sigma\pi_t + (1 - \beta)b_t \\ v_t &= (1 - \beta)(b_{t+1} - b_t) - \sigma(\phi_\pi - 1)\pi_t + \hat{E}_t v_{t+1}, & \hat{E}_t v_{t+1} &= \psi_v^b b_t \\ \pi_t &= (1 - \alpha)\beta\hat{E}_t p_{t+1}^* + \kappa y_t, & \hat{E}_t p_{t+1}^* &= \psi_p^b b_t \end{aligned}$$

After substituting out for expectations and b_{t+1} , the endogenous variables are linear in the state (b_t, z_t) :

$$\begin{aligned} y_t &= y_b b_t + y_z z_t \\ v_t &= v_b b_t + v_z z_t \\ \pi_t &= \pi_b b_t + \pi_z z_t \end{aligned}$$

Let $\rho_z = 0$. We show that $y_b = 0$. In this case $E\pi_{t+1}z_t = Ev_{t+1}z_t = 0$. The least-squares orthogonality conditions simplify to

$$\begin{aligned} Eb_t \left(\frac{\pi_b}{1 - \alpha} b_{t+1} - \psi_p^b b_t \right) &= 0 \Rightarrow \psi_p^b = \frac{A}{1 - \alpha} \pi_b \\ Eb_t (v_b b_{t+1} - \psi_v^b b_t) &= 0 \Rightarrow \psi_v^b = Av_b. \end{aligned}$$

Since

$$v_b = (1 - \beta)(A - 1) - \sigma(\phi_\pi - 1)\pi_b + \psi_v^b$$

we have

$$\begin{aligned}\psi_v^b &= A(1 - \beta) - \sigma(\phi_\pi - 1)A\pi_b + A\psi_v^b \\ &= - \left[A(1 - \beta) + \frac{\sigma(\phi_\pi - 1)A\pi_b}{1 - A} \right].\end{aligned}$$

Now turn to,

$$y_b = (1 - \beta)A + \psi_v^b - \sigma\phi_\pi\pi_b = -\frac{\sigma\pi_b(\phi_\pi - A)}{1 - A}. \quad (14)$$

Now substitute in expressions for y_b, ψ_p^b into the Phillips curve to get

$$\pi_b = \beta A\pi_b + \kappa y_b = \beta A\pi_b - \frac{\kappa\sigma\pi_b(\phi_\pi - A)}{1 - A}.$$

Suppose $\pi_b \neq 0$. Then

$$1 = \beta A - \frac{\kappa\sigma(\phi_\pi - A)}{1 - A} = 1 - \phi_b - \frac{\kappa\sigma(\phi_\pi - A)}{1 - A}.$$

But, since $A = (1 - \phi_b)/\beta < 1$ (recall $1 - \beta < \phi_b < 1$) and $\phi_\pi > 1 > A$ under active monetary policy, we have $\frac{\kappa\sigma(\phi_\pi - A)}{1 - A} \geq 0$ and $\phi_b > 0$, a contradiction. Therefore $\pi_b = 0$. When $\pi_b = 0$, we have $y_b = 0$.

Now let $\rho_z > 0$. Assume $y_b = 0$. Then,

$$\pi_b = (1 - \alpha)\beta\psi_p^b.$$

Furthermore, with serially correlated fiscal shocks

$$\psi_p^b = \frac{1}{1 - \alpha} \frac{E\pi_{t+1}b_t}{Eb_t^2} = \frac{1}{1 - \alpha} \left[\pi_b A + \pi_z \rho_z \frac{Eb_t z_t}{Eb_t^2} \right].$$

Combining,

$$\pi_b = \beta\pi_b A + \pi_z \rho_z \frac{Eb_t z_t}{Eb_t^2}$$

or,

$$(1 - \beta A)\pi_b = \rho_z \pi_z \frac{Eb_t z_t}{Eb_t^2} > 0.$$

Hence, $\pi_b > 0$ implying that $y_b \neq 0$, a contradiction.

Step 2: Now let $\rho_z = 0$ and $n \geq 0$. We showed already that $n = 0$ implies weak Ricardian equivalence. Suppose $n > 0$. With i.i.d. surplus shocks, the debt process is

an exogenous AR(1) with $b_{t+1} = Ab_t + Bz_t$, and z_{t+1} orthogonal to the state (b_t, z_t) . Now we also have ψ^s .

Assume weak Ricardian equivalence, $y_b = 0$. With $\hat{E}_t p_{t+1}^* = n\psi_p^s s_t + (1-n)\psi_p^b b_t$, we have

$$\pi_b = (1-\alpha)\beta [n\phi_b\psi_p^s + (1-n)\psi_p^b].$$

Further, since z_{t+1} is i.i.d., the least-squares orthogonality conditions lead to

$$\begin{aligned}\psi_p^b &= \frac{A}{1-\alpha}\pi_b \\ \psi_p^s &= \frac{Es_t p_{t+1}^*}{Es_t^2} = \frac{K}{1-\alpha}\pi_b\end{aligned}$$

where $K = Es_t b_{t+1}/Es_t^2$. Substituting into the π_b expression, we get

$$\pi_b = (1-\alpha)\beta \left[n\phi_b \frac{K}{1-\alpha}\pi_b + (1-n) \frac{A}{1-\alpha}\pi_b \right] \equiv \beta M \pi_b.$$

Thus, either $\pi_b = 0$ or $M = \beta^{-1}$. Assume $A \in (0, 1)$ and $\phi_\pi \geq A$ (both satisfied under our calibration/policy rules), so that $\frac{\kappa\sigma(\phi_\pi - A)}{1-A} \geq 0$. But, $M = n\phi_b K + (1-n)A$, and it is easy to show that $K < A/\phi_b$ and $M < A < \beta^{-1}$, so it must be that $\pi_b = 0$. (Intuitively, $s_t = \phi_b b_t + z_t$ is a noisy signal of b_t , so projecting b_{t+1} onto s_t attenuates the slope relative to the projection onto b_t .)

To see a contradiction, focus again on

$$y_b = v_b - \sigma\pi_b + 1 - \beta$$

or, if $y_b = 0$, then $v_b = \beta - 1$. With this in hand, we know that

$$\begin{aligned}v_b &= (1-\beta)(A-1) - \sigma(\phi_\pi - 1)\pi_b + \hat{v}_b \\ \hat{v}_b &= n\phi_b\psi_v^s + (1-n)\psi_v^b.\end{aligned}$$

With $v_b = \beta - 1$ and $\pi_b = 0$, $\hat{v}_b = (\beta - 1)A$.

However, the least-squares orthogonality conditions require that

$$\psi_v^s = Kv_b, \quad \psi_v^b = Av_b$$

This implies that

$$\hat{v}_b = n\phi_b K v_b + (1 - n) A v_b = M v_b$$

Using the derived $v_b = (\beta - 1)\hat{v}_b = M v_b$, matching this to the earlier value for $\hat{v}_b$, it follows that $M = A$, which when $n > 0$ implies that $\phi_b K = A$, contradicting $K < A/\phi_b$. Therefore, $n > 0$ is impossible under weak Ricardian equivalence when $\rho_z = 0$.

Ricardian equivalence holds if and only if the knife-edge case $(n, \rho_z) = (0, 0)$. ■

To understand the fragility result it is useful to be explicit about timing and the sense in which the two forecasting models are misspecified. Forecasts are formed at time t given the agents' information set, and the least-squares orthogonality restrictions pin down best linear forecasts of v_{t+1} and p_{t+1}^* given the chosen regressor. Weak Ricardian equivalence fails whenever the chosen regressor is not a sufficient statistic for the conditional expectations of these payoff-relevant variables.

First, suppose $s_b \rightarrow 0$, so that the government budget constraint implies an exogenous law of motion for debt,

$$b_{t+1} = A b_t + B z_t, \quad z_t = \rho_z z_{t-1} + \varepsilon_{z,t}.$$

When $\rho_z = 0$, $z_t = \varepsilon_{z,t}$ is i.i.d. and orthogonal to predetermined b_t , and in particular z_t carries no information about z_{t+1} . Hence, conditional on the belief regime and policy rules, the best linear forecasts of v_{t+1} and p_{t+1}^* can be expressed as functions of b_t alone. In this knife-edge case the debt-based forecasting model is not misspecified in the sense relevant for weak Ricardian equivalence. When $\rho_z > 0$, the omitted fiscal state is persistent and debt is no longer first-order Markov in b_t alone. Eliminating z_t yields the AR(2) representation

$$b_{t+1} = (A + \rho_z) b_t - \rho_z A b_{t-1} + B \varepsilon_{z,t},$$

so a forecasting model that conditions only on b_t forces an AR(1) approximation onto an AR(2) process. This is why the debt-based model becomes misspecified and weak Ricardian equivalence fails when $\rho_z > 0$.¹⁸

The timing logic becomes even more stark once debt is endogenous ($s_b > 0$) and we consider the cull Taylor rule. In that case the government budget constraint takes

¹⁸More generally, if the omitted fiscal state is AR(ℓ), eliminating it yields an AR($\ell+1$) representation for b_t in terms of its own lags.

the form

$$b_{t+1} = \beta^{-1}(b_t - s_b \pi_t - s_t) + s_b i_t, \quad i_t = \phi_\pi \pi_t + \phi_y y_t + w_t,$$

so that the evolution of debt depends on inflation and the output gap, which themselves depend on expectations. When shocks are persistent (i.e., $z_t = \rho_z z_{t-1} + \varepsilon_{z,t}$), one can use the Wold representation $z_t = (1 - \rho_z L)^{-1} \varepsilon_{z,t}$ to write the equilibrium as a reduced form in predetermined variables and innovations. In that reduced form, b_t inherits distributed-lag dependence on past innovations through both the surplus process and the interest/inflation terms in the budget constraint. Consequently, even if debt-based forecasters augment regressors to include finitely many lags $(b_t, b_{t-1}, \dots, b_{t-\ell})$, the conditional expectations $E_t v_{t+1}$ and $E_t p_{t+1}^*$ generally depend on an infinite history when expressed purely in terms of lagged debt. Hence any finite-lag debt forecasting model remains underparameterized once shocks are persistent and $s_b > 0$.

Now consider the surplus-based forecasting model. While $s_t = \phi_b b_t + z_t$ loads on both components of the true state (b_t, z_t) , it still collapses that two-dimensional state into a one-dimensional index. The same observed surplus s_t can be generated by many different decompositions into (b_t, z_t) , and those decompositions imply different future paths for debt and inflation because b_t and z_t propagate differently. Hence conditioning on s_t alone is not sufficient for forecasting v_{t+1} and p_{t+1}^* , which is why the surplus-based model is also misspecified outside knife-edge cases.

3.2 Economic mechanics: the wealth-effect channel ($\sigma = 0$)

To isolate the mechanism through which misspecified fiscal beliefs turn government debt into *perceived net wealth*, we consider the special case $\sigma = 0$. This shuts down the usual intertemporal-substitution channel in the New Keynesian model. In this case, movements in the nominal interest rate matter for consumption allocations only to the extent that they enter the government's intertemporal budget constraint and the perceived fiscal backing of debt. For transparency, we begin with the i.i.d. fiscal innovation benchmark $\rho_z = 0$, in which the pure debt-based forecasting regime ($n = 0$) satisfies weak Ricardian equivalence in the sense of Definition 4.

Restating the Ricardian beliefs condition. With $\sigma = 0$, the (log-linear) government budget constraint is

$$b_{t+1} = \beta^{-1}(b_t - s_b \pi_t - s_t) + s_b i_t, \quad i_t = \phi_\pi \pi_t, \tag{15}$$

where b_t is real government debt, s_t is the primary surplus, and s_b is steady-state debt (relative to output). Define effective (net-of-interest) surpluses

$$\hat{s}_t \equiv s_t - s_b(\beta i_t - \pi_t).$$

Then (15) can be written as the accounting identity

$$b_t = \hat{s}_t + \beta b_{t+1}.$$

Iterating forward and imposing the standard no-bubble condition yields the present-value restriction

$$b_t = \sum_{j \geq 0} \beta^j \hat{s}_{t+j} \equiv PV_t^{\text{true}}(\hat{s}). \quad (16)$$

Under Ricardian equivalence, households perceive fiscal backing correctly, so their perceived present value satisfies

$$PV_t^{\text{beliefs}}(\hat{s}) = b_t. \quad (17)$$

Consumption under restricted perceptions: a wealth term. When expectations are formed using misspecified forecasting models, agents need not impose (16) when evaluating perceived present values. With $\sigma = 0$ and identical households, consumption function (1) can be decomposed into the perceived annuity value of net wealth and a nonfiscal remainder Ξ_t , i.e.,

$$c_t = (1 - \beta) \left[b_t - PV_t^{\text{beliefs}}(\hat{s}) \right] + \Xi_t, \quad (18)$$

where

$$\Xi_t \equiv \sum_{T=t}^{\infty} \beta^{T-t} \hat{E}_t [(1 - \beta)(Y_T - G_T) - \beta(\bar{c}_{T+1} - \bar{c}_T)]$$

collects terms pinned down jointly with (Y_t, π_t, i_t) by the New Keynesian equilibrium conditions (goods market clearing, the IS curve (10), and preference shocks). Importantly, fiscal variables enter (18) *directly* only through the perceived annuity value of net wealth. Consequently, the Ricardian benchmark (17) implies that the perceived annuity value of net wealth is identically zero and there is no wealth-effect channel.

However, in this section, we focus on the direct wealth-effect channel and normalize away the nonfiscal remainder Ξ_t by holding it fixed (which is orthogonal to the present

mechanism). This yields

$$c_t = (1 - \beta) \left[b_t - PV_t^{\text{beliefs}}(\hat{s}) \right]. \quad (19)$$

A wealth effect summary statistic. In the i.i.d. benchmark $\rho_z = 0$, perceived backing in the linear equilibrium can be summarized by a scalar $\chi(n)$ defined by

$$PV_t^{\text{beliefs}}(\hat{s}) = \chi(n) b_t. \quad (20)$$

Combining (19)–(20) gives

$$c_t = (1 - \beta)(1 - \chi(n)) b_t. \quad (21)$$

Ricardian beliefs correspond to $\chi(n) = 1$. When $\chi(n) \neq 1$, outstanding debt is perceived as net wealth, and the sign and magnitude of the wealth effect are summarized by $(1 - \chi(n))$.

How forecasting models map into $\chi(n)$. Consider first the surplus-based forecasting model. Because surplus forecasters project aggregate payoff-relevant variables on s_t alone, perceived backing takes the form

$$PV_t^{(s)}(\hat{s}) = \lambda_s s_t.$$

Now consider an experiment in which the innovation z_t is set to zero after impact so that, along the impulse path, $s_t = \phi_b b_t$ for $t \geq 1$. Then

$$PV_t^{(s)}(\hat{s}) = \lambda_s \phi_b b_t \equiv \chi^{(s)} b_t.$$

Because the s -model does not condition on the full fiscal state (b_t, z_t) , $\chi^{(s)}$ is different from one. In empirically plausible calibrations we obtain $\chi^{(s)} < 1$, so positive debt is perceived as positive net wealth and (21) implies an expansionary wealth effect.

Next consider the debt-based forecasting model. Debt forecasters compress perceived backing into a one-dimensional rule in b_t ,

$$PV_t^{(b)}(\hat{s}) = \lambda_b b_t.$$

In the pure b -model equilibrium ($n = 0$) under $\rho_z = 0$, beliefs satisfy $\lambda_b = 1$, so $\chi(0) = 1$

and there is no wealth effect of inherited debt *in levels*. Nevertheless, the *b*-model can still misperceive the *short-run* response of fiscal backing to a one-time innovation. Model-*b* agents approximate an infinite-horizon object using only contemporaneous debt, so transitory discrepancies between b_t and perceived backing can arise even when predetermined debt does not enter the actual laws of motion for (y_t, π_t) .

A graphical summary. Figure 1 plots $\chi(n)$ as a function of n in the i.i.d. benchmark $\rho_z = 0$ under $\sigma = 0$.¹⁹ The figure provides a compact summary of how increasing the share of surplus forecasters shifts perceived backing away from the Ricardian benchmark $\chi = 1$, thereby strengthening the wealth-effect channel.

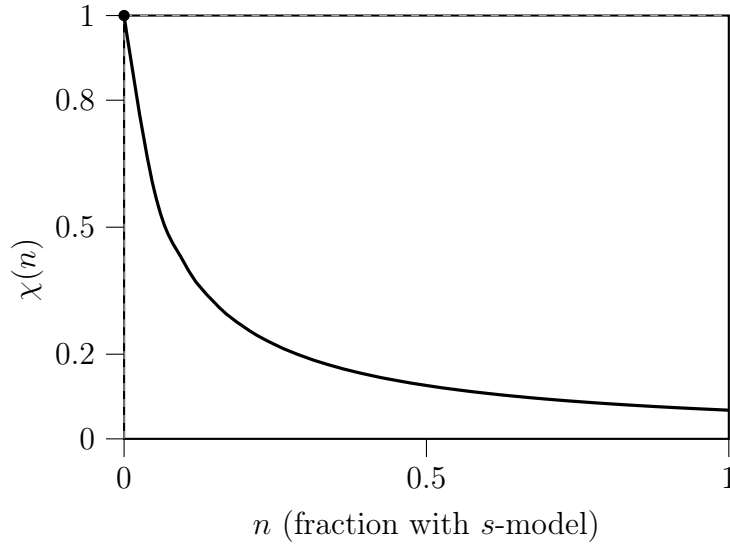


Figure 1: Wealth-effect index $\chi(n)$ as a function of the fraction n of surplus forecasters in the i.i.d. benchmark ($\rho_z = 0$) under $\sigma = 0$. The Ricardian benchmark $\chi = 1$ corresponds to $n = 0$.

Recalling the fragility result in Proposition 1, Figure 1 shows how departures from Ricardian beliefs translate into a nonzero wedge $W_t \equiv b_t - PV_t^{\text{beliefs}}(\hat{s})$ and hence into a direct wealth effect on consumption. The figure highlights two points. First, the pure debt-forecasting regime ($n = 0$) delivers $\chi(0) = 1$, so that $W_t \equiv 0$ and there is no wealth effect of inherited debt in levels (weak Ricardian equivalence). Second, increasing the share of surplus forecasters shifts perceived backing away from the Ricardian benchmark. The statistic $\chi(n)$ falls monotonically in n , so $(1 - \chi(n))$ increases and outstanding debt is perceived as positive net wealth.

¹⁹In this section, all parameters are calibrated according to Table 1 below, unless stated otherwise.

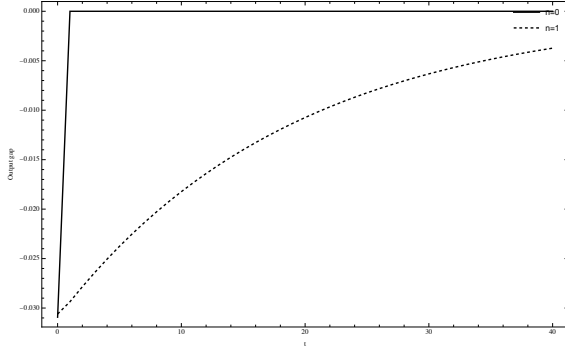
This scalar summary is special to the i.i.d. benchmark. When fiscal shocks are persistent ($\rho_z > 0$), perceived backing is generally not a scalar multiple of predetermined debt because it depends on the full fiscal state (equivalently, on a distributed lag of debt). In that case, the wedge $W_t = b_t - PV_t^{\text{beliefs}}(\hat{s})$ remains the relevant sufficient object for the wealth channel, but it is no longer one-dimensional. This is why, in the next subsection, we move from the index $\chi(n)$ to impulse-response comparisons. Next we use IRFs to trace the dynamic evolution of W_t after a fiscal innovation and show how the resulting wealth effect differs across belief regimes and across model features (σ, s_b, ρ_z) .

3.3 Impulse responses across belief regimes

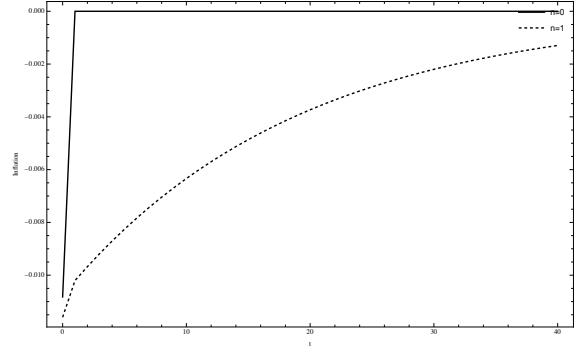
This section deepens further the economic mechanism behind Proposition 1 by comparing impulse responses across the $n = 0$ and $n = 1$ belief regimes. We organize the analysis around four key channels: the share of surplus forecasters $n \in [0, 1]$, persistence in fiscal shocks $\rho_z \in [0, 1]$, the strength of the standard New Keynesian intertemporal-substitution channel parameterized by σ , and the steady-state debt-to-output ratio s_b . Throughout, we focus on a one-time fiscal innovation and report impulse responses of the key aggregate variables.

Pure wealth-effect channel: $\sigma = 0$, i.i.d. fiscal shocks We begin with the i.i.d. fiscal innovation case $\rho_z = 0$ and shut down the standard New Keynesian intertemporal-substitution channel by setting $\sigma = 0$. This isolates the wealth-effect channel emphasized in Section 3.2. Allocations respond to fiscal innovations only through the government budget constraint and perceived fiscal backing. The $n = 0$ equilibrium is weakly Ricardian in this case.

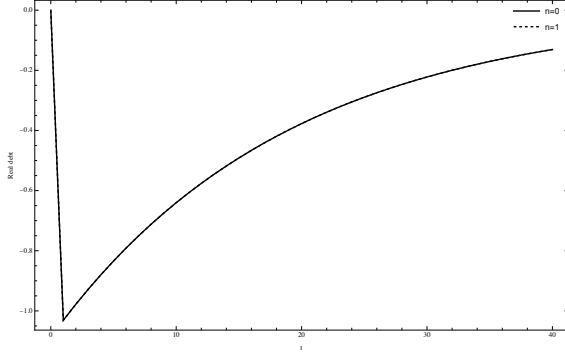
Figure 2 reports impulse responses to a one-time fiscal innovation for two belief regimes: the pure debt-forecasting model ($n = 0$) and the pure surplus-forecasting model ($n = 1$). Consistent with Proposition 1, the $n = 0$ equilibrium is weakly Ricardian in the sense that predetermined debt does not enter the laws of motion for (y_t, π_t) in this benchmark. Nevertheless, the fiscal innovation generates a transitory response through the contemporaneous innovation. In contrast, the $n = 1$ equilibrium exhibits a pronounced non-Ricardian response: debt is perceived as net wealth and output responds more strongly and more persistently.



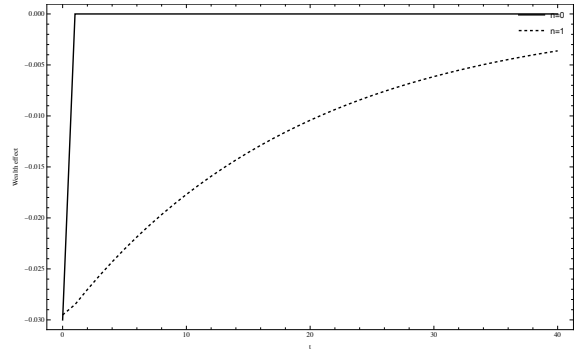
(a) Output gap, y_t



(b) Inflation, π_t



(c) Real debt, b_{t+1}



(d) Wealth effect, c_t

Figure 2: Impulse responses to a one-time fiscal innovation under $\sigma = 0$ and $\rho_z = 0$. Solid lines denote $n = 0$ (debt-based forecasting); dashed lines denote $n = 1$ (surplus-based forecasting).

Turning on persistence: why $\rho_z > 0$ breaks the $n = 0$ weak Ricardian benchmark We next introduce persistence in fiscal shocks, $\rho_z > 0$, holding fixed $\sigma = 0$. Proposition 1 implies that even under the pure debt-based forecasting model ($n = 0$), persistence breaks weak Ricardian equivalence. The key distinction is the way in which persistent fiscal shocks impacts the misspecification of the forecast rules. When $\rho_z = 0$, the fiscal state is effectively one-dimensional. After the initial impact, predetermined debt is a sufficient statistic for the future path of surpluses. Thus, as we observed in the previous subsection, perceived fiscal backing can be summarized as a scalar multiple of b_t and the wealth effect wedge decays back to zero quickly (the $n = 0$ weak-RE benchmark). When z_t is persistent, by contrast, the true fiscal state is two-dimensional. Now, future surpluses depend on both the predetermined debt and the persistent exogenous component z_t . Predetermined debt b_t alone is therefore no longer a sufficient statistic for forecasting payoff-relevant objects, and the wedge

$$W_t \equiv b_t - PV_t^{\text{beliefs}}(\hat{s})$$

is generally nonzero.

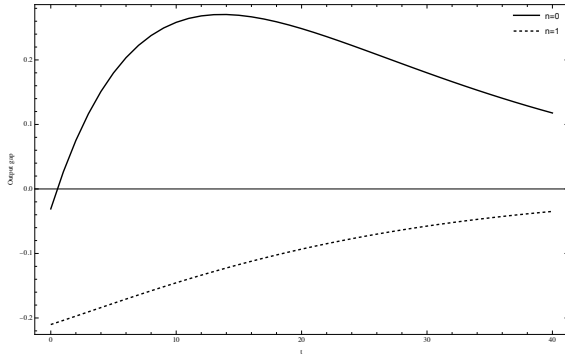
Figure 3 reports impulse responses under $\rho_z > 0$ for $n \in \{0, 1\}$. Relative to the i.i.d. benchmark, persistence amplifies and propagates the responses under both belief regimes. The reason is that persistence turns the fiscal innovation into a long-lived stream of expected future surpluses, so any misspecification in perceived backing affects the present value component over many horizons rather than only at impact.

A useful way to interpret the distinct shapes across n is to focus on how each group forms beliefs about objects that govern perceived backing. In the $n = 0$ regime, agents compress the relevant continuation objects into a one-dimensional rule in past debt,

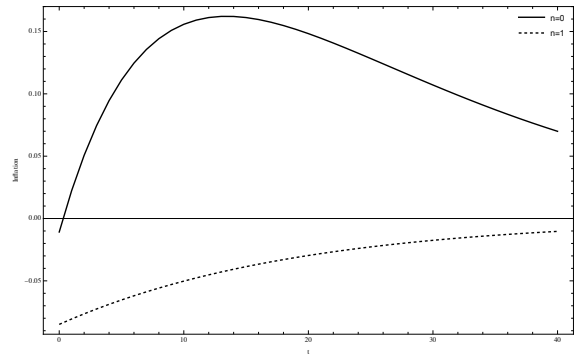
$$\hat{E}_t x_{t+1} = \psi^b b_t, \quad x \in \{p^*, v\},$$

so persistent fiscal shocks are only “observed” through their gradual impact on b_t , a slow moving state variable. In the $n = 1$ regime, agents condition directly on surplus movements, so the persistent component of the innovation is recognized immediately.

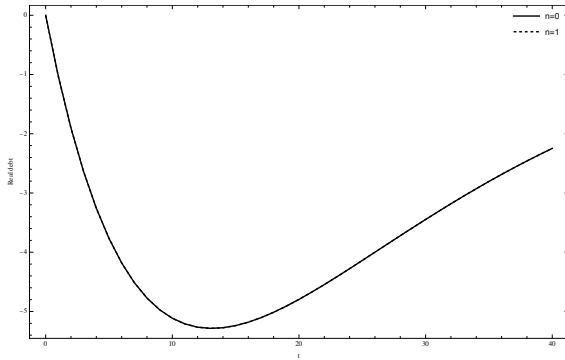
Figure 3 also shows that the qualitative patterns of the wealth effect—and hence the paths of inflation and output—differ sharply across the two belief regimes. Recall that a positive surplus innovation is contractionary: it raises perceived fiscal backing and therefore lowers the net-wealth wedge $W_t = b_t - PV_t^{\text{beliefs}}(\hat{s})$ on impact, implying



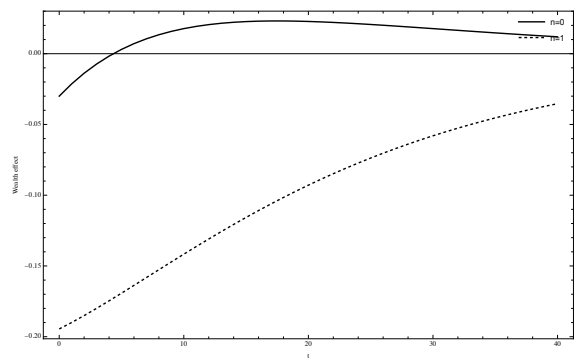
(a) Output gap, y_t



(b) Inflation, π_t



(c) Real debt, b_{t+1}



(d) Wealth effect, W_t

Figure 3: Impulse responses to a one-time positive surplus innovation under $\sigma = 0$ and $\rho_z > 0$. Solid lines denote $n = 0$ (debt-based forecasting); dashed lines denote $n = 1$ (surplus-based forecasting).

a decline in demand, inflation, and the output gap.

In the $n = 1$ regime, persistence mainly strengthens and propagates this same mechanism. Because surplus forecasters condition on the surplus, the variable that directly reflects the persistent innovation at impact, perceived backing rises strongly on impact and then mean-reverts monotonically as the fiscal shock decays. The wealth wedge falls sharply on impact and then returns monotonically toward zero.

The $n = 0$ regime features a different dynamic because agents condition only on existing debt when forecasting aggregate payoff-relevant variables. When $\rho_z = 0$, this restriction is innocuous after impact because the fiscal state becomes effectively one-dimensional along the impulse response path. After the initial shock, debt is a sufficient statistic for future surpluses and the wedge immediately returns to zero (the weak Ricardian benchmark).

When $\rho_z > 0$, however, the true fiscal state is two-dimensional, (b_t, z_t) , and debt is only an imperfect (and lagging) indicator of the persistent innovation z_t . On impact, b_t is predetermined, so the $n = 0$ regime reacts much like in the i.i.d. case. The positive surplus shock raises perceived backing, pushing the wedge down. Subsequently, the persistent innovation gradually works through the debt dynamics and b_t begins to move. Because the one-dimensional forecast rule cannot condition on the missing state, agents revise their perceived backing only with delay as the shock becomes absorbed into debt. This, as seen in the figure, can generate an overshoot. After the initial contractionary shock, perceived backing temporarily falls short of outstanding debt, so the wedge turns positive and becomes hump-shaped before eventually decaying back to zero as both the innovation and debt mean-revert. Since consumption is determined directly by the wedge when $\sigma = 0$, this wedge reversal produces non-monotonic transition paths in the output gap and inflation as both fall on impact, but then exhibit a pronounced hump-shaped expansion back toward steady state as the wedge turns positive.

An equivalent interpretation can be seen in how agents decompose surpluses into systematic and innovation components. Since $s_t = \phi_b b_t + z_t$ and b_t is predetermined at impact, a one-time surplus innovation is primarily a realization of the innovation component z_t . Surplus forecasters condition on s_t and therefore update immediately about the persistent innovation, whereas debt forecasters condition only on b_t and effectively treat the innovation as weakly informative until it is fully incorporated into their observed debt variable. This difference in signal extraction explains why the $n = 1$ wedge mean-reverts monotonically while the $n = 0$ wedge can exhibit delayed

adjustment and hump-shaped dynamics when fiscal shocks are persistent.

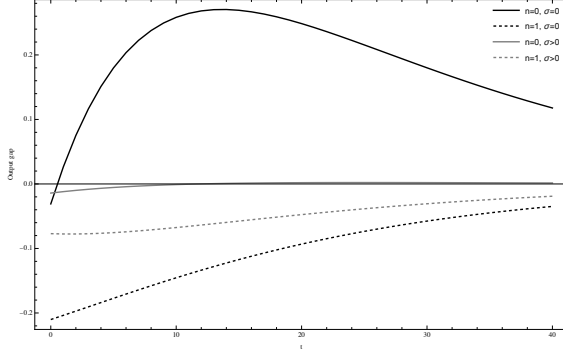
Turning on monetary stabilization ($\sigma > 0$) and the debt-service channel ($s_b \geq 0$). We now compare the impulse responses under $\sigma = 0$ and $\sigma > 0$ holding fixed the fiscal and monetary policy rules. The role of σ is through the usual New Keynesian intertemporal substitution channel. When $\sigma > 0$, the IS curve introduces intertemporal substitution and monetary policy moves the real rate to stabilize demand and inflation. This stabilization can substantially mute the responses of (y_t, π_t) relative to the $\sigma = 0$ case.

Case 1: $s_b = 0$ (exogenous debt). When steady-state debt is zero, the fiscal block is closed:

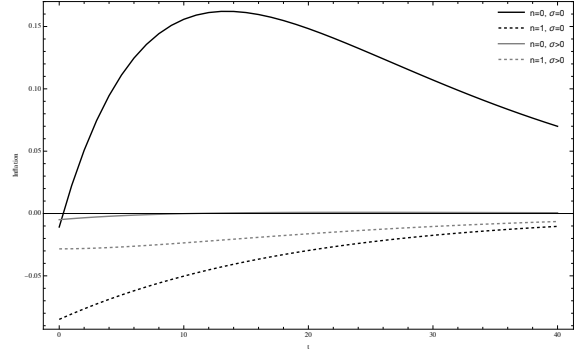
$$b_{t+1} = \beta^{-1}(b_t - s_t), \quad s_t = \phi_b b_t + z_t, \quad z_t = \rho_z z_{t-1} + \varepsilon_{zt},$$

and the payoff-relevant surplus is the primary surplus, $\hat{s}_t = s_t$. In this case the wedge $W_t = b_t - PV_t^{\text{beliefs}}(\hat{s})$ is pinned down entirely by the exogenous fiscal dynamics and the forecasting rules, and, hence, is invariant to σ . This invariance is apparent in the wealth-effect panels in Figure 4. For each belief regime $n \in \{0, 1\}$ the wedge IRF under $\sigma = 0$ and $\sigma > 0$ is essentially identical. Conversely, the responses of inflation and the output gap are sharply attenuated under $\sigma > 0$ because the nominal interest rate rule raises the real rate in response to inflation, offsetting the aggregate demand shift implied by the wealth effect wedge. Put differently, with $s_b = 0$ monetary policy changes how a given wedge transmits to inflation and output gaps, but it does not feed back into the wedge itself.

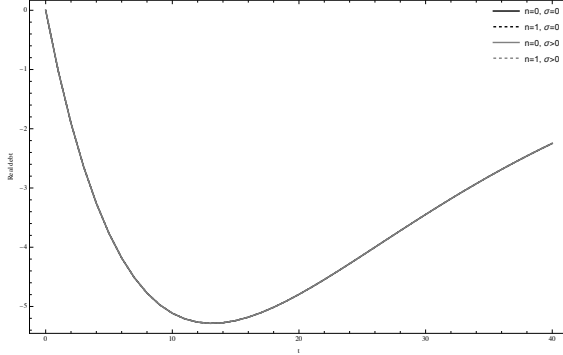
The impact of monetary stabilization is most pronounced in the $n = 0$ regime. The key point is how policy transmission operates rather than beliefs. With $s_b = 0$, the wealth effect wedge $W_t = b_t - PV_t^{\text{beliefs}}(s)$ is pinned down entirely by the fiscal block and the forecasting rules, and is essentially invariant to σ , as shown in the wealth-effect panel of Figure 4. When $\sigma > 0$ it changes how a given wedge maps into the output gap and inflation. The IS equation introduces a real-rate channel, and under a Taylor rule satisfying the Taylor principle the central bank adjusts the real interest rate so as to offset these wealth effects impacts on aggregate demand. Quantitatively, this stabilization is especially effective under $n = 0$ because debt-based forecasting generates a smaller and more delayed wedge disturbance. The policy-induced real interest rate path counters its effects on (y_t, π_t) . In this sense the inflation and output-gap responses



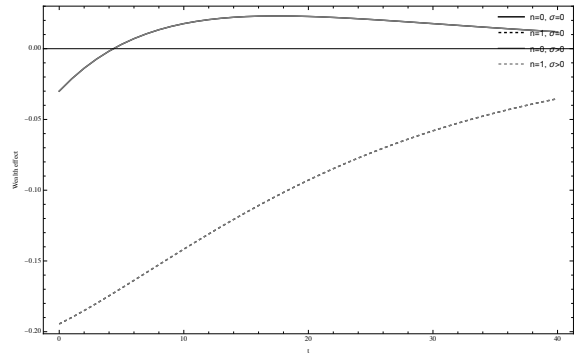
(a) Output gap, y_t



(b) Inflation, π_t



(c) Real debt, b_{t+1}



(d) Wealth effect, W_t

Figure 4: Impulse responses comparing $\sigma = 0$ and $\sigma > 0$ under $\rho_z > 0$. Solid black lines denote $n = 0, \sigma = 0$; dashed black lines denote $n = 1, \sigma = 0$; solid gray lines denote $n = 0, \sigma > 0$; dashed gray lines denote $n = 1, \sigma > 0$.

become “*quantitatively Ricardian*” under $\sigma > 0$ even though the underlying wealth effect does not disappear. Finally, because the real-rate term enters the IS curve with coefficient σ , a higher σ mechanically reduces the sensitivity of the output gap to any wealth effect.

Case 2: $s_b > 0$ (endogenous debt). When steady-state debt is positive, monetary policy enters the government’s effective surplus through the debt-service term,

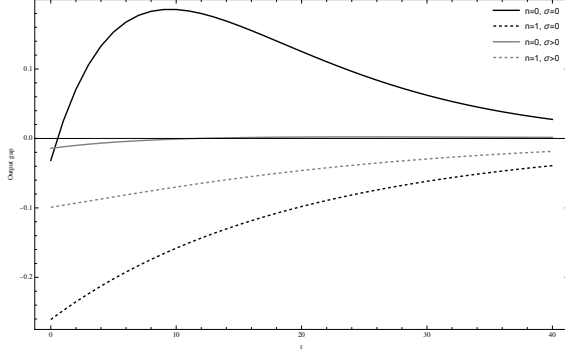
$$\hat{s}_t \equiv s_t - s_b(\beta i_t - \pi_t),$$

and the linearized government budget constraint contains an explicit inflation term. The law of motion for debt is now endogenous. Under $i_t = \phi_\pi \pi_t$, this channel creates a meaningful fiscal–monetary interaction. This case is considered in Figure 5. Now, changing the intertemporal elasticity channel through σ changes the paths of (y_t, π_t, i_t) , which in turn changes the path of the net-of-interest surplus $\hat{s}_t$ and hence the perceived present value $PV_t^{\text{beliefs}}(\hat{s})$. In principle, the wealth effect can depend on σ once $s_b > 0$.

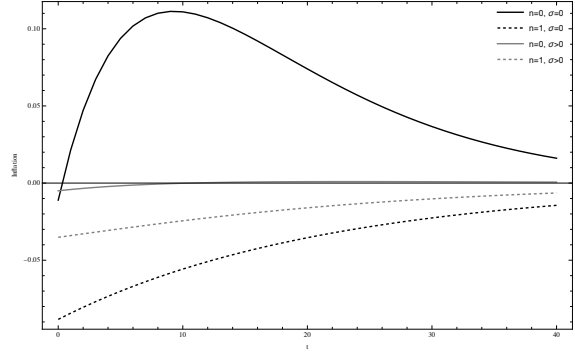
To illustrate this, Figure 5 takes the same policy rules and parameter values as in Figure 4, but sets $s_b = 3.0$. This value for s_b is an order of magnitude larger than the calibrated value in the empirical application. Quantitatively, the effect of s_b is modest and so we set it to an (implausibly) large value for expositional clarity.

The most visible effect of $s_b > 0$ shows up in the debt dynamics. With positive steady-state debt, inflation affects the real value of debt and accumulates through the government budget constraint. This accumulation explains why debt paths can differ noticeably across σ even when wealth effect differences are modest. While the inflation paths are similar to the $s_b = 0$ case, the debt paths are quite different. Notice in the $n = 0$ regime that inflation is positive when $\sigma = 0$ and slightly negative when $\sigma > 0$. Thus, we observe in panel (c) that the debt path is muted in the $\sigma = 0$ case. Similarly, the decreases in inflation in the $n = 1$ regime produce stronger debt reactions.

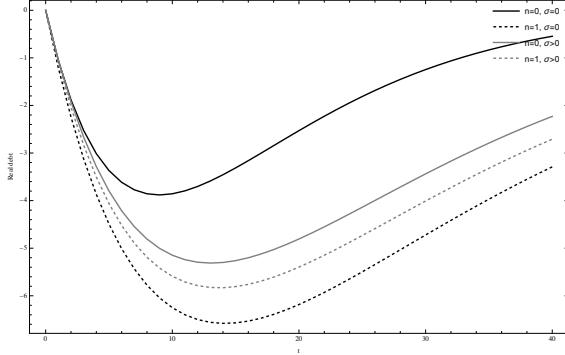
While the endogeneity of debt has modest impacts on inflation paths, we do see a stronger output gap response in the $n = 1$ regime when $s_b > 0$. When $\sigma > 0, s_b > 0$, the perceived wealth effect drops strongly on impact and so does the output gap. While these differences relative to the $\sigma = 0$ case vanish over time, their impacts persist on debt and, hence, the output gap and inflation. Interestingly, monetary stabilization is more effective in the $n = 0$ regime than the $n = 1$ regime. With $\sigma > 0$, the $n = 0$ features negligible output gap and inflation responses to fiscal shocks. This regime is “*quantitatively Ricardian*.”



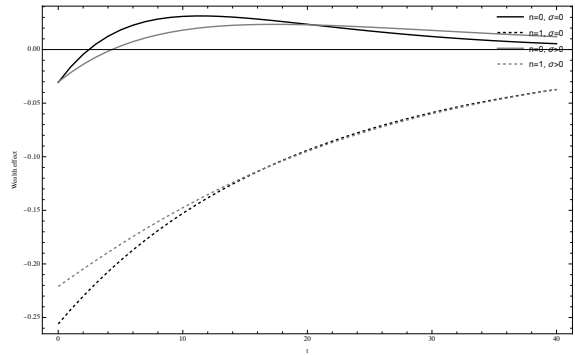
(a) Output gap, y_t



(b) Inflation, π_t



(c) Real debt, b_{t+1}



(d) Wealth effect, W_t

Figure 5: Impulse responses comparing $\sigma > 0$ to $\sigma = 0$ under $s_b > 0$ and $\rho_z > 0$. Solid lines denote $n = 0$ (debt-based forecasting); dashed lines denote $n = 1$ (surplus-based forecasting). The figure shows the case with $s_b = 3.0$.

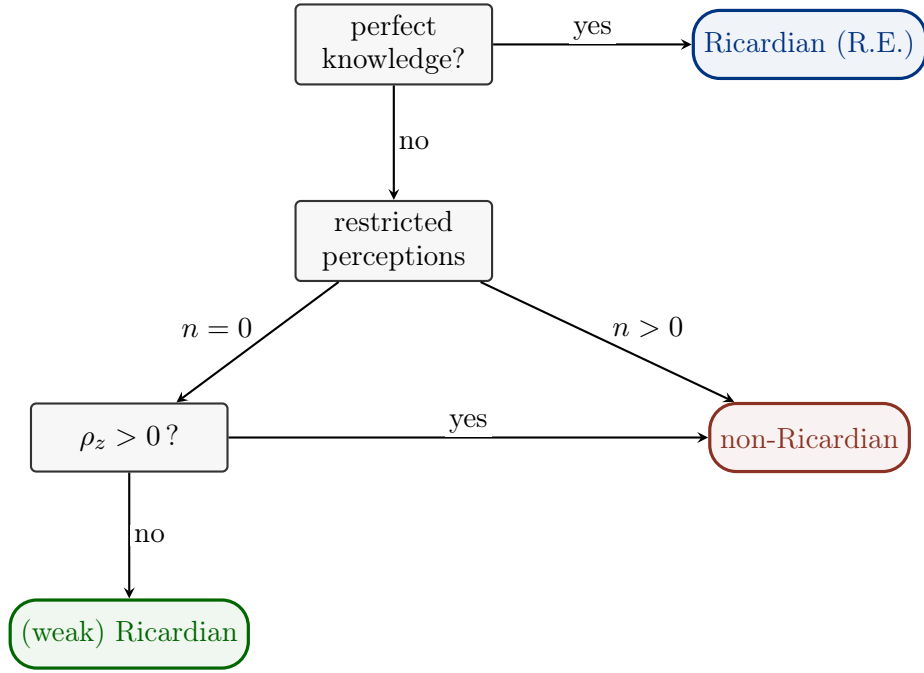


Figure 6: Classification of regimes.

3.4 Regimes

Figure 6 illustrates the full breadth of regimes consistent with the model. Most of the literature focuses on a perfect knowledge/rational expectations setting. The assumed policy setting implies that perfect knowledge produces a Ricardian regime.

This paper focuses on restricted perceptions equilibria. When $n = 0$ and $\rho_z = 0$, the model features a weakly Ricardian regime where predetermined debt does not enter the laws of motion for (y_t, π_t) , though fiscal shocks can still have a transitory impact through the contemporaneous innovation. However, Proposition 1 shows that when $\rho_z > 0$, even the pure debt-forecasting regime ($n = 0$) becomes non-Ricardian because persistence makes the fiscal state two-dimensional, and b_t alone is no longer a sufficient statistic for forecasting aggregate payoff-relevant variables.

The Ricardian regime is fragile in two senses. First, any heterogeneity in beliefs (i.e., $n > 0$) leads to a non-Ricardian regime, even when $\rho_z = 0$. Second, even when all agents use the debt-based forecasting model ($n = 0$), persistence in fiscal shocks ($\rho_z > 0$) breaks weak Ricardian equivalence, as shown in Section 3.3. Moreover, the impulse responses in the previous subsection show that (1.) the qualitative transmission of fiscal shocks to inflation and the output gap differs across belief regimes and (2.)

the $n = 0$ regime in the calibrated model is approximately Ricardian while the $n = 1$ regime is strongly non-Ricardian.

Therefore, a key question is determining n endogenously. A complete characterization of misspecification equilibria is beyond the scope of this paper. However, it is possible to provide a sufficient condition for existence of the non-Ricardian regime.

Proposition 2 *For a sufficiently small intensity of choice ω , there exists a non-Ricardian misspecification equilibrium $0 < \hat{n} < 1$.*

Recall that the population distribution across models is given by the T-map $T_\omega(n)$, which is a function of the relative forecast accuracy of the surplus-based model. With the two forecast models producing forecast errors similar in magnitude (i.e., bounded), then, because of the continuity of T_ω , it is always possible to find an ω such that $0 < n < 1$. Essentially, with modest differences in forecasts between the two models we should expect agents to be distributed across both models. It turns out that this is the empirically relevant case. Our estimates imply an equilibrium value of $\hat{n} \approx 0.48$.

In the case of a large ω , though, the model can feature a range of equilibrium outcomes. Appendix B presents results on the existence of misspecification equilibria when $\omega \rightarrow \infty$ in the special case considered by [Woodford \(2013\)](#). There we see that depending on the structural parameter values of the model, there can exist unique equilibria at $n = 0$ or $n = 1$, and even multiple misspecification equilibria at $n = 0, 1, \hat{n}$ where $0 < \hat{n} < 1$. In other models with negative expectational feedback – not the case here with the specific assumed policy rules – there can even exist a unique misspecification equilibrium at $0 < \hat{n} < 1$. This particular case was coined “intrinsic heterogeneity” by [Branch and Evans \(2006\)](#).

4 (Non-)Ricardian beliefs: estimates

We turn to one of our main interests: measuring the impact of endogenously (non-)Ricardian beliefs on U.S. inflation. We begin with estimating the model's parameters using Bayesian methods. Then, we analyze the estimated path for inflation and the output gap along with a decomposition into (non-)Ricardian effects. The evidence clearly points towards endogenous fluctuations between quantitatively Ricardian and non-Ricardian regimes. We then study what drives these fluctuations by identifying the “most likely unlikely” sequence of shocks that can trigger an escape from a non-Ricardian to a quantitatively Ricardian equilibrium. Counterfactual experiments deepen our understanding of the interaction between macroeconomic outcomes and beliefs.

4.1 Empirical results

Our main interest is an empirical assessment of the role played by endogenously (non-)Ricardian beliefs. To that end, we first describe the Bayesian methods used to estimate the model's key parameters and assess model fit *vis a vis* a restriction to rational expectations or only Ricardian beliefs.

4.1.1 Empirical methodology

After plugging in the policy rules, expectations, and recursive updating equations for the learning rules, the model, in non-linear state space form, is:

$$\begin{aligned} X_t &= g(X_{t-1}, \Theta) + Q(X_{t-1}, \Theta)\nu_t \\ \mathbb{W}_t &= f(X_t, v_t), \end{aligned}$$

where the state vector is

$$X'_t = (b_{t+1}, \pi_t, y_t, v_t, s_t, g_t, u_t, w_t, z_t, n_t, MSE_{st}, MSE_{bt}, \text{vec}(\psi_t^s), \text{vec}(\psi_t^b)),$$

$\text{vec}(\cdot)$ is the vectorization operator, the observation variables are

$$\mathbb{W}'_t = (y_t, \pi_t, s_t, b_{t+1}, i_t),$$

and the parameter vector is

$$\Theta' = (\kappa, \alpha, \phi_\pi, \phi_y, \phi_b, \rho_g, \rho_u, \rho_w, \rho_z, \sigma_g, \sigma_u, \sigma_w, \sigma_z, \omega, \gamma_1, \gamma_2).$$

The measurement and state disturbances are v_t, ν_t respectively. Our sample for the observed variables is 1955.1-2007.3.²⁰ We measure y_t as the log difference between output and the CBO’s measure of potential output. We measure π_t from the PCE index. We compute b_t and s_t as the debt-GDP ratio and primary surplus-GDP ratio, respectively. All variables are deviations from the mean and annualized.

The empirical exercise aims to identify reasonable values for the parameters in Θ to explain the U.S. economy over the sample period. We are especially interested in measuring the extent of (non-)Ricardian beliefs over the period and how it impacts inflation and output gap dynamics. To this end, we adopt Bayesian methods and use a simulation-based technique called the particle filter to approximate the likelihood function $p(Y_t|\Theta)$. The endogenous predictor selection and learning render the state-space non-linear, making the analytic calculation of the likelihood intractable. In place of a Kalman Filter, we adopt the Bootstrap particle filter as described in [Herbst and Schorfheide \(2015\)](#). The particle filter is a way to produce recursive approximations to the distribution of the latent state variables X_t . Our algorithm samples the posterior distribution through an adapted Random-Walk Metropolis-Hastings (RWMH) MCMC technique with the particle-filter-based estimate of the likelihood function. A detailed discussion of the MCMC-particle filter algorithm, the priors, and its implementation is in [Appendix C](#).

4.1.2 Parameter estimates

[Table 1](#) reports the means and the fifth and ninety-fifth percentiles of the marginal posterior distributions of the parameters. In the estimation, we fixed the parameters $\beta = 0.99$, $s_b = 0.30$, and $\sigma = 2.00$. The parameter estimates are mostly in line with previous estimates in the literature, with a few notable exceptions described below.

Because estimation is Bayesian and likelihood-based in a nonlinear state-space model, identification is joint and conditional on the model structure and priors. In this setting, posterior inference reflects how parameter values improve fit to the joint dynamics of inflation, output, debt, surplus, and the policy rate. Persistence and volatility patterns are particularly informative for exogenous shock parameters, co-movement between policy instruments and macro variables is informative for policy-rule coefficients, and learning/model-selection parameters are inferred from their contribution to time variation in persistence, volatility, and co-movement in observables.

²⁰We end the sample before the ZLB episode as incorporating an effective lower bound on interest rates is beyond the scope of the present paper. Below, we consider the model’s performance in an out of sample forecasting exercise.

Table 1: Posterior distribution of parameters

	90% credible region		
Parameter	Mean	5%	95%
<i>Structural parameters</i>			
κ	0.392	0.351	0.433
α	0.584	0.538	0.630
<i>Policy parameters</i>			
ϕ_π	1.854	1.665	2.042
ϕ_y	0.119	0.105	0.132
ϕ_b	0.062	0.034	0.091
<i>Exogenous shocks</i>			
ρ_g	0.050	0.007	0.093
ρ_u	0.343	0.296	0.389
ρ_w	0.305	0.262	0.347
ρ_z	0.860	0.814	0.907
$100\sigma_g$	0.631	0.429	0.833
$100\sigma_u$	0.141	0.002	0.281
$100\sigma_w$	1.440	1.359	1.526
$100\sigma_z$	0.235	0.117	0.353
<i>Learning parameters</i>			
ω	6.301	4.428	8.174
γ_1	0.002	0.000	0.003
γ_2	0.091	0.079	0.103

We briefly discuss the key parameters associated with endogenously (non-)Ricardian beliefs. The “intensity of choice” parameter, ω , has a mean estimate of 6.301. This estimate is close to the value of 5.04 estimated by [Cornea-Madeira et al. \(2017\)](#) from a benchmark New Keynesian model with heterogeneous expectations. The speed of adjustment in estimating the relative forecasting accuracies of the two models, γ_2 , has a mean estimate of 0.091, implying that model selection averages past forecast errors with a geometric decay of 0.909. The interpretation of this value is that the memory size for model selection is approximately 11 quarters. The 5/95% intervals imply a range of memory from 9.7-12.7 quarters. The mean estimated gain for coefficient updating, γ_1 , is 0.002. This value is on the lower end of previously reported estimates. For instance, [Eusepi and Preston \(2018\)](#) reports a value of 0.035, closer to the high end, but for a different learning model and without the forecast model selection. The relative values for γ_1, γ_2 imply that these estimates suggest fast-slow learning dynamics. Model selection takes place faster than coefficient estimation in response to past data. As we see later, these gain parameters imply recurrent, but persistent, belief changes.

The estimated Taylor-rule policy parameters are in line with previous estimates (e.g., [Eusepi and Preston \(2018\)](#), [Justiniano et al. \(2011\)](#)) and suggest an active monetary policy. While the estimated reaction coefficients are higher than these earlier papers, the coefficients are smaller than the active monetary/passive fiscal regime in [Bianchi and Ilut \(2017\)](#). The fiscal policy reaction coefficient is close to the mean value reported in [Eusepi and Preston \(2018\)](#). The estimated shock processes are also similar to existing estimates, but exhibit less auto-correlation relative to rational expectations models. The latter is a common finding in the literature (e.g., [Milani, 2011](#); [Slobodyan and Wouters, 2012](#)). Relative to standard RE benchmarks, this shift is expected. In our particular model, part of macro persistence is propagated through learning and model selection rather than exogenous shock persistence alone, which shifts posterior mass for some traditional parameters. One difference is that the estimated process for the aggregate demand shock is less volatile and persistent than typical estimates. However, we include serially correlated monetary policy shocks, whose innovations have the highest variance among the set of shocks considered. The estimated fiscal shocks are also the most persistent.

Two parameters whose estimates differ somewhat from many estimates in the literature involve the slope of the Phillips curve and the degree of price rigidity in the economy. For the latter, we estimate that prices update, on average, every 2.4 quarters

more frequently than the typical range of 4-7 quarters. This estimate is in line with Klenow and Kryvtsov (2008) and Nakamura and Steinsson (2008) and also consistent with Milani (2006), who estimates a NK model under learning and finds that the degree of price rigidity is inversely related to the auto-correlation of shocks. Finally, consistent with the relatively low estimated degree of price stickiness, the slope of the Phillips curve is on the high side and considerably higher than in Eusepi and Preston (2018). In our estimation, κ is treated as a reduced-form Phillips-curve slope parameter rather than being pinned down by the full structural restriction linking κ to price-setting primitives.

We can assess the empirical relevance of the endogenous non-Ricardian belief mechanism via the Bayes factor comparing our model to a version that fixes the fraction n at the temporary (non-)Ricardian equilibrium, i.e., $n_t = \{0, 1\}$ in all periods. When we make this comparison, we find that the difference in log marginal likelihoods is in favor of the model with endogenously (non-)Ricardian beliefs and whose difference totals 5.7180 when $n = 0$ and 7.026 when $n = 1$. The Bayes factor takes the product of the ratios of the marginal likelihoods and the prior probabilities for each of the two models. The empirical evidence favoring the model with endogenously (non-)Ricardian beliefs is substantial. The data will prefer that model over the quantitatively Ricardian or fully non-Ricardian versions for any prior ratio below 304.29 and 1,125.40, respectively. Thus, a researcher would have to be *a priori* more confident in a quantitatively Ricardian (i.e., self-confirming expectations) model by a factor of over 300 to reject the model with endogenous (non-)Ricardian beliefs.²¹

4.2 Endogenously (non-)Ricardian beliefs in the U.S.

In this section, we provide compelling empirical evidence to support the main thesis of our paper: non-Ricardian beliefs play a significant role in explaining macroeconomic dynamics, and the endogenous transition between non-Ricardian and (nearly) Ricardian regimes can account for both periods of disinflation and low, stable inflation. We begin by estimating the (one step ahead) predicted state path $E(X_{t+1}|\mathbb{W}_t, \Theta)$ via the particle filter. From that predicted path, we can compute a decomposition of each data series into (non-)Ricardian effects. Our findings are presented in Figure 7 comprising six separate plots, each highlighting different aspects of U.S. macroeconomic

²¹Of course, this model comparison is just on those beliefs within this paper’s economic environment. One might have a subjective prior for a different environment and need to compute the associated marginal likelihood.

data from 1960 to 2007. In the first two panels, dotted lines represent U.S. data, dashed lines denote model-predicted values, and solid lines measure the non-Ricardian effect by differencing model predicted values from a counterfactual with Ricardian beliefs (arising under rational expectations). The shaded regions in each panel form the 60% confidence intervals, providing a clear visualization of the uncertainty around the estimates.

Inflation and output gap dynamics. The first panel of the figure displays actual U.S. inflation rates, the model’s predicted inflation rate (computed as the one-step-ahead forecast from the particle filter), and a series that aids in decomposing the contribution to inflation from non-Ricardian effects. The non-Ricardian effect is measured by taking the difference between the predicted inflation rate and a counterfactual where the private-sector has full information rational expectations. This panel provides strong evidence that non-Ricardian effects significantly influenced inflation during the 1970s, with high inflation in the early 1970s being mostly non-Ricardian, and the high inflation at the end of the 1970s being estimated to be entirely non-Ricardian (because the dashed and solid line overlap). In contrast, inflation during the 1960s was essentially quantitatively Ricardian. The early 1980s disinflation was mostly Ricardian, while the late 1980s, a period of large budget deficits, saw modest non-Ricardian inflation. The 1990s, characterized by substantial budget surpluses, exhibited weak non-Ricardian effects that contributed to low inflation.

Overall, the model’s predictions capture the salient features of U.S. inflation rates. The model does not completely capture the volatility in inflation rates, particularly between 1960-1973. However, the non-Ricardian effect series also shows that learning and endogenously non-Ricardian beliefs captures substantially more of the variation in inflation than would occur, with the same sequence of shocks, under quantitatively Ricardian beliefs.

The second panel presents the output gap, the model’s predicted output gap, and the counterfactual between the model’s prediction and rational expectations. The model does well predicting actual output gaps, with the solid line measuring the extent to which non-Ricardian beliefs contributed to the measured output gap. During the 1970s, non-Ricardian predicted output gaps were mildly contractionary. Notably, the non-Ricardian effect is significantly negative at the beginning of the 1980 disinflation. In the late 1980s, the decomposition indicates that most of the predicted output gap

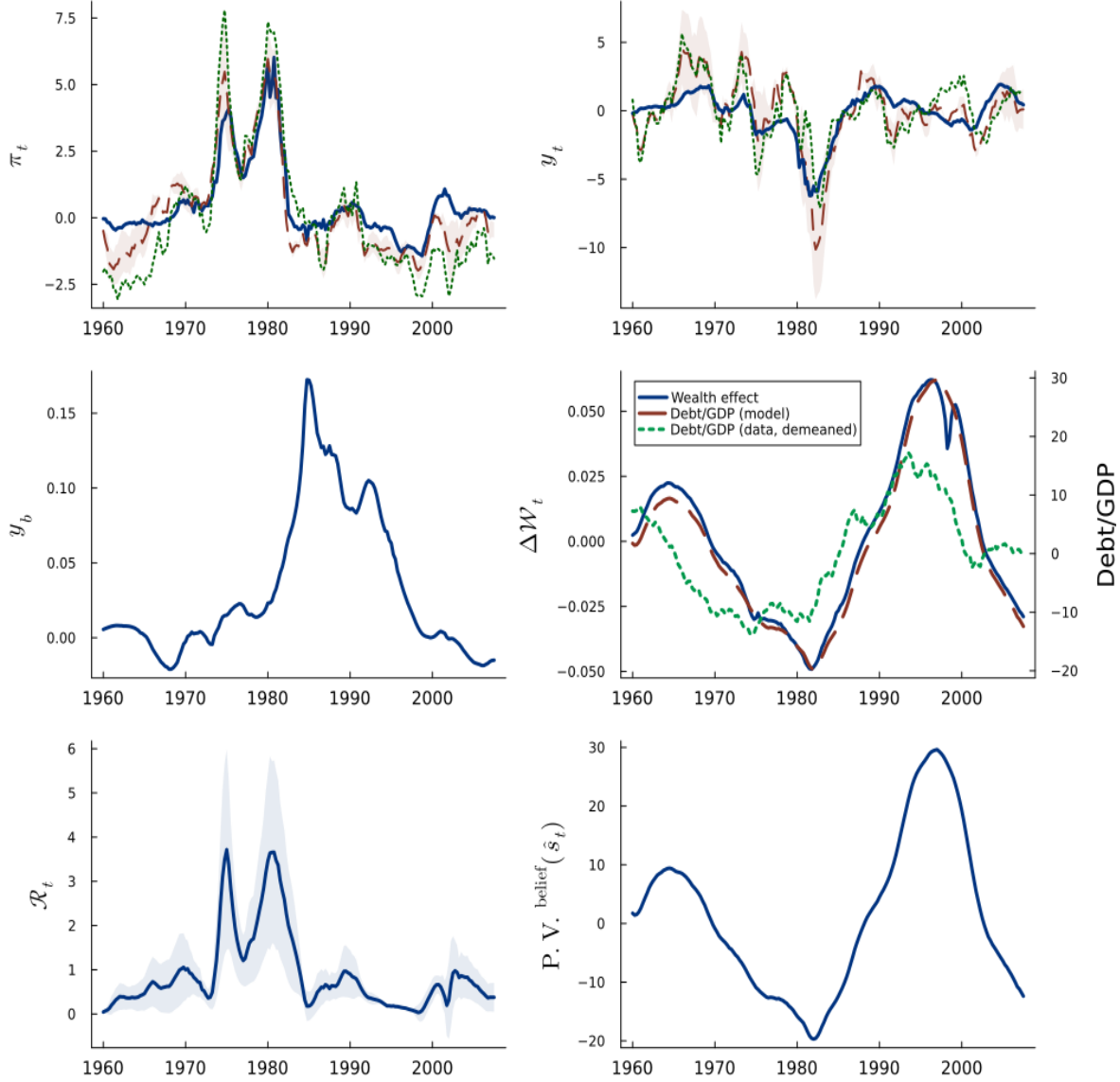


Figure 7: Estimated state dynamics. Dotted lines are U.S. data. Dashed lines are model-predicted values. Solid lines measure the non-Ricardian effect by differencing model-predicted values from a counterfactual with quantitatively Ricardian beliefs. Shaded areas form the 60% credible interval.

resulted from non-Ricardian effects, which are strongly expansionary. However, during the 1990s, the model's output gap does not adequately explain U.S. output gaps. The early 2000s saw a strong non-Ricardian component in the output gap.

The Ricardian wedge. When $n = 0$ – everyone forecasts with model- b – beliefs are generically non-Ricardian (see Section 3). However, Figure 4 showed that for $n = 0$ beliefs are quantitatively Ricardian in the calibrated model. For any $n > 0$, beliefs are non-Ricardian. When agents are learning, next to the persistence in fiscal shocks, ρ_z , there are two sources for deviating from Ricardian beliefs: the choice of model, summarized by n_t , and coefficient updating $\psi_{j,t}^k, j = \{v, p\}, k = \{s, b\}$. Estimated parameters in Table 1 show that these components evolve at different rates γ_1, γ_2 .

Thus, we define a function $\mathcal{R}(n, \{\psi_j^k\}, X_t)$ that provides a distance measure between aggregate beliefs and Ricardian beliefs. We define the *Ricardian wedge* by the function

$$\mathcal{R}(n, \{\psi_j^k\}_{j,k}, X_t) = \left(\hat{E}_t p_{t+1}^* - E_t^r p_{t+1}^* \right)^2 + \left(\hat{E}_t v_{t+1} - E_t^r v_{t+1} \right)^2,$$

where E^r denotes the expectations that would arise in a $n = 0$ world.²² The Ricardian wedge is an appropriate distance measure for the impact of non-Ricardian beliefs on expectations about all payoff relevant aggregate variables. While the function is defined with n as an argument, in the quantitative analysis it is evaluated at the endogenous, time-varying share n_t along with the time-varying PLM coefficients $\psi_{j,t}^k$.

The southwest (SW) panel, displaying the estimated Ricardian wedges, corroborates our findings. The 1960s exhibit a wedge very close to zero, while a large wedge emerges from the 1970s until the early 1980s. Another significant wedge appears in the late 1980s and after the 2001 recession. When the Ricardian wedge is large, indicating that private-sector beliefs depart significantly from full information rational expectations, the panels indicate that predicted non-Ricardian components are also large.

The mechanics of the non-Ricardian effect. The middle panels ($y_b, \mathcal{W}$) and the southeast (SE) panel ($P.V.(\hat{s}_t)$) reveal the mechanics and link back to the intuition built in Section 3. The left middle panel plots y_b , the linear coefficient marking the

²²Below, it will be apparent that this wedge is isomorphic to one that calculates the distance to what a zero-mass fully rational agent would expect given the actual temporary equilibrium learning path.

direct effect of predetermined debt b_t on the output gap. The right middle panel plots the *perceived* non-Ricardian wealth gap (measured on the left axis). The perceived wealth gap comes from the aggregate Ricardian belief condition (2):

$$\mathcal{W}_t = b_t - \sum_{T \geq t} \beta^{T-t} \hat{E}_t \{s_T - s_b [\beta i_T - \pi_T]\}.$$

The annuitized value $\mathcal{W}_t/(1-\beta)$ contributes directly to aggregate consumption. Under Ricardian beliefs, the perceived wealth gap is equal to zero. So the expression provides a direct measure of deviations in the perceived fiscal backing of debt (i.e., the present value of future surpluses) and the wealth effect arising from non-Ricardian beliefs. This panel also plots the model-implied and actual values for the debt-to-GDP ratio (measured on the right axis). The SE panel plots the second term in the perceived wealth effect i.e. the perceived present value of (net of interest) surpluses. The wealth effect is the difference between the model-implied debt-to-GDP ratio and the perceived present-value of surpluses. The left middle plot of y_b measures the loading of that wealth effect onto the output gap. Together, these three panels reveal the mechanics of the non-Ricardian effect on the output gap.

Perceived wealth effects are modest and generally track slow-moving trends in the debt-to-GDP ratio. However, the actual transmission onto macroeconomic outcomes depends on y_b , and as seen in Section 3, this effect depends on the extent of non-Ricardian beliefs. The left middle panel shows that non-Ricardian effects are smallest in the 1960s and late 1990s-early 2000s. The effects increase along with the Ricardian belief wedge $\mathcal{R}_t$. While perceived wealth gaps are fairly large during the 1990's (as is actual debt-to-GDP), these effects do not transmit strongly to the output gap. The 1990's are a time of nearly Ricardian beliefs (SE panel) and declining loading of debt onto the output gap (left middle panel).

A story emerges that departures from full information rational expectations are important for explaining high inflation during the 1970s. At the onset of the disinflation, we observe strong non-Ricardian beliefs, negative perceived wealth effects, and modest pass-through of these pessimistic beliefs, helping to generate the disinflation. Conversely, during the 1990s, we observe large perceived wealth effects, but beliefs are close to rational. The pass-through effect of these perceived wealth effects are declining and help answer why large wealth gaps did not produce significant inflation during the 1990s.

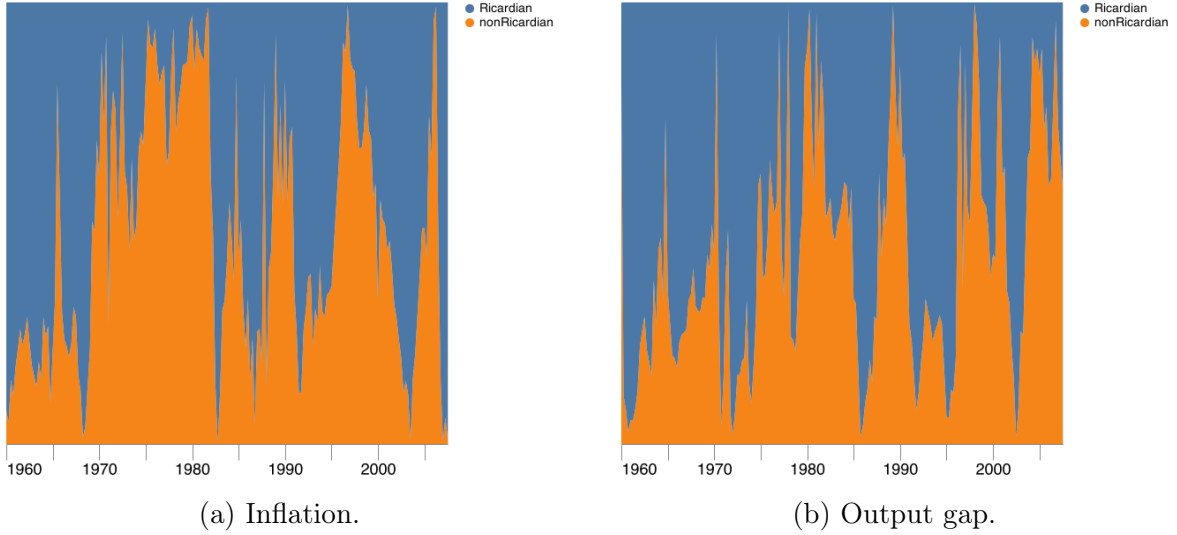


Figure 8: Decomposing (non-)Ricardian effects on inflation and output gap. Height of the non-Ricardian bars represent the fraction attributable to non-Ricardian effects.

Decomposing (non-)Ricardian effects. Figure 8 helps visualize the non-Ricardian decomposition in Figure 7. In each panel, the box decomposes the fraction of the data series, in a given time period, explained by the estimated non-Ricardian effect. Figure 8a shows that the 1970s are strongly influenced by non-Ricardian effects, whereas during the subsequent disinflation these effects account for only a very small share of overall predicted inflation. Thus, inflation and the share of non-Ricardian effects co-move during this period. Non-Ricardian effects on inflation appear again in the late 1980s and late 1990s. Figure 8b similarly estimates regimes with (non-)Ricardian effects. These effects are not as strong or persistent as for inflation.

The most compelling part of these figures is that fluctuations between Ricardian and non-Ricardian regimes occur endogenously, in response to economic shocks, and without policy change. The next section turns to what shocks drive beliefs to fluctuate between belief regimes.

By closely examining the figure, readers can clearly see how the non-Ricardian effects have a significant impact on inflation dynamics, while endogenous transitions between non-Ricardian and Ricardian regimes help explain periods of disinflation and low, stable inflation. This strong evidence in favor of our thesis is further reinforced by the consistency between the estimated Ricardian wedges and the non-Ricardian wealth gap.

4.3 What drives Ricardian beliefs?

The model estimates indicate a non-Ricardian beliefs equilibrium with $\hat{n} \approx 0.48$. This equilibrium is a limit point to the learning process. In finite time, though, learning dynamics can dislodge the economy from its stable equilibrium towards another point. The estimated paths in Figure 7 show that the economy periodically escapes its non-Ricardian equilibrium and a period of (quantitatively) Ricardian beliefs emerge. According to the model estimates, non-Ricardian beliefs are the usual outcome, so the natural question is what drives beliefs to be temporarily (quantitatively) Ricardian?

This question is similar in spirit to the one posed by Sargent (1999) and Cho et al. (2002). In their framework, there is no trade-off between inflation and unemployment. The central bank, however, entertains the possibility of a Phillips curve and chooses policy optimally given their estimates of the slope in their statistical Phillips curve. The high inflation time-consistent Kydland-Prescott outcome is a restricted perceptions equilibrium and stable under central bank learning. However, the right sequence of shocks can lead the central bank's estimates to the low inflation optimal policy. Eventually, time consistency re-emerges. Cho et al. (2002) and Williams (2019) develop tools to identify the “most likely unlikely sequence of shocks” that drives learning to escape the basin of attraction of the stable equilibrium.

We can apply a similar approach here to give insight into what types of shocks drive the non-Ricardian beliefs to become quantitatively Ricardian, particularly for the period following the 1970s inflation. Following Cho et al. (2002) and Branch and Evans (2011), we replace the exogenous shocks with trinomial approximations. In particular, we assume that

$$\begin{aligned} g_t &\in \{-\sigma_g, 0, \sigma_g\} \\ u_t &\in \{-\sigma_u, 0, \sigma_u\} \\ w_t &\in \{-\sigma_w, 0, \sigma_w\} \\ z_t &\in \{-\sigma_z, 0, \sigma_z\} \end{aligned}$$

Then, we draw deterministic sequences of shocks from this set; these are the “unlikely sequences.” Each sequence applies the same vector of shocks in every period. This generates a deterministic, persistent shock that is unlikely under the stationary distribution but reveals which of these unlikely shock patterns can dislodge beliefs from their equilibrium. For each sequence, initialize the model in equilibrium, and simulate

Table 2: Escape paths to Ricardian beliefs.

escape time	shocks
13	$(\sigma_g, \sigma_u, 0, \sigma_z)$
21	$(\sigma_g, \sigma_u, \sigma_w, -\sigma_z)$
22	$(0, \sigma_u, \sigma_w, \sigma_z)$
22	$(0, -\sigma_u, -\sigma_w, -\sigma_z)$
22	$(-\sigma_g, -\sigma_u, -\sigma_w, \sigma_z)$
26	$(0, 0, \sigma_w, \sigma_z)$
26	$(\sigma_g, 0, \sigma_w, 0)$
26	$(-\sigma_g, 0, -\sigma_w, 0)$
28	$(0, -\sigma_u, -\sigma_w, 0)$
43	$(\sigma_g, -\sigma_u, \sigma_w, -\sigma_z)$

the model for $T = 200$ periods. An escape (from the non-Ricardian equilibrium) occurs when n_t drops below 0.25, a value sufficiently far away from the equilibrium.²³ The escape time is the number of periods, for a given shock sequence, until an escape occurs. Essentially, we look for unusual sequences of shocks that lead the learning dynamics to escape the basin of attraction of the equilibrium, which is the asymptotic limit point of the learning dynamics. That first passage time is the escape time. Table 2 reports the results for those shocks with the lowest escape times. The dominant escape path is the sequence of shocks that results in the shortest escape time.

The dominant escape path – repeated positive demand, cost-push, and surplus shocks – generates a path in which actual inflation and surpluses increase while real government debt falls. This disrupts the negative correlation between surpluses and future inflation that sustains the forecasting advantage of model- s .

Recall that model- s forecasts future inflation as a function of the current surplus s_t . However, while model- s predicts decreasing inflation in response to positive surpluses, actual inflation is rising, generating systematic forecast errors. In contrast, model- b , conditioning on b_t , interprets falling debt as lower expected surpluses and, hence, forecasts higher inflation. Since inflation is in fact increasing, model- b 's forecasts are more accurate. Consequently, the dominant escape path is the one that makes model- b 's explanatory variable informative while reducing that of model- s .

Figure 9's bottom panel tracks the relative forecast accuracy $F(n_t) = EU^s - EU^b$, $EU^k = -(MSE_{v,t}^k + MSE_{p^*,t}^k)$, along each escape path. Positive values of F indicate that model- s is more accurate than model- b . Under the dominant sequence (thick

²³The dominant path features $n_t \rightarrow 0$, so this threshold is not limiting.

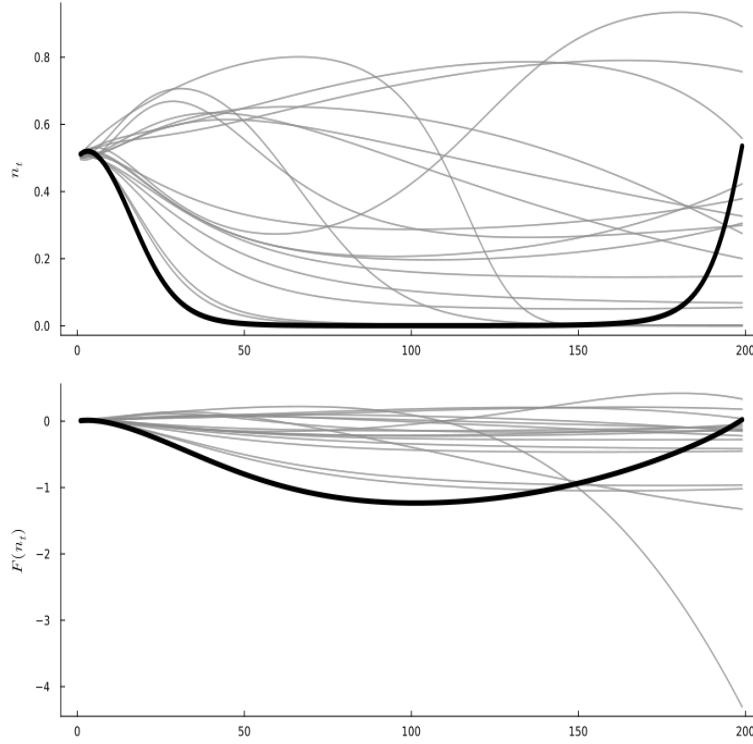


Figure 9: Escape dynamics to Ricardian equilibrium. Thick line denotes the dominant escape path (Table 2).

line), F crosses below zero after approximately 10 periods. At this point, model- b begins outperforming model- s .

With the model estimates, the gain parameter controlling the discounting in updating forecast model accuracy is stronger than for updating forecast coefficients. Because model selection adjusts faster than coefficient estimation ($\gamma_2 \gg \gamma_1$), the share of agents using model- s , n_t , responds to the shift in relative accuracy before forecast coefficients ψ_j^k have time to adapt. The top panel of the figure shows n_t decreasing toward zero along almost all escape paths. As n_t falls, the wealth-effect channel identified in Section 3.2 weakens. The weight $\chi(n_t)$ increases toward one (Figure 1), the coefficient y_b that measures the impact of perceived wealth onto the output gap shrinks, and the economy enters the quantitatively Ricardian regime identified in Section 3.3.

The escape is self-reinforcing but temporary. As n_t approaches zero, the non-Ricardian feedback loop – surplus-based forecasts generate inflation responses to fiscal variables which, in turn, confirm those beliefs – shrinks. With this feedback loop inactive, model- s loses its forecasting advantage. However, Section 3.4 showed that

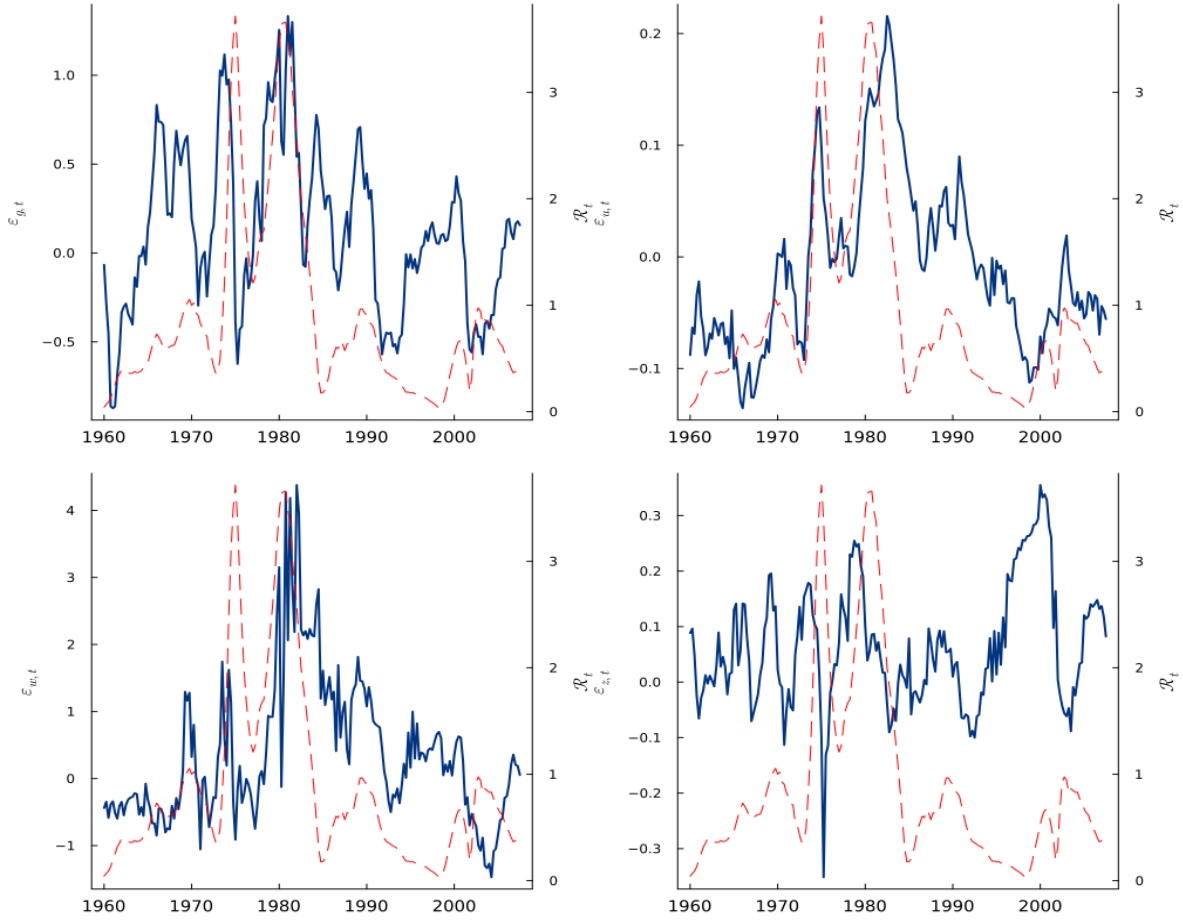


Figure 10: Estimated shocks and escape paths to Ricardian beliefs. Solid line are shocks, dashed line is the non-Ricardian wedge.

the $n = 0$ equilibrium is not a limit point for finite ω . Eventually, the usual (mean) learning dynamics reassert itself and restore a positive F (seen in the bottom panel of Figure 9), model- s regains its forecasting accuracy advantage, and n_t drifts back toward its equilibrium value, $\hat{n}$.

Examining the relationship between the Ricardian wedge and estimated shock sequences shows the dominant escape path at work in the estimated model. Figure 10 plots the predicted paths for the four shock sequences over the sample. The right-hand vertical axis measures the Ricardian wedge, overlayed in the panels as the dashed line.

Look, in particular, at the period during the late 1970's to very early 1980's, a period estimated to be non-Ricardian. Here there is a sustained period of higher than average demand, supply and fiscal shocks. Table 2 and Figure 9 predict a persistent

period of this shock pattern – the “most likely unlikely” sequence of shocks – will trigger an escape to the (quantitatively) Ricardian equilibrium. That is, in fact, exactly what happens. A similar dynamic plays out during the 1990s, a period of sustained fiscal consolidation and budget surpluses that is consistent with the dominant escape path.

It is worth emphasizing that these regime-like fluctuations arise entirely from the endogenous evolution of private-sector beliefs. The baseline model holds the monetary and fiscal policy rules fixed throughout the sample. Unlike explanations that attribute the disinflation of the early 1980s to an exogenous shift in the monetary policy regime, the mechanism here requires no change in the policy stance. The dominant escape path identifies the shock patterns that disrupt the non-Ricardian feedback loop, and the estimated shock decomposition confirms that these patterns were present during the late 1970s and the 1990s.

4.4 Out-of-sample predictions

In this section, we conduct an out-of-sample exercise that offers valuable insights into our model’s performance. We base our estimates on the 1960-2007 period, intentionally excluding the zero lower bound period from 2007.4 to 2015. Incorporating the zero lower bound would introduce unnecessary complexity to our already computationally challenging model and extend beyond the scope of this study. While we could employ a shadow fed funds rate to circumvent the zero lower bound, we find it more prudent to focus on out-of-sample predictions as an effective means of evaluating our model.

The out-of-sample exercise is conducted as follows: first we fix structural parameters at their median posterior value; second, we expand the sample of observables from 2007.4 to 2015.4 and filter the latent state variables through 2015.4; finally, we utilize the particle filter to generate one-period-ahead out-of-sample forecasts for inflation, the output gap, and the Ricardian wedge.

Figure 11 showcases the results of our out-of-sample exercise. The top panel depicts out-of-sample predictions for inflation, offering a clear representation of our model’s accuracy. Meanwhile, the bottom panel provides a glimpse into the predicted Ricardian wedge. Overall, our model demonstrates a strong alignment with the qualitative features of inflation throughout this period. It is worth noting, however, that the model overestimates inflation between 2018 and 2020.

The bottom panel reveals an interesting aspect of our model’s predictions: the pe-

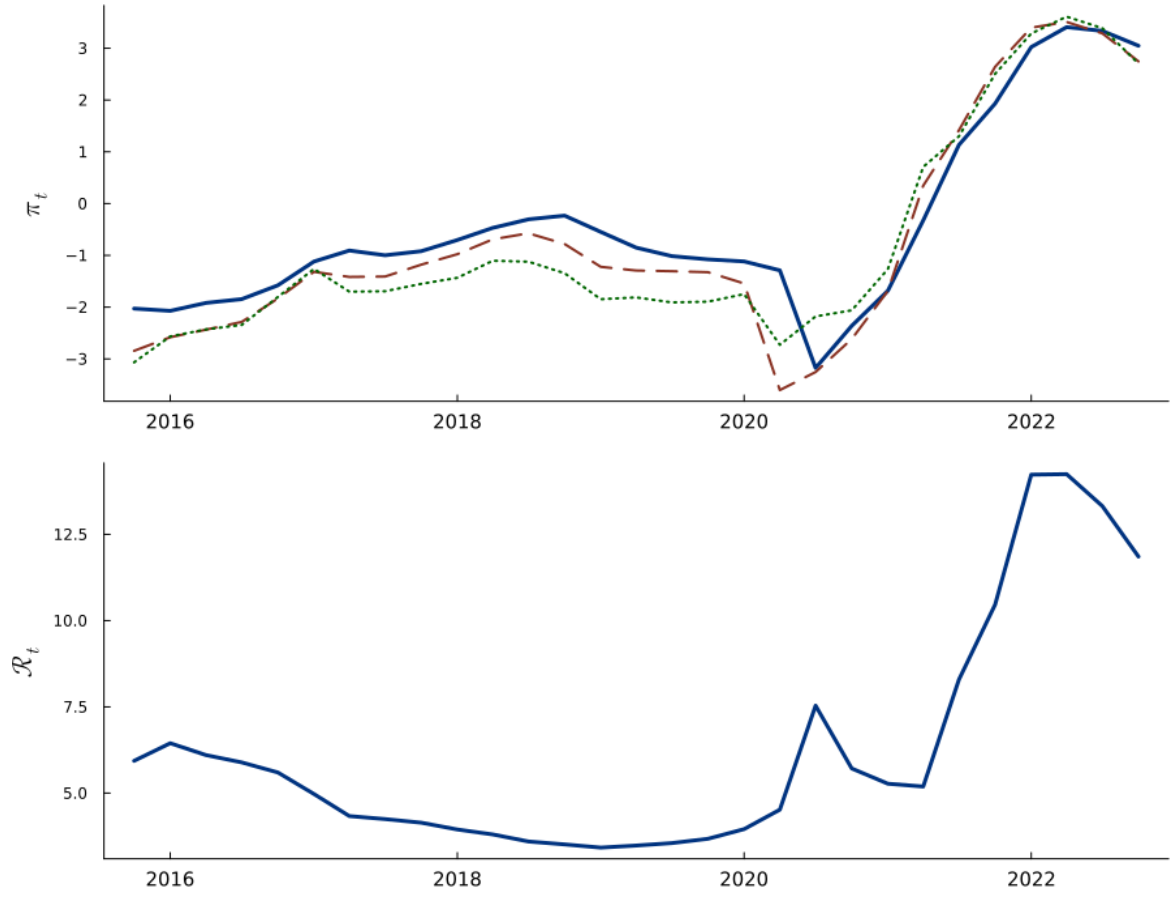


Figure 11: Out-of-sample model predictions 2016-2022. Dotted lines are U.S. data. Dashed lines are model-predicted values. Solid lines measure the non-Ricardian effect by differencing model-predicted values from a counterfactual with quantitatively Ricardian beliefs.

riod between 2017 and 2020 adheres to rather quantitatively Ricardian beliefs. A sharp increase in the Ricardian wedge emerges in 2021, signifying a strong shift towards non-Ricardian beliefs. Thus, our analysis of the inflation plot indicates that the deflation experienced during the onset of the COVID pandemic was not driven by non-Ricardian beliefs.

Nevertheless, our out-of-sample forecasts identify a compelling development: starting in 2021, the significant increase in inflation over the 2021-2022 period appears to be largely attributable to endogenously non-Ricardian beliefs. The timing of this surge in non-Ricardian beliefs corresponds with the implementation of the American Rescue Plan’s fiscal policies. This observation underscores the influential role non-Ricardian beliefs play in shaping macroeconomic outcomes and highlights the critical need to comprehend their interactions with fiscal and monetary policies.

4.5 Counterfactuals

4.5.1 Policy change

The results thus far assume policy rules with time and state-invariant coefficients. However, a line of research argues that the stance of monetary and fiscal policy evolves over time. [Davig and Leeper \(2006\)](#), [Bianchi \(2013\)](#), and [Bianchi and Ilut \(2017\)](#) identify regime-switching policy rules that alternate between active/passive and passive/active monetary/fiscal policies. Estimating regime-switching policy and beliefs is beyond this paper’s scope, but we present rolling-window estimates of time-varying non-Ricardian beliefs alongside time-varying monetary and fiscal policies.

To make the analysis computationally feasible, we shut down endogenous selection of beliefs and estimate the fraction n as a structural parameter.²⁴ We consider rolling 20-year windows and use Bayesian techniques to estimate structural parameters, including n , over each rolling sample, advancing the window by eight quarters. This provides a rough estimate of policy and belief evolution.

Figure 12 shows the rolling window estimates for monetary policy rule reaction coefficients, which are consistent with previous findings. The period ending in 1981 featured a less active monetary policy, while subsequent windows were near two but declined during the late 1990s. The reaction coefficient to the output gap is estimated to be below 0.10 for most windows. Panel (c) plots the fiscal policy reaction coefficient ϕ_b ,

²⁴We can analytically evaluate the likelihood function with the Kalman Filter by shutting down the real-time predictor selection and focusing on the RPE in each window.

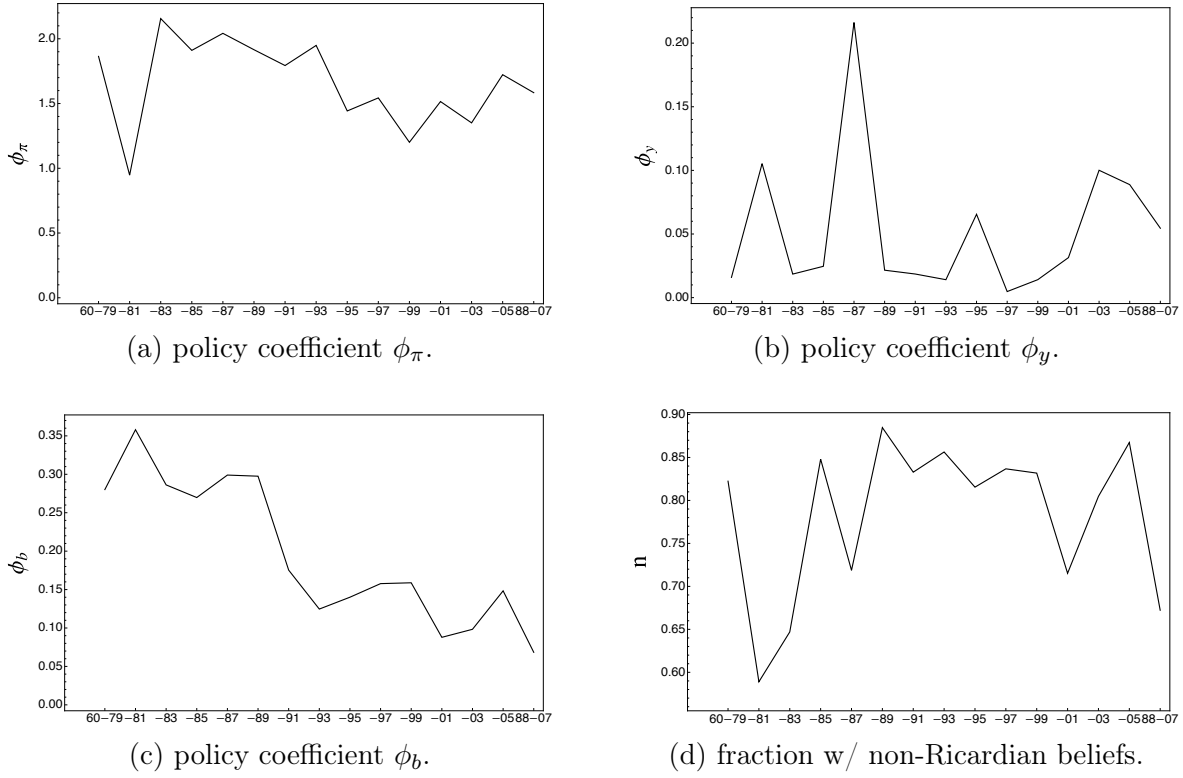


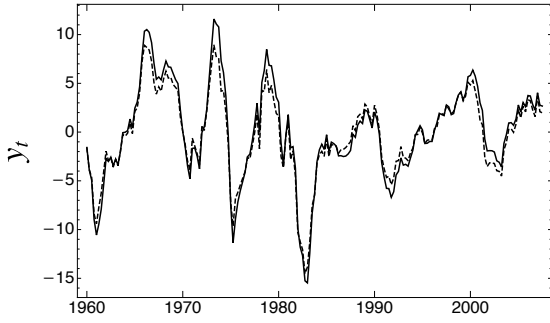
Figure 12: Rolling window estimation results. Panels (a)-(c) coefficient estimates. Panel (d) displays estimated beliefs n_t .

which increases through the mid-1980s before declining in later windows, corresponding to a relatively more active fiscal policy.

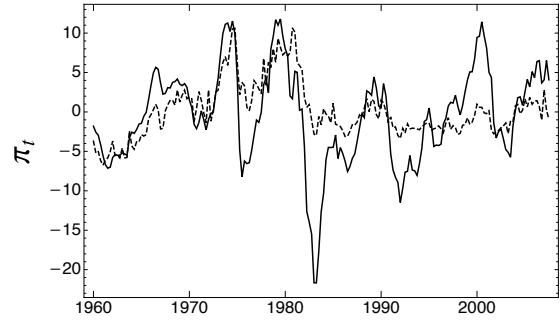
Turning to panel (d) of Figure 12, the average degree of non-Ricardian beliefs is significantly higher than in the benchmark estimation. Over the windows estimated with declining ϕ_π and ϕ_b , there is a downward trend in n . During the 1970s, policy coefficients moved ambiguously regarding their theoretical impact on n . Estimates are consistent with less active monetary policy at the beginning of the Great Inflation period, followed by an increasing role of non-Ricardian beliefs in maintaining high inflation. In sum, endogenous (non-)Ricardian beliefs persist if we allow policy to evolve.

4.5.2 Learning speed

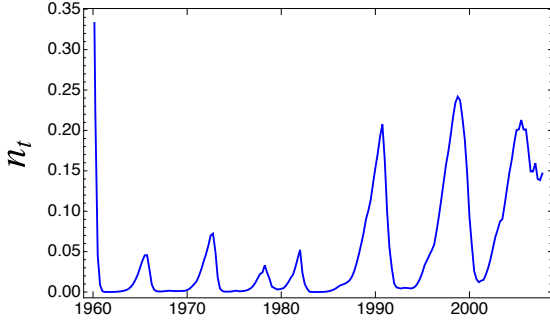
We use two other counterfactuals to disentangle the role played by real-time learning. The first counterfactual, plotted in Panels 13a and 13b, fixes the shocks and the parameters at their mean values. However, this counterfactual undoes the estimated fast-slow learning dynamic and considers the implications for the output gap and in-



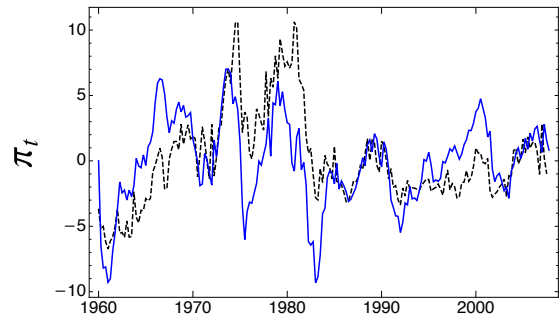
(a) fast learning ($\gamma_1 = 0.03$): output gap.



(b) fast learning ($\gamma_1 = 0.03$): inflation.



(c) no-learning: fraction non-Ricardian.



(d) no-learning: inflation.

Figure 13: Counterfactuals. Panels (a)-(b) from counterfactual with higher learning gain $\gamma_1 = 0.03$. Panels (c) and (d) from counterfactual with no learning. Solid lines are the counterfactual paths, dashed lines are the estimated paths.

flation if coefficient learning is fast by setting $\gamma_1 = 0.03$. This gain controlling the speed of updating beliefs $\psi_{j,t-1}^k$ in (12) is 15 times larger than the estimated posterior mean. The right panel shows that inflation volatility is counterfactually high. Inflation is predicted in this alternate economy to be greater during the 1960s, with vast over-shooting during the 1970s: the counterfactual implication is a dramatic collapse from the 1970s inflation. The counterfactual scenario also predicts that faster learning would have led to a second high inflation episode during the late 1990s.

Panels 13c and 13d construct a counterfactual scenario that shuts down learning entirely. Parameters and shocks are held fixed at their estimated mean values, allowing n to evolve endogenously, but for each value of n_t , the model coefficients equal their RPE values. The left panel plots the counterfactual fraction with non-Ricardian beliefs, while the right panel is the corresponding inflation path. Clearly, in this scenario, beliefs tend to be more quantitatively Ricardian but with more volatility in the later sample periods. The counterfactual scenario substantially under-predicts inflation dur-

ing the 1970s and early 1980s and over-predicts during the 1990s. The model with no learning does well to explain inflation during the late 1980s and 2000s.

Evidently, the combination of real-time learning in beliefs (slow) *and* model selection (fast) is required to explain the data.

5 Related literature

This paper is related to a large literature that examines monetary policy design when replacing rational expectations with an adaptive learning rule. Key contributions include [Bullard and Mitra \(2002\)](#), [Evans and Honkapohja \(2003\)](#), and [Preston \(2005\)](#). The first to characterize fiscal and monetary policy interaction under adaptive learning are [Evans and Honkapohja \(2007\)](#) and [Eusepi and Preston \(2012\)](#), while [Gasteiger \(2018\)](#) is the first to extend this analysis to a setting with heterogeneous expectations. [Evans et al. \(2012\)](#) examine the conditions under which Ricardian equivalence holds or fails under adaptive learning. The theory of restricted perceptions is related to a wide variety of applications of misspecified models, e.g., [Sargent \(1999\)](#), [Adam \(2005\)](#), [Branch and Evans \(2006\)](#), [Sargent \(2008\)](#), [Branch and McGough \(2018\)](#), and [Cho and Kasa \(2015\)](#). Misperceptions and learning about policy rules are introduced by [Cogley et al. \(2015\)](#) and [Hollmayr and Matthes \(2015\)](#). [Honkapohja and Mitra \(2004\)](#) show that E-instability of both fundamental and non-fundamental equilibria can generate persistent learning dynamics that sustain high inflation; our mechanism is related but distinct, as the non-Ricardian restricted perceptions equilibrium in our model is stable under learning, with occasional escapes driven by unusual shock sequences. This paper builds on insight from [Woodford \(2013\)](#), where an example of a restricted perceptions equilibrium leads to a failure of Ricardian equivalence. In short, our paper extends the theory of forecast misspecification in [Branch and Evans \(2006\)](#) into the [Eusepi and Preston \(2018\)](#) environment with fiscal and monetary policy interaction and generalizing the beliefs in [Woodford \(2013\)](#).

Our paper is also related to a long-standing tradition of constructing equilibria with the property that inflation is (partly) driven by fiscal policy. In the original contribution by [Leeper \(1991\)](#), an active fiscal policy, combined with a monetary policy not committed to price stability, generates inflation driven by fiscal variables, i.e., the “fiscal theory of the price-level.” See also, [Sims \(1994\)](#), [Cochrane \(2001\)](#) and [Woodford \(2001\)](#). A separate strand attributes pre-Volcker inflation to indeterminacy under passive monetary policy ([Clarida et al., 2000](#)). Recent research explains post-war U.S. inflation via recurrent change between non-Ricardian and Ricardian policy regimes. Examples include [Davig and Leeper \(2006\)](#), [Sims \(2011\)](#), and [Bianchi and Ilut \(2017\)](#). These papers also derive their results from an important role given to non-Ricardian beliefs. When agents assign a positive probability to changes from the Ricardian policy regime to the non-Ricardian policy regime, then the beliefs imply failure of Ricardian

equivalence, and inflation is also a fiscal phenomenon. Non-Ricardian beliefs, though, in these settings require that agents anticipate recurrent changes in the coordination and objectives of fiscal and monetary policymakers. Indeed, there is not an empirical consensus for economically meaningful time-variation in policy rules (c.f. [Sims and Zha, 2006](#); [Primiceri, 2005](#); [Liu et al., 2011](#)). The question addressed by our paper is whether fiscal shocks can matter at times, even without policy regime change. Including the possibility of regime change is an interesting topic to be explored. We note, however, that our results do not suggest that policy regime change is an unimportant part of the inflation story. More subtle changes within the Ricardian policy regime can generate belief-driven regime change. We explored this point by taking rolling window estimates of the policy coefficients and non-Ricardian beliefs. We found that the estimated fraction of non-Ricardian beliefs evolves along with policy rules as predicted by theory. While beyond the scope of the present study, we leave open quantifying policy-regime change versus belief-regime change.

Finally, our theory here is inspired by and builds on [Eusepi and Preston \(2018\)](#), who show that replacing rational expectations with an adaptive learning rule produces temporary equilibrium dynamics that feature departures from Ricardian equivalence. In addition, their paper illustrates how the maturity structure of government debt has important implications for inflation in a non-Ricardian belief economy. They also estimate a quantitative version of their model and conduct counterfactual analyses demonstrating that perceived net wealth may be a significant factor in high-debt economies.

6 Conclusion

In this paper, we have studied the role of non-Ricardian beliefs in shaping macroeconomic outcomes, particularly inflation and output gap dynamics. We developed a model incorporating agents with Ricardian and non-Ricardian beliefs and provided a novel learning mechanism for agents to choose between these beliefs endogenously. Our approach sheds light on the interaction between agents' expectations, fiscal policy, and monetary policy, offering insights into the evolution of inflation and output gap dynamics over time.

Our central theoretical and quantitative finding is that Ricardian equivalence is inherently fragile. It obtains if and only if all agents adopt a debt-based forecasting model *and* fiscal shocks are serially uncorrelated. The introduction of even an arbitrarily small share of surplus-based forecasters, *or* any degree of persistence in fiscal

shocks, is sufficient to overturn Ricardian beliefs. This fragility implies that a non-Ricardian equilibrium is a plausible outcome of a learning process. At the same time, least-squares learning can give rise to endogenous fluctuations between Ricardian and non-Ricardian regimes.

Our quantitative analysis shows that non-Ricardian beliefs played a significant role in generating higher inflation during the 1970s and anchoring expectations during the 1990s. We demonstrated that the presence of non-Ricardian beliefs can lead to more substantial economic reactions to fiscal shocks. Moreover, we found that the specific reaction coefficients in monetary and fiscal policies can vary over time and have an impact on the extent of non-Ricardian beliefs in the economy.

This paper contributes to the literature by offering a richer understanding of the relationship between agents' expectations, fiscal policy, and monetary policy in shaping macroeconomic outcomes. Future research could extend the analysis by incorporating additional factors, such as incorporating heterogeneous agents with different types of non-Ricardian beliefs. For instance, [Branch et al. \(2022\)](#) show that restricted perceptions equilibria are often accompanied by similar equilibria that include a volatile sentiments, or sunspot, component. Another potential direction for future research is to explore the implications of our findings for optimal policy design, considering the interactions between monetary and fiscal policies and agents' endogenous belief formation.

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A Appendix

A.1 Computation of the restricted perceptions equilibrium

For a given distribution of PLMs, n , for all versions of the model the RPE can be computed in a similar way. First, we can re-organize the ALM to obtain

$$y_t = \delta_0 b_{t+1} + \delta_1 b_t + \delta_2 s_t + \delta_3 u_t \quad (\text{A.1.1})$$

$$b_{t+1} = \xi_1 b_t + \xi_2 s_t + \xi_3 u_t. \quad (\text{A.1.2})$$

Moreover, we can aggregate (6.3) and combine it with (6), (6.5), (A.1.1) and (A.1.2) to obtain

$$v_t = \mu_{v,1} b_t + \mu_{v,2} s_t + \mu_{v,3} u_t, \quad (\text{A.1.3})$$

and (6.5), (5), (A.1.1) and (A.1.2) imply that

$$p_t^* = \mu_{p,1} b_t + \mu_{p,2} s_t + \mu_{p,3} u_t. \quad (\text{A.1.4})$$

We combine (A.1.3) and (A.1.4) to

$$\mathbf{z}_t = \boldsymbol{\mu}_b b_t + \boldsymbol{\mu}_s s_t + \boldsymbol{\mu}_u u_t, \quad (\text{A.1.5})$$

where $\mathbf{z}_t \equiv (v_t, p_t^*)'$, $\boldsymbol{\mu}_b \equiv (\mu_{v,1}, \mu_{p,1})'$, $\boldsymbol{\mu}_s \equiv (\mu_{v,2}, \mu_{p,2})'$, and $\boldsymbol{\mu}_u \equiv (\mu_{v,3}, \mu_{p,3})'$.

Next, recall that PLMs are given by

$$\mathbf{z}_t = \psi^{s'} s_{t-1} + \eta_t^s \quad (\text{A.1.6})$$

$$\mathbf{z}_t = \psi^{b'} b_{t-1} + \eta_t^b, \quad (\text{A.1.7})$$

where $\psi^s \equiv (\psi_v^s, \psi_p^s)$, $\psi^b \equiv (\psi_v^b, \psi_p^b)$, $\eta_t^s \equiv (\eta_{v,t}^s, \eta_{p,t}^s)'$, and $\eta_t^b \equiv (\eta_{v,t}^b, \eta_{p,t}^b)'$. In this special case, ψ_v^k, ψ_p^k are scalars. This implies four orthogonality conditions that can be written as

$$0 \stackrel{!}{=} E[s_{t-1} \eta_t^s] = E[s_t \eta_{t+1}^s] \quad (\text{A.1.8})$$

$$0 \stackrel{!}{=} E[b_{t-1} \eta_t^b] = E[b_t \eta_{t+1}^b]. \quad (\text{A.1.9})$$

Now, plug the PLM (A.1.6) and ALM (A.1.5) into (A.1.8), i.e.,

$$\begin{aligned} 0 &\stackrel{!}{=} E[s_t \eta_{t+1}^s] = E[s_t(\mathbf{z}_{t+1} - \psi^{s'} s_t)] \\ &\Leftrightarrow \psi^{s'} E[s_t^2] = E[s_t \mathbf{z}_{t+1}]. \end{aligned}$$

Equation by equation, we obtain

$$\begin{aligned} &\Leftrightarrow \psi_v^s E[s_t^2] = E[s_t(\mu_{v,1} b_{t+1} + \mu_{v,2} s_{t+1} + \mu_{v,3} u_{t+1})] \\ &\psi_v^s E[s_t^2] = \mu_{v,1} E[s_t b_{t+1}] + \mu_{v,2} E[s_t s_{t+1}] + \mu_{v,3} E[s_t u_{t+1}] \\ &\Leftrightarrow \psi_v^s = \mu_{v,1} \frac{E[s_t b_{t+1}]}{E[s_t^2]} + \mu_{v,2} \frac{E[s_t s_{t+1}]}{E[s_t^2]} + \mu_{v,3} \frac{E[s_t u_{t+1}]}{E[s_t^2]} \quad \text{and (A.1.10)} \end{aligned}$$

$$\begin{aligned} &\Leftrightarrow \psi_p^s E[s_t^2] = E[s_t(\mu_{p,1} b_{t+1} + \mu_{p,2} s_{t+1} + \mu_{p,3} u_{t+1})] \\ &\psi_p^s E[s_t^2] = \mu_{p,1} E[s_t b_{t+1}] + \mu_{p,2} E[s_t s_{t+1}] + \mu_{p,3} E[s_t u_{t+1}] \\ &\Leftrightarrow \psi_p^s = \mu_{p,1} \frac{E[s_t b_{t+1}]}{E[s_t^2]} + \mu_{p,2} \frac{E[s_t s_{t+1}]}{E[s_t^2]} + \mu_{p,3} \frac{E[s_t u_{t+1}]}{E[s_t^2]}. \quad \text{(A.1.11)} \end{aligned}$$

Likewise plug the PLM (A.1.7) and ALM (A.1.5) into (A.1.9), i.e.,

$$\begin{aligned} 0 &\stackrel{!}{=} E[b_t \eta_{t+1}^b] = E[b_t(\mathbf{z}_{t+1} - \psi^{b'} b_t)] \\ &\Leftrightarrow \psi^{b'} E[b_t^2] = E[b_t \mathbf{z}_{t+1}]. \end{aligned}$$

Again, equation by equation, we obtain

$$\begin{aligned} &\Leftrightarrow \psi_v^b E[b_t^2] = E[b_t(\mu_{v,1} b_{t+1} + \mu_{v,2} s_{t+1} + \mu_{v,3} u_{t+1})] \\ &\psi_v^b E[b_t^2] = \mu_{v,1} E[b_t b_{t+1}] + \mu_{v,2} E[b_t s_{t+1}] + \mu_{v,3} E[b_t u_{t+1}] \\ &\Leftrightarrow \psi_v^b = \mu_{v,1} \frac{E[b_t b_{t+1}]}{E[b_t^2]} + \mu_{v,2} \frac{E[b_t s_{t+1}]}{E[b_t^2]} + \mu_{v,3} \frac{E[b_t u_{t+1}]}{E[b_t^2]} \quad \text{and (A.1.12)} \end{aligned}$$

$$\begin{aligned} &\Leftrightarrow \psi_p^b E[b_t^2] = E[b_t(\mu_{p,1} b_{t+1} + \mu_{p,2} s_{t+1} + \mu_{p,3} u_{t+1})] \\ &\psi_p^b E[b_t^2] = \mu_{p,1} E[b_t b_{t+1}] + \mu_{p,2} E[b_t s_{t+1}] + \mu_{p,3} E[b_t u_{t+1}] \\ &\Leftrightarrow \psi_p^b = \mu_{p,1} \frac{E[b_t b_{t+1}]}{E[b_t^2]} + \mu_{p,2} \frac{E[b_t s_{t+1}]}{E[b_t^2]} + \mu_{p,3} \frac{E[b_t u_{t+1}]}{E[b_t^2]}. \quad \text{(A.1.13)} \end{aligned}$$

The next step is to compute the moments. For this purpose, it is convenient to

combine (A.1.2) and (7) in a VAR(1), i.e.,

$$\begin{aligned} \begin{bmatrix} 1 & -\xi_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} b_{t+1} \\ s_t \end{bmatrix} &= \begin{bmatrix} \xi_1 & 0 \\ \phi_b & 0 \end{bmatrix} \begin{bmatrix} b_t \\ s_{t-1} \end{bmatrix} + \begin{bmatrix} \xi_3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_t \\ z_t \end{bmatrix} \\ \Leftrightarrow \mathcal{Y}_t &= \mathbf{A}\mathcal{Y}_{t-1} + \mathbf{C}\varepsilon_t, \end{aligned} \quad (\text{A.1.14})$$

where $\mathcal{Y}_t \equiv (b_{t+1}, s_t)'$ and $\varepsilon_t \equiv (u_t, z_t)'$.

Define the variance-covariance matrix $\mathbf{\Omega} \equiv E[\mathcal{Y}_t\mathcal{Y}_t']$ and likewise $\mathbf{\Sigma} \equiv E[\varepsilon_t\varepsilon_t']$. Then we can compute

$$\begin{aligned} \mathbf{\Omega} &= E[(\mathbf{A}\mathcal{Y}_{t-1} + \mathbf{C}\varepsilon_t)(\mathbf{A}\mathcal{Y}_{t-1} + \mathbf{C}\varepsilon_t)'] = \mathbf{A}E[\mathcal{Y}_{t-1}\mathcal{Y}_{t-1}']\mathbf{A}' + \mathbf{C}E[\varepsilon_t\varepsilon_t']\mathbf{C}' \\ \mathbf{\Omega} &= \mathbf{A}\mathbf{\Omega}\mathbf{A}' + \mathbf{C}\mathbf{\Sigma}\mathbf{C}' \\ \Leftrightarrow \text{vec}(\mathbf{\Omega}) &= [\mathbf{I} - \mathbf{A} \otimes \mathbf{A}]^{-1} (\mathbf{C} \otimes \mathbf{C}) \text{vec}(\mathbf{\Sigma}) \end{aligned}$$

Moreover, the auto-covariance matrix is defined as $E[\mathcal{Y}_t\mathcal{Y}_{t-1}']$, thus

$$E[\mathcal{Y}_t\mathcal{Y}_{t-1}'] = E[(\mathbf{A}\mathcal{Y}_{t-1}\mathcal{Y}_{t-1}' + \mathbf{C}\varepsilon_t\mathcal{Y}_{t-1}')] = \mathbf{A}E[\mathcal{Y}_{t-1}\mathcal{Y}_{t-1}'] = \mathbf{A}\mathbf{\Omega}.$$

Notice that

$$\mathbf{\Omega} = \begin{bmatrix} E[b_{t+1}^2] & E[b_{t+1}s_t] \\ E[s_tb_{t+1}] & E[s_t^2] \end{bmatrix}, \quad \mathbf{A}\mathbf{\Omega} = \begin{bmatrix} E[b_{t+1}b_t] & E[b_{t+1}s_{t-1}] \\ E[s_tb_t] & E[s_ts_{t-1}] \end{bmatrix}.$$

Recall definitions $\Gamma_b^s \equiv E[b_{t+1}s_t]/E[s_t^2]$ and $\Gamma_b^b \equiv E[b_{t+1}b_t]/E[b_t^2]$ as well as $E[b_{t+1}s_t] = E[s_tb_{t+1}]$, $E[b_{t+1}b_t] = E[b_tb_{t+1}]$, $E[s_{t+1}s_t] = E[s_ts_{t+1}]$, and that $E[s_tu_{t+1}] = E[b_tu_{t+1}] = 0$. Moreover, recall that (7) implies that $E[s_ts_{t+1}] = \phi_b E[s_tb_{t+1}]$ and that $E[b_ts_{t+1}] = \phi_b E[b_tb_{t+1}]$. Thus, we can rewrite (A.1.10), (A.1.11), (A.1.12) and (A.1.13) as

$$\begin{aligned} \psi_v^s(n) &= \mu_{v,1}\Gamma_b^s + \mu_{v,2}\phi_b\Gamma_b^s \\ \psi_p^s(n) &= \mu_{p,1}\Gamma_b^s + \mu_{p,2}\phi_b\Gamma_b^s \\ \psi_v^b(n) &= \mu_{v,1}\Gamma_b^b + \mu_{v,2}\phi_b\Gamma_b^b \\ \psi_p^b(n) &= \mu_{p,1}\Gamma_b^b + \mu_{p,2}\phi_b\Gamma_b^b. \end{aligned}$$

These conditions can be solved for $\psi_v^s(n)$, $\psi_p^s(n)$, $\psi_v^b(n)$, and $\psi_p^b(n)$. In case for $s_b > 0$, this can only be achieved numerically as matrices $\mathbf{A}$ and $\mathbf{C}$ in (A.1.14) also

depend on these coefficients.

A.2 Computation of the misspecification equilibrium

Recall the objective (11). Moreover, we have

$$E^s[\mathbf{z}_t^s] = \psi^{s'}(n)s_t, \quad \text{and} \quad (\text{A.2.1})$$

$$E^b[\mathbf{z}_t^b] = \psi^{b'}(n)b_t. \quad (\text{A.2.2})$$

Thus, we can use (A.1.5) and (A.2.1) to compute

$$(\mathbf{z}_t - E^s[\mathbf{z}_t^s]) = \boldsymbol{\mu}_b b_t + \boldsymbol{\mu}_s s_t + \boldsymbol{\mu}_u u_t - \psi^s(n)' s_t.$$

Under the assumption $E[b_t u_t] = E[s_t u_t] = 0$, it follows that

$$E[(\mathbf{z}_t - E^s[\mathbf{z}_t^s])'(\mathbf{z}_t - E^s[\mathbf{z}_t^s])] = E[[b_t' \boldsymbol{\mu}_b' + s_t' \boldsymbol{\mu}_s' + u_t' \boldsymbol{\mu}_u' - s_t \psi^s(n)] [\boldsymbol{\mu}_b b_t + \boldsymbol{\mu}_s s_t + \boldsymbol{\mu}_u u_t - \psi^s(n)' s_t]]$$

$$\begin{aligned} E[(\mathbf{z}_t - E^s[\mathbf{z}_t^s])'(\mathbf{z}_t - E^s[\mathbf{z}_t^s])] &= (\boldsymbol{\mu}_b' \boldsymbol{\mu}_b) E[b_t^2] \\ &\quad + [\boldsymbol{\mu}_s' \boldsymbol{\mu}_s + \psi^s(n) \psi^s(n)' - \boldsymbol{\mu}_s' \psi^s(n)' - \psi^s(n) \boldsymbol{\mu}_s] E[s_t^2] \\ &\quad + (\boldsymbol{\mu}_u' \boldsymbol{\mu}_u) E[u_t^2] + [\boldsymbol{\mu}_b' \boldsymbol{\mu}_s - \boldsymbol{\mu}_b' \psi^s(n)' + \boldsymbol{\mu}_s' \boldsymbol{\mu}_b - \psi^s(n) \boldsymbol{\mu}_b] E[b_t s_t]. \end{aligned}$$

In consequence, we obtain

$$\begin{aligned} EU^s &= - [(\boldsymbol{\mu}_b' \boldsymbol{\mu}_b) E[b_t^2] + [\boldsymbol{\mu}_s' \boldsymbol{\mu}_s + \psi^s(n) \psi^s(n)' - \boldsymbol{\mu}_s' \psi^s(n)' - \psi^s(n) \boldsymbol{\mu}_s] E[s_t^2] \\ &\quad + (\boldsymbol{\mu}_u' \boldsymbol{\mu}_u) E[u_t^2] + [\boldsymbol{\mu}_b' \boldsymbol{\mu}_s - \boldsymbol{\mu}_b' \psi^s(n)' + \boldsymbol{\mu}_s' \boldsymbol{\mu}_b - \psi^s(n) \boldsymbol{\mu}_b] E[b_t s_t]]. \end{aligned}$$

Likewise, we can use (A.1.5) and (A.2.2) to compute

$$(\mathbf{z}_t - E^b[\mathbf{z}_t^b]) = \boldsymbol{\mu}_b b_t + \boldsymbol{\mu}_s s_t + \boldsymbol{\mu}_u u_t - \psi^b(n)' b_t.$$

Therefore it follows that

$$E[(\mathbf{z}_t - E^b[\mathbf{z}_t^b])'(\mathbf{z}_t - E^b[\mathbf{z}_t^b])] = E[[b_t' \boldsymbol{\mu}_b' + s_t' \boldsymbol{\mu}_s' + u_t' \boldsymbol{\mu}_u' - b_t \psi^b(n)] [\boldsymbol{\mu}_b b_t + \boldsymbol{\mu}_s s_t + \boldsymbol{\mu}_u u_t - \psi^b(n)' b_t]].$$

Again we use the assumption $E[b_t u_t] = E[s_t u_t] = 0$ to obtain

$$\begin{aligned} E[(\mathbf{z}_t - E^b[\mathbf{z}_t^b])'(\mathbf{z}_t - E^b[\mathbf{z}_t^b])] &= [\boldsymbol{\mu}'_b \boldsymbol{\mu}_b + \psi^b(n) \psi^b(n)' - \boldsymbol{\mu}'_b \psi^b(n)' - \psi^b(n) \boldsymbol{\mu}_b] E[b_t^2] + (\boldsymbol{\mu}'_s \boldsymbol{\mu}_s) E[s_t^2] \\ &\quad + (\boldsymbol{\mu}'_u \boldsymbol{\mu}_u) E[u_t^2] + [\boldsymbol{\mu}'_b \boldsymbol{\mu}_s + \boldsymbol{\mu}'_s \boldsymbol{\mu}_b - \boldsymbol{\mu}'_s \psi^b(n)' - \psi^b(n) \boldsymbol{\mu}_s] E[b_t s_t]. \end{aligned}$$

In consequence

$$\begin{aligned} EU^b &= - [[\boldsymbol{\mu}'_b \boldsymbol{\mu}_b + \psi^b(n) \psi^b(n)' - \boldsymbol{\mu}'_b \psi^b(n)' - \psi^b(n) \boldsymbol{\mu}_b] E[b_t^2] + (\boldsymbol{\mu}'_s \boldsymbol{\mu}_s) E[s_t^2] \\ &\quad + (\boldsymbol{\mu}'_u \boldsymbol{\mu}_u) E[u_t^2] + [\boldsymbol{\mu}'_b \boldsymbol{\mu}_s + \boldsymbol{\mu}'_s \boldsymbol{\mu}_b - \boldsymbol{\mu}'_s \psi^b(n)' - \psi^b(n) \boldsymbol{\mu}_s] E[b_t s_t]]. \end{aligned}$$

Finally, one can define $F(n) : [0, 1] \rightarrow \mathbb{R}$ as $F(n) \equiv EU^s - EU^b$, thus

$$\begin{aligned} F(n) &= [\psi^b(n) \psi^b(n)' - \boldsymbol{\mu}'_b \psi^b(n)' - \psi^b(n) \boldsymbol{\mu}_b] E[b_t^2] \\ &\quad + [\boldsymbol{\mu}'_s \psi^s(n)' + \psi^s(n) \boldsymbol{\mu}_s - \psi^s(n) \psi^s(n)'] E[s_t^2] \\ &\quad + [\psi^s(n) \boldsymbol{\mu}_b + \boldsymbol{\mu}'_b \psi^s(n)' - \psi^b(n) \boldsymbol{\mu}_s - \boldsymbol{\mu}'_s \psi^b(n)'] E[b_t s_t]. \end{aligned}$$

B Proofs

B.1 Proof of Proposition B.1.1 (Existence)

Proposition B.1.1 *Let $N_*(\omega) = \{n_* \mid n_* = T_\omega(n_*)\}$ denote the set of misspecification equilibria. As $\omega \rightarrow \infty$, N_* has one of the following properties:*

1. *If $F(0) < 0$ and $F(1) < 0$ then $n_* = 0 \in N_*$.*
2. *If $F(0) > 0$ and $F(1) > 0$ then $n_* = 1 \in N_*$.*
3. *If $F(0) < 0$ and $F(1) > 0$ then $\{0, \hat{n}, 1\} \subset N_*$, where $\hat{n} \in (0, 1)$ is such that $F(\hat{n}) = 0$.*
4. *If $F(0) > 0$ and $F(1) < 0$ then $n_* = \hat{n} \in N_*$, where $\hat{n} \in (0, 1)$ is such that $F(\hat{n}) = 0$.*

Remark B.1.1 *Proposition B.1.1 relies only on the continuity of $F(n)$ and $T_\omega(n)$. If $F(n)$ is monotonic then a stronger statement is possible: Proposition B.1.1 then identifies the full set of misspecification equilibria. In most parameterizations, $F(n)$ is monotonic. When $F(n)$ is non-monotonic it is theoretically possible for there to exist*

multiple interior equilibria, though, in all of the numerical cases examined we found at most 3 misspecification equilibria.

The proof to Proposition B.1.1 is straightforward, but relies on the existence of a unique restricted perceptions equilibrium for an open set of n . The following Lemma provides the necessary and sufficient conditions for a unique RPE to exist.

Before stating the proposition, note first that the temporary equilibrium equations can be written in the form of an expectational difference equation:

$$X_t = \check{A} \begin{bmatrix} b_t \\ s_t \end{bmatrix} + \check{B} \hat{E}_t X_{t+1} + \check{C} \hat{\epsilon}_t$$

where $X'_t = (s_t, b_t)$ and $\hat{\epsilon}_t$ is a vector of white noise shocks and $\check{A}, \check{B}, \check{C}$ are conformable. Further, denote $EX_t X'_t = \Omega$, $\Gamma_1 = E \begin{bmatrix} b_t \\ s_t \end{bmatrix} \begin{bmatrix} b_{t-1} \\ s_{t-1} \end{bmatrix}'$, and e_j is a (1×2) unit vector with a 1 in the j th element.

Lemma 1 *A unique restricted perceptions equilibrium exists for all n if and only if*

$$\Delta \equiv \det \left(I_4 - P' \otimes \check{B} \right) \neq 0$$

where

$$P = \Gamma'_1 \left[n e'_1 (e_1 \Omega e'_1)^{-1} e_1 + (1 - n) e'_2 (e_2 \Omega e'_2)^{-1} e_2 \right]$$

Proof. In an RPE

$$E e_j \begin{bmatrix} b_{t-1} \\ s_{t-1} \end{bmatrix} \left(X_t - \psi^{k'} e_j \begin{bmatrix} b_{t-1} \\ s_{t-1} \end{bmatrix} \right)' = 0$$

After plugging in for aggregate expectations into the expectational difference equation

$$X_t = \xi \begin{bmatrix} b_t \\ s_t \end{bmatrix} + \check{C} \hat{\epsilon}_t$$

where

$$\xi = \check{A} + n \check{B} \psi^{s'} e_1 + (1 - n) \check{B} \psi^{b'} e_2$$

Using this notation,

$$\begin{aligned}\psi^k &= (e_j \Omega e'_j)^{-1} E e_j \begin{bmatrix} b_{t-1} \\ s_{t-1} \end{bmatrix} X'_t \\ &= (e_j \Omega e'_j)^{-1} e_j \Gamma_1 \xi'\end{aligned}$$

After plugging in for ψ^s, ψ^b into ξ :

$$\begin{aligned}\xi &= \check{A} + \check{B} \xi \Gamma'_1 \left[n e'_1 (e_1 \Omega e'_1)^{-1} e_1 + (1-n) e'_2 (e_2 \Omega e'_2)^{-1} e_2 \right] \\ \Leftrightarrow \xi &= \check{A} + \check{B} \xi P\end{aligned}$$

It follows that

$$\text{vec}(\xi) = \text{vec}(\check{A}) + \left(P' \otimes \check{B} \right) \text{vec}(\xi)$$

Finally, the RPE coefficient is given by

$$\text{vec}(\xi) = \left(I_4 - \left(P' \otimes \check{B} \right) \right)^{-1} \text{vec}(\check{A})$$

and the stated conditions provides necessary and sufficient conditions for a unique ξ .

■

Proof of Proposition B.1.1.

The existence of a set of fixed points $n_* = T_\omega(n_*)$ follow directly from applying Brouwer's theorem, since $T_\omega : [0, 1] \rightarrow [0, 1]$ and $F(n)$ is continuous provided there exists an RPE. Lemma 1 provides the requisite necessary and sufficient conditions. To complete the proof, we simply require establishing the existence of a unique RPE for an open set of n . This is straightforward as for $n = 0$ or $n = 1$ implies that $\Delta = 0$ and ξ is continuous in n .

■

C Bayesian estimation details

Recall the nonlinear state-space model:

$$\begin{aligned}X_t &= g(X_{t-1}, \Theta) + Q(X_{t-1}, \Theta) \nu_t \\ \mathbb{W}_t &= f(X_t, v_t),\end{aligned}$$

where the state vector is

$$X'_t = (b_{t+1}, \pi_t, y_t, v_t, s_t, g_t, u_t, w_t, z_t, n_t, MSE_{st}, MSE_{bt}, \text{vec}(\psi_t^s), \text{vec}(\psi_t^b)),$$

$\text{vec}(\cdot)$ is the vectorization operator, the observation variables are

$$\mathbb{W}'_t = (y_t, \pi_t, s_t, b_{t+1}, i_t),$$

and the parameter vector is

$$\Theta' = (\kappa, \alpha, \phi_\pi, \phi_y, \phi_b, \rho_g, \rho_u, \rho_w, \rho_z, \sigma_g, \sigma_u, \sigma_w, \sigma_z, \omega, \gamma_1, \gamma_2).$$

The measurement and state disturbances are v_t, ν_t respectively. The 4 exogenous shocks follow a linear transition equation with a diagonal matrix whose diagonals are the respective AR(1) coefficients.

In brief, the particle filter, like the Kalman filter, operates in both a prediction and update steps. The first step, given the previous period's particle approximation, is to draw a large number of innovations and then iterate on the state transition equation to yield a predicted next-period state. The predicted particles are then re-weighted based on the most recent data observation, this is the updating step. The updated weights are used to directly compute the likelihood value.

The Bootstrap particle filter, while conceptually straightforward, introduces several computational challenges. First, a stable approximation of the likelihood function requires a large number of particles. Even with efficient parallelization and vectorization each computation of the likelihood approximation is time-consuming. In our application, this is especially true since the state-vector consists of 34 variables. Adopting stochastic gradient learning versus constant gain learning lowers the computational cost considerably. With constant gain least-squares, the particle filter algorithm would require inverting the regressor covariance matrix 10^{12} times. Second, measurement noise in the observation equation is necessary for an accurate particle filter approximation of the likelihood function. The consequence is that the parameter estimates are estimated with greater uncertainty. We follow [Herbst and Schorfheide \(2015\)](#) in defining measurement error so that $\Sigma_v = 0.25 \times \text{diag}[\text{Var}(\mathbb{W}_T)]$.

To approximate the likelihood function we use the Bootstrap Particle Filter, as developed in [Herbst and Schorfheide \(2015\)](#):

Algorithm 1 *Bootstrap Particle Filter.*

1. Initialization Draw the initial particles $\nu_0^j \stackrel{iid}{\sim} p(X_0)$ and let $W_0^j, j = 1, \dots, M$.

2. Recursion For $t = 1, \dots, T$:

(a) Forecasting X_t . Iterate on the state-transition equation:

$$\hat{X}_t^j = g(X_{t-1}^j, \Theta) + Q(X_{t-1}^j, \Theta) \nu_t^j, \quad \nu_t^j \sim F_\nu(\cdot, \Theta)$$

(b) Forecasting $\mathbb{W}_t$. Define the weights

$$\hat{w}_t^j = p(\mathbb{W}_t | \hat{X}_t^j, \Theta)$$

where

$$p(\cdot | \cdot) \approx \frac{1}{M} \sum_{j=1}^M \hat{w}_t^j W_{t-1}^j$$

and

$$\hat{w}_t^j = (2\pi)^{-n/2} |\Sigma_v|^{-1/2} \times \exp \left\{ -\frac{1}{2} \left[\mathbb{W}_t - f(\hat{X}_t^j) \right]' \Sigma_v^{-1} \left[\mathbb{W}_t - f(\hat{X}_t^j) \right] \right\}$$

where Σ_v is the covariance matrix for the measurement errors.

(c) Updating. Normalize weights:

$$\hat{W}_t^j = \frac{\hat{w}_t^j W_{t-1}^j}{\frac{1}{M} \sum_{j=1}^M \hat{w}_t^j W_{t-1}^j}$$

(d) Selection. Define $\hat{m} = M / \left(M^{-1} \sum_{j=1}^M (\hat{W}_t^j)^2 \right)$ and let r_t be an indicator variable whenever $\hat{m} < M/2$.

Case 1: $r_t = 1$ Let $X_t^j, j = 1, \dots, M$ denote M iid draws from a multinomial distribution with support points/weights $\{\hat{X}_t^j, \hat{W}_t^j\}$ and set $W_t^j = 1, \forall M$.

Case 2: $r_t = 0$ Then set $X_t^j = \hat{X}_t^j$ and $W_t^j = \hat{W}_t^j$.

3. Likelihood approximation:

$$\ln \hat{p}(\mathbb{W}^T | \Theta) = \sum_{t=1}^T \ln \left(M^{-1} \sum_{j=1}^M \hat{w}_t^j W_{t-1}^j \right)$$

Finally, we use a Metropolis-Hastings algorithm to construct the posterior distribution:

$$p(\Theta|\mathbb{W}_T) \propto p(\mathbb{W}_T|\Theta) \times p(\Theta)$$

where $p(\Theta)$ is the prior distribution and $p(\Theta|\mathbb{W}_T)$ is the object of interest. Our algorithm samples from the posterior distribution through an adapted Random-Walk Metropolis Hastings (RWMH) MCMC technique, with the particle-filter based estimate of the likelihood function. Convergence properties of the algorithm are discussed in [Herbst and Schorfheide \(2015\)](#). One challenge for the RWMH is identifying a good candidate distribution to draw from, and to be sure that the chain samples from the entire support of the distribution. We do this in two ways. First, we use an adaptive approach to recursively update the candidate distribution combined with a long transient burn-in period. Second, in 99% of the time we draw from a mixture distribution, each proportional to the recursively updated candidate that is centered on the previously accepted draw from the RWMH. The remaining time the draws come naively from the candidate distribution. This ensures that the algorithm is not concentrated near a local maxima and that convergence occurs relatively quickly. We also used a tempering procedure in the early stages of the burn-in period.

We can now describe the Metropolis-Hastings algorithm.

Algorithm 2 *Metropolis-Hastings.* For $i = 1, \dots, N$

1. Draw a candidate θ from $q(\theta|\Theta^{i-1})$
2. Set $\Theta^i = \theta$ with probability

$$\alpha(\theta|\Theta^{i-1}) = \min \left\{ 1, \frac{\hat{p}(\mathbb{W}^T|\theta)p(\theta)/q(\theta|\Theta^{i-1})}{\hat{p}(\mathbb{W}^T|\Theta^{i-1})p(\Theta^{i-1})/q(\Theta^{i-1}|\theta)} \right\}$$

where $\hat{p}(\cdot|\cdot)$ is computed using Algorithm 1, and $p(\cdot)$ is the prior.

For the candidate density we use a variant on a random-walk Metropolis Hastings. With probability $\delta \approx 1$ we set

$$q(\cdot|\Theta^{i-1}) = N(\Theta^{i-1}, c\hat{\Sigma}_i)$$

where we use a recursive adaptive algorithm to compute $\hat{\Sigma}$:

$$\begin{aligned}\bar{\Theta}_i &= \frac{i+1}{100+i+1} \left[\frac{i}{1+i} \bar{\Theta}_{i-1} + \frac{1}{1+i} \Theta^{i-1} \right] + \frac{100}{100+i+1} \Theta^0 \\ \hat{\Sigma}_i &= \frac{i}{100+i} \left[\frac{i-1}{i} \hat{\Sigma}_{i-1} + \left(i \bar{\Theta}_{i-1} \bar{\Theta}_{i-1}' - (i+1) \bar{\Theta}_i \bar{\Theta}_i' \Theta^{i-1} \Theta^{(i-1)'} \right) / i \right] + \frac{100}{100+i} \hat{\Sigma}_0\end{aligned}$$

and with complementary probability

$$q(\cdot | \Theta^{i-1}) = N(\bar{\Theta}_i, c \hat{\Sigma}_i)$$

In the estimation, we set $M = 120,000$ and $N = 200,000$. We use a burn-in period of 80,000 draws. We ran a tuning run to initialize $\hat{\Sigma}$ during which we also used a tempering schedule. The priors are specified below.

Table A1: Prior distribution of parameters

Parameter	Dist.	Para(1)	Para(2)
<i>Structural parameters</i>			
κ	Beta	0.1	0.2
α	Beta	0.6	0.2
<i>Policy parameters</i>			
ϕ_π	Normal	1.50	0.25
ϕ_y	Normal	0.125	0.05
ϕ_b	Inv. Gamma	0.15	0.05
<i>Exogenous shocks</i>			
ρ_g	Beta	0.50	0.20
ρ_u	Beta	0.50	0.20
ρ_w	Beta	0.50	0.20
ρ_z	Beta	0.50	0.20
$100\sigma_g$	Inv. Gamma	0.01	2.00
$100\sigma_u$	Inv. Gamma	0.01	2.00
$100\sigma_w$	Inv. Gamma	0.01	2.00
$100\sigma_z$	Inv. Gamma	0.01	2.00
<i>Learning parameters</i>			
ω	Gamma	5.00	2.00
γ_1	Beta	0.015	0.015
γ_2	Beta	0.03	0.03

Many of our priors are the same as in [Eusepi and Preston \(2018\)](#) and [Herbst and Schorfheide \(2015\)](#). We set the prior for the intensity of choice in line with survey estimates provided in [Branch \(2004\)](#). Our prior for the gain parameters are informed by previous estimates provided in the literature. The observation equation also includes measurement errors, as discussed in the main text. We set these, following [Herbst and Schorfheide \(2015\)](#), as $\Sigma_v = 0.25 \times \text{diag} [\text{Var} (\mathbb{W}^T)]$.

Supplementary Appendix

Additional model details

This Appendix provides additional details on the model and its derivations. The reader is referred to [Woodford \(2013\)](#) for more complete details. All parameters and variables are explained in the paper above.

FIRMS. A firm j that can optimally reset price $p_t^*(j)$ will do so to satisfy the first-order condition

$$p_t^*(j) = (1 - \alpha\beta) \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} (E_t^j p_T^{\text{opt}} - p_{t-1}),$$

where $E_t^j p_T^{\text{opt}}$ is the perceived optimal price in period T . This condition can be written recursively:

$$\begin{aligned} p_t^*(j) &= (1 - \alpha\beta) (E_t^j p_t^{\text{opt}} - p_{t-1}) + (\alpha\beta) E_t^j p_{t+1}^*(j) + (\alpha\beta) \pi_t, \quad \text{where (6.1)} \\ E_t^j p_{t+1}^*(j) &\equiv (1 - \alpha\beta) \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} (E_t^j p_{T+1}^{\text{opt}} - p_t). \end{aligned}$$

TEMPORARY EQUILIBRIUM WITH HETEROGENEOUS BELIEFS. Recall the consumption function recursion:

$$\begin{aligned} c_t^i &= (1 - \beta)b_t^i + (1 - \beta)(Y_t - \tau_t) - \beta[\sigma - (1 - \beta)s_b]i_t - (1 - \beta)s_b\pi_t + \beta\bar{c}_t + \beta E_t^i v_{t+1}^i \\ v_t^i &= (1 - \beta)(Y_t - \tau_t) - [\sigma - (1 - \beta)s_b](\beta i_t - \pi_t) - (1 - \beta)\bar{c}_t + \beta E_t^i v_{t+1}^i. \end{aligned}$$

The latter two can be written as

$$c_t^i = (1 - \beta)b_t^i + \bar{c}_t - \sigma\pi_t + v_t^i,$$

which, together with $b_t \equiv \int b_t^i di$, $v_t \equiv \int v_t^i di$ and (9) yields aggregate demand

$$Y_t = \bar{c}_t + G_t + (1 - \beta)b_t + v_t - \sigma\pi_t. \quad (6.2)$$

To express the aggregate demand equation in explicit dependence of expectations,

we apply (8) and (6.2) to the v_t^i recursion and obtain

$$v_t^i = (1 - \beta)v_t + (1 - \beta)\beta(b_{t+1} - b_t) - \beta\sigma(i_t - \pi_t) + \beta E_t^i v_{t+1}. \quad (6.3)$$

Averaging over expectations in (6.3) and plugging into (6.2) yields (10). Thus, because the continuation variable v_t^i consists of aggregate variables that are beyond the household's control, and the agents understand their optimal consumption plan and perceived budget constraints, the aggregation result in the main text follows immediately. The ease with which the heterogeneous beliefs aggregate follows from the infinite-horizon learning consumption, which depends on household i 's subjective *forecasts* of aggregate variables beyond their control. An example of where aggregation of heterogeneous beliefs is made more difficult by higher-order beliefs is provided by Branch and McGough (2009).

Next, as in Woodford (2013, Section 2.3), in equilibrium the optimal price in this model can be expressed as

$$p_t^{\text{opt}} = p_t + \zeta (Y_t - Y_t^n) + \mu_t, \quad (6.4)$$

where $\zeta > 0$ is a composite term of structural parameters measuring the output elasticity of a firm's optimal price.²⁵ The exogenous random variable Y_t^n is the natural level of output that captures exogenous demand shocks and μ_t represents exogenous disturbances to the desired markup over marginal cost.

As the firm's price is a decision variable, it is natural to impose that $E_t^j p_t^{\text{opt}} = p_t^{\text{opt}}$. It follows, then, from plugging (6.4) and (5) into (6.1) that

$$p_t^*(j) = (1 - \alpha)p_t^* + (1 - \alpha\beta) [\zeta y_t + \mu_t] + \alpha\beta E_t^j p_{t+1}^*(j).$$

Again averaging across firms, defining the output gap as $y_t \equiv Y_t - Y_t^n$, parameter $\kappa \equiv [(1 - \alpha)(1 - \alpha\beta)\zeta]/\alpha$, and the cost-push supply shock as $u_t \equiv \{[(1 - \alpha)(1 - \alpha\beta)]/\alpha\}\mu_t$, yields the New Keynesian Phillips curve

$$\pi_t = (1 - \alpha)\beta \int E_t^j p_{t+1}^*(j) dj + \kappa y_t + u_t. \quad (6.5)$$

²⁵The term is defined in Woodford (2003, ch.3-4).

As for the households, after applying the law of iterated expectations a firm j sets

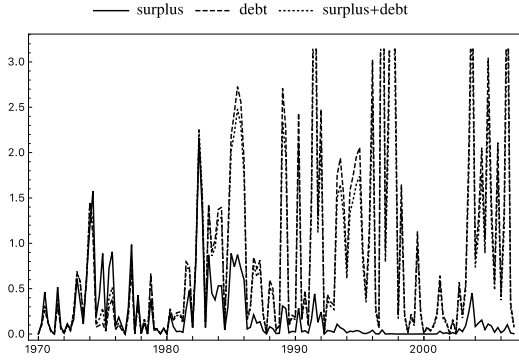
$$p_t^*(j) = (1 - \alpha)p_t^* + (1 - \alpha\beta) [\zeta y_t + \mu_t] + \alpha\beta E_t^j p_{t+1}^*$$

and, an aggregate New Keynesian Phillips Curve results after averaging across all firms:

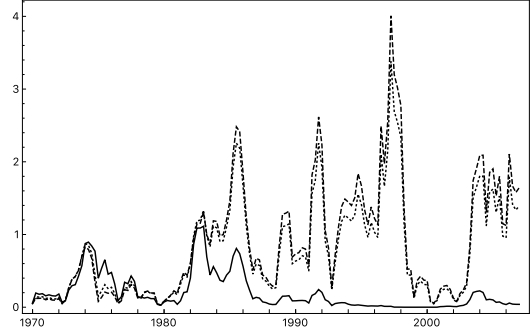
$$\pi_t = (1 - \alpha) \beta \hat{E}_t p_{t+1}^* + \kappa y_t + u_t.$$

Restricted perceptions vs. rational expectations

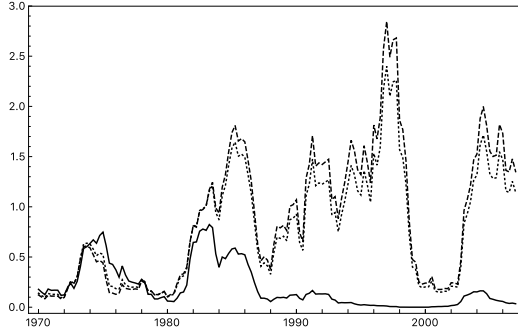
Since $n = 0$, in the estimated model, is quantitatively close to the rational expectations model, our estimation results present an econometric test of our model of restricted perceptions against rational expectations. The results tell us that the data prefer the specification of the model with a non-Ricardian equilibrium over most of the sample. To better assess the plausibility of these results, we present (informal) evidence from the Survey of Professional Forecasters (SPF). Although beyond the scope of this paper, a complete empirical analysis would use the empirical framework in [Branch \(2004\)](#) to analyze the probability that an individual-level survey forecast was made by a simple model with a restricted set of fiscal variables. As a first pass of providing some external evidence, we compute the statistical scores of the median SPF forecast across three different possible sets of forecasting model regressors: one that includes the primary surplus only, one that includes the debt, and one that includes both. Specifically, we compute moving averages of the statistical score $Ex_{j,t-1}(\pi_t - \pi_{t-1,t}^e)$ where $x_{j,t} \in \{s_t, b_t\}$, π_t is the PCE inflation rate, and $\pi_{t-1,t}^e$ is the one-step ahead median SPF survey forecast. The results are plotted in [Figure A1](#). Notice that within a restricted perceptions equilibrium, the (time-)average score should be zero. Thus, if the surplus model leads to a lower, and near zero, score then this provides indirect evidence in favor of a restricted perceptions equilibrium with non-Ricardian beliefs. The results in the scores-Figure shows that, beginning in the late 1980's, the median SPF is consistent with a greater share of forecasters using the primary surplus as the fiscal variable. In fact, in the late 1990's that score vector is near zero, as predicted by a non-Ricardian restricted perceptions equilibrium.



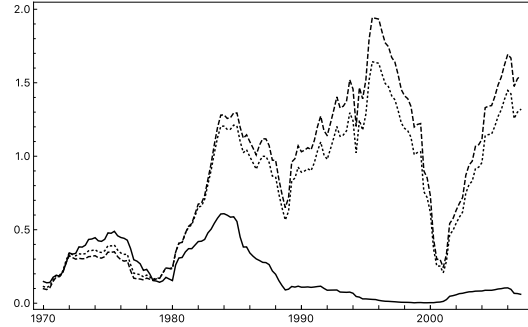
(a) Raw scores.



(b) 4-Qtr. Moving-average.



(c) 8-Qtr. Moving-average.



(d) 5-Yr. Moving-average.

Figure A1: Measured scores for different sets of predictors in the Survey of Professional Forecasters. Each panel computes the scores with different moving average lengths. A score close to zero is consistent with a restricted perceptions equilibrium.

Connection to rational expectations/transitional learning dynamics: details

It is natural to wonder how strong the misspecification assumptions are and whether a learning model with a correctly specified forecasting model may still feature endogenously non-Ricardian beliefs. In this Appendix, we address this concern through the lens of econometric learning, e.g., [Evans and Honkapohja \(2001\)](#), and relax the parsimony assumption by assuming that agents form their expectations via a correctly specified model

$$v_t = \psi^s s_t + \psi^b b_t.$$

We continue to maintain the imperfect knowledge assumptions, including not *a priori* imposing Ricardian beliefs, and further assume that the belief coefficients ψ^s, ψ^b are real-time estimates from a constant gain learning model, a form of discounted least-squares. With this perceived law of motion, the actual law of motion implied by these beliefs can be written as

$$v_t = S(\psi^s, \psi^b)' \begin{bmatrix} s_t \\ b_t \end{bmatrix} - (1/(1 + \sigma^{-1}\phi_y^{-1})) g_t,$$

where

$$S(\psi^s, \psi^b) = \frac{1}{1 + \sigma\phi_y} \begin{bmatrix} -\beta^{-1}(\psi^s\phi_b + \psi^b + 1 - \beta) \\ \beta^{-1}(\psi^s\phi_b + \psi^b) + (1 - \beta)(\beta^{-1} - 1) - \sigma\phi_y(1 - \beta) \end{bmatrix}.$$

The S -map has the usual interpretation: given a perceived law of motion with coefficients $(\psi^s\psi^b)'$ the corresponding coefficients in the actual law of motion implied by these beliefs are $S(\psi^s, \psi^b)$. A rational expectations equilibrium is a fixed point of the “ S -map”, i.e., $\Theta^* = S(\Theta^*)$, $\Theta' = (\psi^s, \psi^b)$.

We can solve for the “mean dynamics” associated to the constant gain learning dynamics as a (small gain) approximation to the expected transitional learning dynamics. Adapting the stochastic recursive approximation results in [Evans and Honkapohja \(2001\)](#) it is possible to show that, across sequences of increasingly smaller gain parameters, the learning dynamics weakly converge to the expected path for Θ given by the following system of ordinary differential equations (O.D.E.’s)

$$\begin{aligned} \dot{\Theta} &= R^{-1}M(S(\Theta) - \Theta) \\ \dot{R} &= M - R, \end{aligned}$$

where

$$M = E \begin{bmatrix} s_t \\ b_t \end{bmatrix} \begin{bmatrix} s_t & b_t \end{bmatrix}.$$

The mean dynamics are the solution path, for a given initial condition $\Theta(0)$, to this system of O.D.E.’s.

The mean dynamics are useful for understanding the qualitative nature of learning dynamics. Standard results in the literature show that constant gain learning dynamics are distributed asymptotically normal with a mean equal to the rational expectations

equilibrium values and a variance that is proportional to the size of the gain parameter. Over time, one can expect with high probability to see coefficient estimates Θ that fluctuate around Θ^* . The response of Θ to a particular sequence of unlikely shocks is described by the “escape dynamics”, which provide the “most likely unlikely” path away from the rational expectations equilibrium, and then the mean dynamics describe the transition path back to the equilibrium.²⁶ The escape dynamics, therefore, can be thought of as re-initializing the mean dynamics. We can use different starting values for the mean-dynamics to characterize the type of learning paths that we might actually observe.

We use these insights to show that the learning dynamics in the case of fully specified perceived laws of motion will be drawn, for a finite stretch of time, towards the $n = 1$ restricted perceptions equilibrium. The mean dynamics are derived from a continuous time approximation of the real-time learning dynamics and the application of a law of large numbers, however, it is straightforward to convert the notional time in the O.D.E. to actual discrete time according to $t = \gamma^{-1}\tau^\gamma$, so that a small constant gain, γ , corresponds to a long stretch of real time.

Figure A2 plots the mean dynamics for a particular illustrative parameterization: $\phi_y = 0.5$, $\phi_b = 0.9$, $\sigma = 2$, $\sigma_z^2 = 1$.²⁷ We then choose initial values for the ψ^s, ψ^b that are both above their rational expectations equilibrium values. The mean dynamics O.D.E. is then solved and Figure A2 plots the expected learning path (ψ^s shown). The experiment is to imagine an “escape” that has driven beliefs above their rational expectations equilibrium values and use the solution to the mean dynamics O.D.E. to trace out how the economy is most likely to respond.

The figure plots (solid line) the expected transition path for ψ^s while the dashed line is the value in an $n = 1$ restricted perceptions equilibrium, that is the value for ψ^s and ψ^b that would arise in an $n = 1$ RPE. The learning dynamics are expected to eventually converge to the rational expectations equilibrium. However, with a small constant gain, that speed of convergence can be quite slow. Interestingly, along the transition path for ψ^s , the beliefs hover for a finite stretch of time at its $n = 1$ RPE. This coincides with a path for ψ^b (not shown) that is also drifting down towards its RPE value. As the path for ψ^b continues to transition towards its REE value, this draws ψ^s

²⁶See Williams (2019) for details and a comprehensive set of results and toolkit on escape dynamics in constant gain learning models.

²⁷For expositional ease, we present an example where the RPE and REE values are starkly far apart.

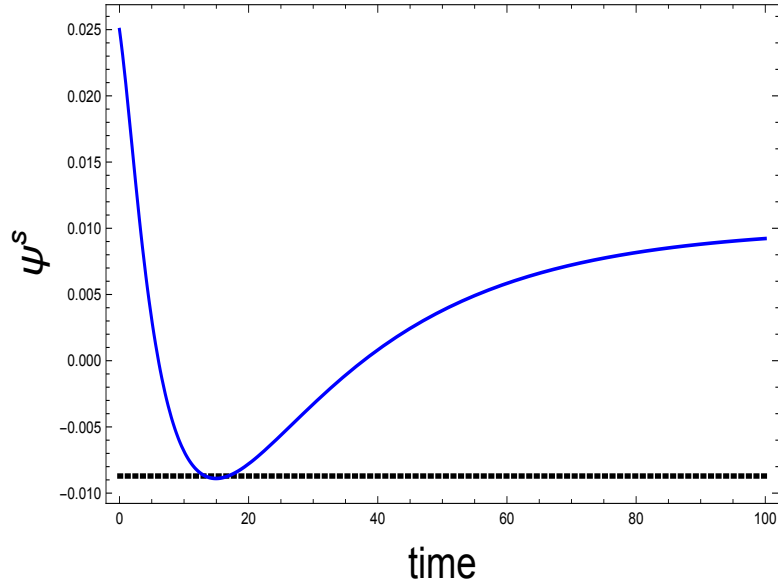


Figure A2: Expected learning dynamics for a correctly specified forecast model

away from its RPE value and back towards the rational expectations equilibrium.

We conclude from Figure A2 that even if agents did not face any computational/cognitive constraints the RPE is a relevant concept as we can expect recurrent escapes near a non-Ricardian equilibrium even when all agents in the economy form forecasts from a correctly specified model. Moreover, for small gains γ , the economy will persist near the RPE for long stretches of time. These dynamics are reminiscent of [Cho and Kasa \(2017\)](#).

The mean dynamics in Figure A2 also help to better understand the connection between this paper and [Eusepi and Preston \(2018\)](#). In their model, beliefs nest the rational expectations equilibrium however the agents attempt to learn about the long-run stances of fiscal and monetary policy. They show how learning dynamics can generate fluctuations with non-Ricardian effects. These non-Ricardian effects are strengthened in economies with a high steady state debt/output ratio. The theory of expectation formation here emphasizes restricted perceptions which require the agents to estimate the relevant auto- and cross-covariances which, in combination, gives scope for escape dynamics. Therefore, the theory in this paper is complementary to theirs, while providing an equilibrium explanation for the phenomenon of non-Ricardian beliefs.

Additional theoretical results

In the special case of iid fiscal shocks, $\sigma = 0, \kappa = 0, \phi_y = 0$ (the case considered in [Woodford \(2013\)](#)), additional theoretical results are possible. We present them here for completeness.

The following result provides necessary and sufficient conditions for the existence of multiple misspecification equilibria in the limiting case $\omega \rightarrow \infty$.

Theorem 1 *Consider the special parametric case of the model. Let $\omega \rightarrow \infty$ and $\beta > 2/3$. There exists multiple misspecification equilibria, $n^* \in \{0, \hat{n}, 1\}$, if and only if*

$$\underline{\phi}(\beta) < \phi_b < \bar{\phi}(\beta),$$

where

$$\begin{aligned}\underline{\phi}(\beta) &= \max \left\{ 1 - \beta, \frac{1}{4} (4 - 2\beta - 3\beta^2) \right\} \\ \bar{\phi}(\beta) &= \frac{1}{4} \left[(2 - 3\beta - 2\beta^2) + \sqrt{4 + 4\beta + 5\beta^2 - 4\beta^3} \right].\end{aligned}$$

We can similarly characterize the necessary and sufficient conditions for unique (non-)Ricardian equilibria.

Corollary 6.0.1 *Let $\omega \rightarrow \infty$. The following results hold.*

i. *A unique misspecification equilibrium $n^* = 1$ exists if and only if*

$$1 - \beta < \phi_b \leq \frac{1}{4} (4 - 2\beta - 3\beta^2).$$

ii. *A unique misspecification equilibrium $n^* = 0$ exists if and only if*

$$\bar{\phi}(\beta) < \phi_b < 1.$$

Corollary 6.0.2 *For finite $\omega > 0$ and $2/3 < \beta < 1$, $\exists \underline{\phi}(\beta)$ such that for all $\underline{\phi}(\beta) < \phi_b < \bar{\phi}(\beta)$ there exists three interior non-Ricardian misspecification equilibria n_l^*, n_h^* where $0 < n_l^* < \hat{n} < n_h^* < 1$.*

Remark 6.0.1 *The existence of the unique non-Ricardian equilibrium, $n^* = 1$, requires that $\beta < 2/3 \Leftrightarrow 1 - \beta < \frac{1}{4} (4 - 2\beta - 3\beta^2)$. Thus, the special case non-Ricardian*

equilibria is likely to arise in empirically plausible models in the form of multiple equilibria.

Figure A3 illustrates the results in Theorem 1 and Corollary 6.0.1. In the graph, are the combinations of (β, ϕ_b) consistent with multiple or unique equilibria. The large unshaded area in the lower half of the plot corresponds to active fiscal policy, i.e., $(1 - \beta) < \phi_b$. The restriction to Ricardian policy rules out equilibria in this region. Then, moving outward from the origin, the shaded area with a dashed-boundary consists of the pairs of (β, ϕ_b) consistent with a unique non-Ricardian equilibrium. The next shaded area, with grid lines, corresponds to the existence of multiple equilibria. Finally, the outermost shaded area is where a self-confirming Ricardian equilibrium, $n^* = 0$, is the unique misspecification equilibrium.

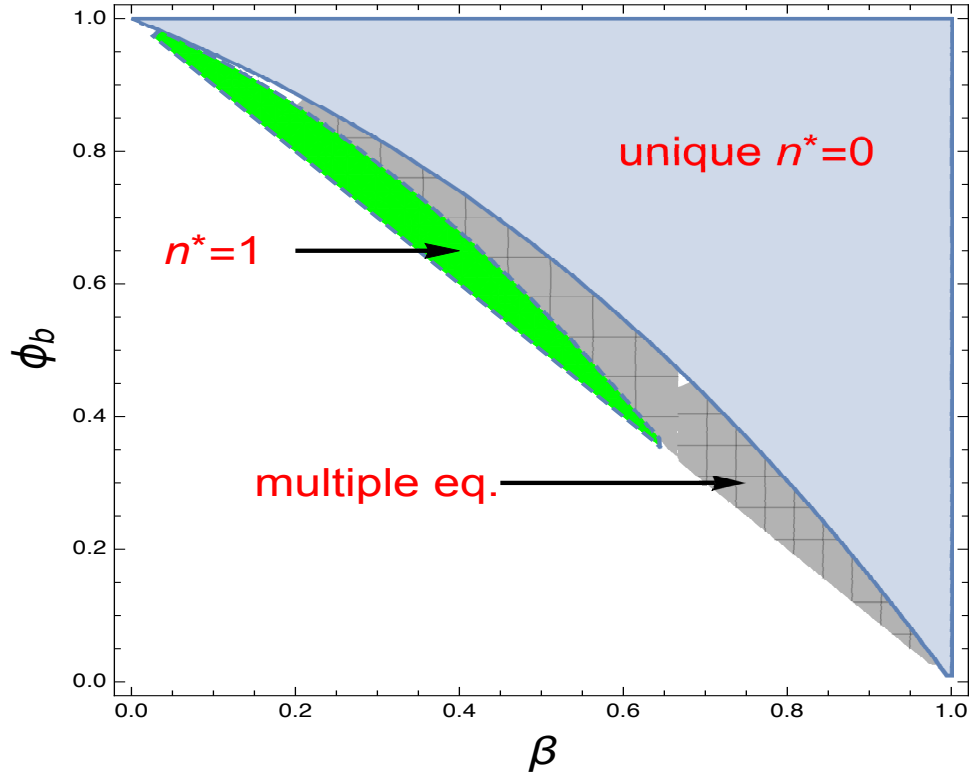


Figure A3: Equilibrium existence

Theorem 1, and its corollaries, is the main theoretical result of the paper: even though fiscal policy is passive, non-Ricardian beliefs can emerge endogenously. For ϕ_b within a certain range $[\underline{\phi}, \bar{\phi}]$ then the non-Ricardian outcome can be sustained in a misspecification equilibrium. Most interestingly, for these fiscal policy rules there

exists multiple misspecification equilibria with existence also of a Ricardian equilibrium $n^* = 0$. As we discuss below the case of multiple equilibria leads to interesting model dynamics that offer an alternative interpretation to regime-switching non-Ricardian policy effects. As an example, Figure A4 plots the T-map $T_\omega(n)$ and the relative predictor fitness function $F(n)$ when $\beta = 0.99$, $\phi_b = 0.015$, and $\sigma_z = 1$. In the bottom plot, it is evident that $F(0) < 0$ and $F(1) > 0$, which implies the existence of both Ricardian and non-Ricardian equilibria, respectively. The top panel plots the T-map for various values of n when $\omega \rightarrow \infty$. This figure clearly indicates the three misspecification equilibria. In the quantitative analysis, the interior misspecification equilibrium is unstable, and so the learning dynamics can feature recurrent switching between the basins of attraction of the two (non-)Ricardian equilibria.

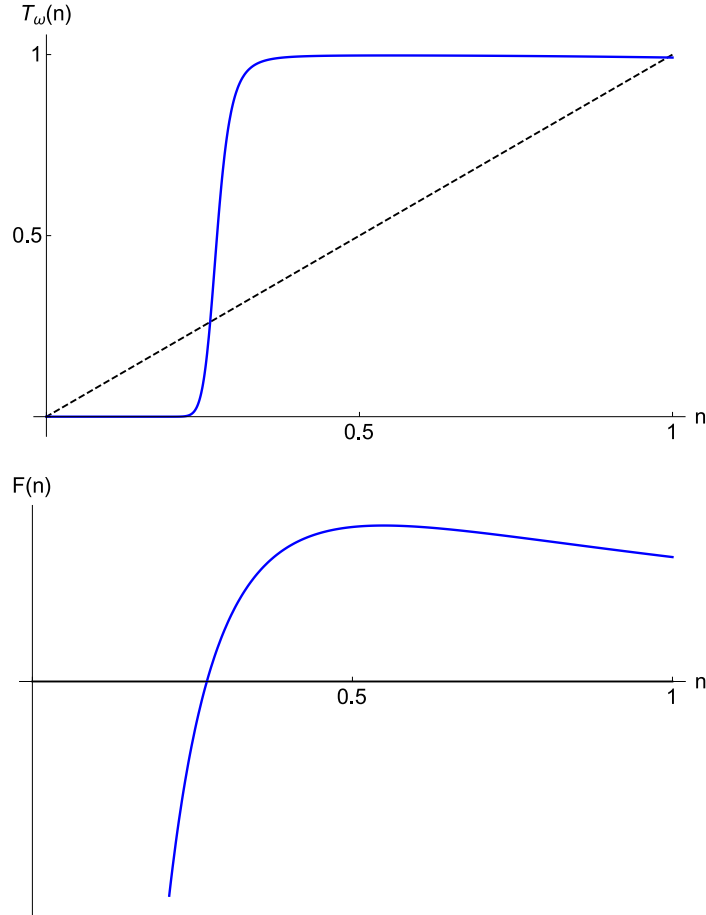


Figure A4: Multiple (non-)Ricardian equilibria in the special case.

Why would individuals ever prefer the non-Ricardian forecasting model? An closer

examination of $F(n)$ provides the intuition. The predictor fitness measures, in the simple case, are

$$\begin{aligned} -EU^s &= \mu_{v1}^2 Eb_t^2 + (\mu_{v2} - \psi^s)^2 Es_t^2 + 2\mu_{v1} (\mu_{v2} - \psi^s) Eb_t s_t \\ -EU^b &= (\mu_{v1} - \psi^b)^2 Eb_t^2 + \mu_{v2} Es_t^2 + 2\mu_{v2} (\mu_{v1} - \psi^b) Eb_t s_t. \end{aligned}$$

Thus, a given model's predictor fitness depends, essentially, on three components. First, the distance between the belief parameter and the corresponding coefficient in the actual law of motion. Second, how volatile the missing component is from their forecasting model. Third, a term that is best interpreted as the omitted variable bias component of the prediction error. These distances are all weighted by the corresponding equilibrium covariances of the state variables.

After calculating the differences between these predictor fitness functions leads to

$$F(n) = [\psi^b (\psi^b - 2\mu_{v1}) - \psi^s (\psi^s - 2\mu_{v2}) \phi_b^2] Eb_t^2 + 2 [\mu_{v1} \psi^s - \mu_{v2} \psi^b] Eb_t s_t - \psi^s [\psi^s - 2\mu_{v2}] \sigma_z^2.$$

The fraction of agents who use the surplus-model then depends a balancing of how well the surplus model captures the serial correlation of the debt process and the additional predictive power from the surplus model conditioning indirectly on the z_t innovation. For small values of ϕ_b , the surplus- and debt-models are weakly correlated and so $n^* = 1$ can emerge as the unique equilibrium as it best captures the contemporaneous z_t innovations. Such an equilibrium is self-fulfilling in the sense that with $n = 1$ the indirect effect of z_t is strengthened through the self-referential features of the model. For larger values of ϕ_b the agents will always mass onto the Ricardian predictor as the surplus and debt models become more correlated the potential forecasting advantage of the surplus model is reduced. Finally, when ϕ_b takes middling values between these two extremes, then either (non-)Ricardian equilibrium can emerge. Now the debt-model is sufficiently correlated with the surplus-model that an agent will prefer it if $n = 0$. Similarly, the debt and surplus-models are not too closely correlated so that when $n = 1$, the agents will prefer the surplus model that captures the contemporaneous innovation in its forecast.

To sum up: for empirically plausible parameterizations of the special case, a Ricardian self-confirming equilibrium is always supported as a misspecification equilibrium. For a range of fiscal policy parameters there can also exist a non-Ricardian equilibrium. The non-Ricardian equilibrium is the unique outcome for ϕ_b sufficiently small and close

to the border between active and passive fiscal policy.